

**DESIGN AND DEVELOPMENT OF A COMMUNICABLE FAULT
PASSAGE INDICATOR FOR LOW VOLTAGE DISTRIBUTION
NETWORKS IN BANGLADESH**

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Declaration

It is hereby declared that

- The Final Year Design Project (FYDP) submitted is my/our own original work while completing degree at BRAC University.
- The Final Year Design Project (FYDP) does not contain material previously published or written by a third party, except where this is appropriately cited through full and accurate referencing.
- The Final Year Design Project (FYDP) does not contain material which has been accepted, or submitted, for any other degree or diploma at a university or other institution.
- I/We have acknowledged all main sources of help.

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Ethics Statement

The project titled “Design and Development of a Communicable Fault Passage Indicator for Low Voltage Distribution Lines in Bangladesh” was done adhering to all the requirements and the highest possible ethical standards. The entire team made a collective effort to complete the task in the utmost sincere manner. The interaction between the team members was polite, fair and every decision was taken keeping all the stakeholders in mind. The literature review was done accordingly and all the works were cited using IEEE Format. The Turnitin Plagiarism and AI detector was used and the similarity index of this report came out to be 10% and AI percentage was below 35%.

Abstract/ Executive Summary

Bangladesh's low-voltage distribution network suffers from prolonged fault restoration times, typically three to four hours, due to a heavy reliance on manual line patrolling for fault localisation. This project presents the design and development of a communicable Fault Passage Indicator (FPI) that addresses this gap through a working, field-tested prototype. The system uses an ESP8266 microcontroller with a Gravity analog AC current sensor to continuously monitor line current and classify fault events single line-to-ground, line-to-line, double line-to-ground, and three-phase faults while a u-blox NEO-6M GPS module tags each event with location data. Detected faults are indicated locally through dedicated LEDs and transmitted wirelessly via ESP-NOW to an ESP32 receiver, which feeds a live dashboard for remote monitoring. The prototype is powered by a rechargeable battery pack with solar-charging provision. Bench testing confirmed reliable classification across all four fault types, and the system was successfully demonstrated at the final project showcase, where it was awarded second place. The completed work validates a low-cost, locally buildable approach to reducing fault detection and restoration time in Bangladesh's LV distribution networks.

Keywords: Fault Passage Indicator; Distribution Network; fault detection; LoRa communication; smart grid

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Chapter 1: Introduction- [CO1, CO2, CO10]

1.1 Introduction

The electric power distribution network forms the final stage of the electrical power system, delivering electricity from distribution substations to residential, commercial, and industrial consumers. As electricity demand continues to rise due to rapid urbanization, industrial growth, and economic development, the reliability and efficiency of distribution networks have become increasingly important. While Bangladesh has achieved remarkable progress in expanding its electricity generation capacity and increasing nationwide electrification, maintaining uninterrupted power delivery remains a significant challenge because of frequent faults occurring in low-voltage (LV) distribution feeders. These faults not only interrupt electricity supply but also reduce system reliability, increase operational costs, and negatively affect both utility providers and consumers.

Low-voltage distribution systems in Bangladesh are particularly vulnerable to electrical faults arising from aging conductors, equipment deterioration, excessive loading, illegal power connections, insulation failures, adverse weather conditions, and accidental conductor contacts. Following the occurrence of a fault, conventional distribution systems primarily depend on protective devices such as circuit breakers, fuses, and overcurrent relays to isolate the faulty section. However, while these protection devices successfully disconnect the affected feeder, they generally provide little or no information regarding the exact location of the fault. Consequently, utility personnel must manually inspect distribution lines to identify the fault location before restoration activities can begin. This manual process significantly increases outage duration, operational expenditure, and customer dissatisfaction while adversely affecting power quality indices such as the System Average Interruption Duration Index (SAIDI) and Energy Not Supplied (ENS).

To improve distribution reliability and reduce restoration time, modern utilities have increasingly adopted Fault Passage Indicators (FPIs). FPIs are intelligent field devices installed along distribution feeders that continuously monitor line conditions and identify the passage of abnormal fault currents. By indicating the section through which a fault current has passed, these devices significantly reduce the time required to locate faults and isolate damaged network segments. Recent advances in embedded systems, wireless communication, Internet of Things (IoT) technologies, and low-power sensing hardware have enabled FPIs to evolve from simple visual indicators into intelligent monitoring devices capable of transmitting real-time fault information to remote monitoring centers.

Various sensing technologies have been investigated for fault detection applications, including conventional current transformers (CTs), Rogowski coils, Hall-effect current sensors, capacitive probes, and non-contact magnetic sensors. Conventional CT-based protection systems remain widely deployed because of their maturity and standardization; however, they often suffer from large physical dimensions, installation complexity, saturation

under high fault currents, and relatively high deployment costs, particularly when extensive feeder coverage is required. Similarly, Rogowski coils provide excellent high-frequency response but require additional signal conditioning and integration circuits. Capacitive and electric-field sensors offer attractive non-contact measurement capabilities but continue to face challenges related to calibration stability, environmental robustness, and long-term field reliability. Hall-effect current sensors have emerged as an attractive alternative because they provide galvanic isolation, compact size, low power consumption, and relatively simple installation while maintaining sufficient measurement accuracy for low-voltage monitoring applications.

In parallel with developments in sensing technologies, wireless communication has become an essential component of modern distribution automation systems. Communication technologies such as Wi-Fi, GSM, LoRaWAN, LTE-M, Power Line Communication (PLC), and short-range peer-to-peer protocols have enabled remote monitoring of electrical assets with significantly reduced infrastructure requirements. For low-cost embedded applications, ESP-NOW communication has gained considerable attention due to its low latency, low power consumption, and direct device-to-device communication capability without requiring an external wireless access point. When combined with cloud-based IoT platforms, wireless communication enables maintenance personnel to receive real-time fault notifications together with system status information, thereby facilitating faster decision-making and improved network management.

Considering these technological developments and the operational challenges experienced by Bangladesh's distribution utilities, this project presents the design, development, and implementation of a communicable Fault Passage Indicator (FPI) for low-voltage distribution networks. The developed prototype employs a WCS1800 Hall-effect current sensor integrated with an ESP32 microcontroller to continuously monitor feeder current and detect abnormal operating conditions. Once a fault is detected, the system transmits fault information together with geographical coordinates obtained from a GPS module to an ESP8266-based receiver using the ESP-NOW wireless communication protocol. The receiver subsequently forwards the received information to a cloud-based monitoring platform using the Blynk IoT application, enabling both local and remote monitoring of distribution feeder conditions. This integrated sensing and communication architecture provides a practical, economical, and scalable solution for improving fault localization and reducing restoration time in low-voltage distribution systems.

Unlike many commercially available Fault Passage Indicators that are primarily intended for medium-voltage distribution systems and require expensive sensing and communication infrastructure, the proposed system focuses on developing a cost-effective solution specifically suitable for low-voltage distribution networks commonly found in Bangladesh. By combining low-cost sensing hardware, embedded processing, wireless communication, and IoT-based remote monitoring within a compact platform, the developed prototype demonstrates the feasibility of implementing intelligent fault monitoring without requiring

major modifications to the existing distribution infrastructure. Furthermore, the proposed architecture establishes a foundation for future smart distribution networks through its compatibility with cloud-based monitoring platforms and potential integration with advanced distribution management systems.

The remainder of this chapter discusses the specific problem addressed by this research, reviews the existing state of fault detection technologies and distribution automation practices, identifies the research gaps that motivated this work, and highlights the relevance of intelligent Fault Passage Indicator systems for the current and future development of Bangladesh's electrical power distribution network.

1.1.1 Problem Statement

Bangladesh's low-voltage (LV) distribution network which serves mainly residential and commercial consumers face frequent faults. Bangladesh's distribution network experiences technical (10–15%) and non-technical (5–10%) losses (According to DPDC) despite sufficient generation capacity, indicating inefficiencies at the distribution level. The LV networks remain largely unmonitored, relying on manual inspections and line patrols for fault localization. This leads to long restoration times of about 3-4 hours on average [1]. These outages are caused due to old conductors, overloading, illegal connections, high-impedance faults and environmental disturbances [2]. Additionally, the lack of accurate load forecasting and real-time monitoring makes it challenging for utilities to predict faults and optimize grid management. Prolonged fault detection contributes to high SAIDI (System Average Interruption Duration Index) values (12–15 hours annually), particularly in rural distribution networks. (DPDC Annual Report 2023-2024). It significantly increases ENS (Energy Not Supplied) and the associated economic cost in \$/kWh of unserved energy, often dominating the annual reliability penalties imposed on utilities [3], [4].

1.1.2 Background Study

From a hardware standpoint, many MV (medium voltage) and LV feeders still rely on sparse protection relays and a limited number of field devices, so a single missed or false indication can expand the isolation area, raising the number of affected customers and the total ENS [3], [4], [8]. Reviews of distributed-generation-integrated networks emphasize that hardware-based protection schemes must operate correctly under wide variations in short-circuit level; otherwise, protection blinding and unnecessary trips increase both direct interruption costs and regulatory penalties [9]. As utilities seek to minimize capital and operational expenditure, there is strong motivation to deploy low-power, environmentally robust fault-sensing hardware with high mean-time-between-failures (MTBF), modest per-unit cost and dependable fault indication to support fast, selective restoration [3], [4], [8], [9]. Conventional LV protection relies heavily on iron-cored current transformers (CTs) feeding overcurrent and earth-fault relays. While mature and widely standardized, CTs are bulky, require primary circuit interruption for installation and can saturate under high fault

currents or DC offset. This causes measurement distortion and compromised fault detection and location [3], 2, [8]. CT errors of several percent in amplitude and phase under transient or high-burden conditions are common, limiting the achievable accuracy of impedance-based or directional elements that depend on precise current phasors [3], [4].

Despite significant advances in sensing technologies and intelligent monitoring systems, several challenges associated with existing fault detection systems remain unresolved. Conventional CT-based protection systems require extensive installation work and provide limited fault localization capability. High-precision monitoring devices often involve substantial capital investment and increased power consumption, restricting large-scale deployment across low-voltage distribution networks. Many advanced sensing technologies have been validated only under laboratory conditions, with limited evidence regarding their long-term reliability under harsh outdoor environments involving temperature variation, humidity, dust, vibration, ultraviolet exposure, and electromagnetic interference. Furthermore, sophisticated multi-sensor monitoring platforms generally assume the availability of extensive communication infrastructure and dense measurement coverage, conditions that are rarely available in developing countries such as Bangladesh. These limitations collectively demonstrate the need for compact, low-cost, energy-efficient, and communicable fault monitoring systems capable of providing dependable fault indication while remaining suitable for widespread deployment in practical low-voltage distribution networks.

1.1.3 Literature Gap

Fault passage indicators (FPIs) are dedicated field devices that monitor local line conditions to indicate the passage of fault currents. Modern FPIs incorporate various sensing hardware: clamp-on CTs or Rogowski coils for current, capacitive or electric-field probes for voltage and integrated microcontrollers with low-power communication modules [3], [8], [13]. In resistor-grounded MV networks, earth faults are frequent but can have relatively low zero-sequence currents, challenging classical FPI pickup criteria. An adaptive zero-sequence overcurrent criterion for FPIs demonstrated that, by adjusting pickup thresholds and time settings based on network conditions, detection of earth faults, especially high-impedance faults, can be improved by more than 40% compared with classical fixed-threshold criteria [16]. This highlights that even with conventional sensors, intelligent signal processing in FPI hardware can significantly improve detection sensitivity. Alternative non-contact FPI-like concepts based on capacitive probes mounted near conductors have been proposed to detect zero-sequence voltage components and asymmetry without direct electrical connection [13]. Laboratory prototypes have shown that suitably arranged capacitive probes can detect earth faults with sensitivity comparable to conventional VT-based systems [13]. However, moving from prototypes to outdoor hardware requires addressing issues like stable calibration under temperature and humidity, robust mechanical design, adequate IP rating for rain and dust and ensuring sufficient immunity to overvoltages and electromagnetic interference [3], [13], [16]. Hall-effect current sensors (HECSs) and related integrated magnetic sensors offer galvanic

isolation, wide bandwidth, and compact form factors, with typical linearity on the order of $\pm 0.5\text{--}1\%$ and dynamic ranges from a few amperes to kiloampere levels [10]. Implementation studies highlight their advantages in terms of low power consumption (tens of mW for integrated CMOS Hall front-ends) and suitability for embedded or retrofit applications in power electronics and smart-grid environments [10]. However, Hall sensors are sensitive to temperature and external fields; temperature drift and offset errors can degrade accuracy, especially when operating over industrial ranges of $-40\text{ }^{\circ}\text{C}$ to $+85\text{ }^{\circ}\text{C}$ [10]. Shielding and calibration circuits increase complexity and cost, and long-term stability and MTBF under outdoor MV conditions (humidity, vibration, pollution) remain less documented in field studies compared with traditional CTs [3], [4], [10]. In resource-constrained utilities, the need to balance per-unit cost, installation complexity, insulation coordination, and accuracy has slowed the systematic replacement of CT-based hardware, particularly on overhead rural feeders [3], [8]. Rogowski coils demonstrated high signal-to-noise ratio and reliable measurement of high-frequency components, outperforming high-frequency current transformers for partial discharge (PD) detection in MV/HV air-insulated switchgear [11]. Being lightweight and flexible, Rogowski coils can be retrofitted around existing conductors without interrupting service and their inherently low output voltage contributes to intrinsic safety. However, Rogowski coils measure only the derivative of current and require integrating electronics, which must maintain linearity and low noise over large dynamic ranges, including from low load currents up to tens of kA of fault current [11]. Sensitivity to mutual coupling and external electromagnetic interference complicates installation in compact MV switchgear and cable terminations [11], [12]. Non-contact magnetic sensors, such as external loop antennas or D-dot/E-field probes, have been explored for PD and transient detection. These sensors offer fully non-intrusive operation and good high-frequency response but typically provide only relative or uncalibrated measurements without careful calibration, limiting their use for quantitative fault-current estimation or impedance-based location [11], [12]. For field deployment on overhead lines and outdoor equipment, environmental protection (e.g., IP67 ingress rating) and mechanical robustness under wind and contamination are essential, yet many research prototypes are characterized only in laboratory or indoor test bays [11]–[13]. Fault-sensing hardware research is geographically diverse. In Europe, work on capacitive probes for LV earth-fault detection and on improved FPI criteria for resistor-grounded networks has focused on overhead lines typical of Central and Eastern European grids [13], [16]. Hybrid PD sensing for LV switchgear has been driven by European utilities and manufacturers concerned with asset health and life extension [11]. In developing countries and emerging smart cities, research highlights the mismatch between sophisticated sensor/communication solutions and the financial capabilities of local utilities; low-cost, low-power, easily deployable devices are repeatedly identified as a priority, especially for rural MV feeders with few existing measurement points [3], [4], [5], [9], [18]. Across hardware categories, several recurring limitations constrain accurate fault detection and location in MV networks. Conventional CT/VT-based solutions require outages and significant installation work [3], [8], [12]. This limits widespread deployment on existing networks. High-precision sensors offer better

linearity and bandwidth but at per-unit costs and power budgets that are difficult to justify for dense field deployment [8], [10], [11], [17]. Many experimental capacitive and non-contact sensors are validated only in laboratories. Long-term MTBF, drift and failure modes under outdoor MV conditions (pollution, UV, vibration, $-40\text{ }^{\circ}\text{C}$ to $+85\text{ }^{\circ}\text{C}$) are seldom reported [11]–[13], [16], [18]. Next, rural and suburban feeders often have very few measurement points. Sophisticated multi-sensor or PMU-based platforms assume an instrumentation density and communication infrastructure that do not exist in these grids [3], [4], [5], [18]. Pole-top or line-mounted devices must operate on constrained power budgets, sometimes from batteries or line energy harvesting while supporting reliable communication (e.g., PLC, sub-GHz RF, LTE-M) over long distances. Many high-performance sensing concepts have not been engineered under such constraints [3], [5], [12], [18]. These gaps collectively point to the need for compact, low-power, environmentally hardened sensing hardware capable of providing dependable fault passage and direction information without the complexity and cost of full PMU-class instrumentation or dense multi-sensor arrays. Furthermore, electricity theft, including practices like meter tampering, meter bypassing, and illegal line hooking remains a significant challenge that manual inspection systems are ill-equipped to detect. In Bangladesh, these illegal activities not only result in financial losses but also contribute to the unavailability of power for legitimate consumers due to overloading caused by unauthorized connections. The Bangladesh Energy Regulatory Commission (BERC) reports that around 15% of electricity consumption is unaccounted for due to theft, costing the economy around \$600 million annually.

1.2 Objectives, Requirements, Specification and constraints

1.2.1. Objectives

- Design and develop an intelligent Fault Passage Indicator (FPI) device capable of detecting and identifying fault currents.
- Develop communication capabilities to interface the FPI device with a central monitoring server.
- Verify system performance by conducting laboratory tests across multiple fault conditions to assess reliability and accuracy against standard benchmarks.

1.2.2 Functional and Nonfunctional Requirements

a) Functional Requirements

- Detect Fault Currents: Detect abnormal current patterns such as fault currents, high/low impedance faults, and sudden fluctuations. Reduces fault detection identification time.
- Real-Time Monitoring: Perform continuous current monitoring using non-intrusive Hall-effect sensors.
- Distinguish Fault-induced from Normal Load Variation: Identify fault-induced current signatures separately from normal load variations.

- **Communicate Fault Alerts:** Send fault alerts to a central server via LoRaWAN, GSM, or Wi-Fi communication protocols.
- **Provide Local Indication:** Indicate faults locally with LEDs or a buzzer during fault events. Sends real time status alerts.
- **Current Theft:** Detects Current theft by comparing the flow of current in between 2 consecutive poles.
- **Support Reset Options:** Allow both automatic and manual reset after fault events.

b) Non-Functional Requirements

- **Manual Installation:** The device should be easy to install manually with minimal tools.
- **Fault type :** The device detects the type of faults.
- **Repeated Faults:** Demonstrate detection performance across repeated fault events under identical operating conditions.
- **Response time:** It should be less than 0.2s to 0.5s.
- **Low Power Consumption:** Operate efficiently with a rechargeable battery and optional solar charging.
- **Compact and Weather-Resistant:** Must be compact, weather-resistant (IP55+), and easy to mount on poles or distribution boxes.
- **Communication Range:** Maintain a communication range of 1–2 km (LoRaWAN) or continuous connectivity via LTE.
- **Low Latency:** Ensure less than 150 ms latency from fault detection to server update.
- **Reliability and Accuracy:** Provide reliable and accurate fault detection under various conditions.
- **External Parameters:** Analyses the wire condition and considers weather conditions.

1.2.3 Specifications

- The system operates at a voltage level of **230V/440V**, designed specifically for low-voltage distribution lines.
- The device monitors currents within a **range of 0–100 A**, ensuring comprehensive fault detection across typical load conditions.
- The fault detection system uses Hall-effect sensors with a current measurement range of **±100 A**, offering high accuracy with a resolution of **0.1 A** and an accuracy tolerance of **±1%**.
- The microcontroller, single-board computer (SBC), or FPGA used in the system is capable of handling the required processing power and communication tasks with an operating clock speed of **≥ 100 MHz** and supporting interfaces for communication.
- Communication is facilitated through LoRaWAN, GSM, or Wi-Fi, ensuring reliable data transmission over **distances of 1–2 km** for LoRaWAN or continuous LTE connectivity.

- The power system uses a 3.7V Li-ion or LiFePO₄ battery, with an optional **5W–10W solar module** to ensure long-term efficiency and sustainability.
- The system ensures real-time fault alerts with a latency of **less than 150 ms** from event detection to server update.
- Designed to be compact and easy to install, the device can be mounted on poles or distribution boxes, providing ease of use in field installations.

1.2.4 Technical and Non-technical consideration and constraint in design process

The design and implementation of the proposed communicable Fault Passage Indicator (FPI) required careful consideration of several technical and non-technical factors to ensure reliable operation, cost-effectiveness, and practical deployment within low-voltage (LV) distribution networks. Since the prototype was developed as a low-cost solution for Bangladesh's power distribution system, the selection of hardware components, communication technology, sensing devices, and system architecture was influenced by both engineering requirements and real-world implementation constraints. The major considerations and constraints encountered during the design process are summarized below.

- **Fault Detection Challenges:** Detecting faults in low-voltage distribution systems is more challenging than in medium-voltage networks because normal load variations and transient current fluctuations can resemble fault conditions. Therefore, appropriate current thresholds and filtering techniques were implemented to minimize false fault detection while maintaining adequate sensitivity.
- **Fault Detection Time:** To distinguish actual faults from temporary transients or switching disturbances, the system was designed to recognize a fault only when the abnormal current condition persisted for approximately **40 ms to 100 ms**. This improves detection reliability while preventing unnecessary fault indications.
- **Communication Reliability:** Reliable transmission of fault information is essential for remote monitoring. The proposed prototype utilizes **ESP-NOW** communication for low-latency wireless data transfer between the ESP32 transmitter and ESP8266 receiver. However, communication performance may be affected by transmission distance, physical obstructions, radio-frequency interference, and environmental conditions. Future large-scale deployment may require communication technologies such as **LoRaWAN**, **LTE-M**, or **NB-IoT** to achieve wider coverage.
- **Power Supply Considerations:** The monitoring device is intended for continuous operation and therefore requires a stable and efficient power source. Proper voltage regulation was necessary to ensure reliable operation of the ESP32, ESP8266, GPS module, and Hall-effect current sensor while minimizing power consumption.
- **Sensor Accuracy and Calibration:** The WCS1800 Hall-effect current sensor and Gravity Analog AC sensor was selected because of its compact size, electrical isolation, and ease of installation. However, Hall-effect sensors are susceptible to offset drift, temperature variation, and external magnetic interference. Consequently,

calibration and offset compensation were incorporated before current measurement to improve detection accuracy.

- **Safety Requirements:** As the device is intended for installation near energized distribution conductors, electrical isolation, operator safety, and electromagnetic compatibility must be considered during the design process. The system should comply with relevant electrical safety practices, and the surrounding electromagnetic field intensity should remain within acceptable limits to minimize measurement errors and communication interference.
- **Environmental Considerations:** Practical deployment requires the monitoring device to operate reliably under varying environmental conditions, including high temperature, humidity, dust, rainfall, and mechanical vibration. These factors influence enclosure design, component selection, and long-term operational reliability.
- **Budget Constraints:** The project was developed within the financial limitations of an undergraduate final-year design project. Consequently, commercially available industrial-grade Fault Passage Indicators and high-cost sensing equipment were not considered. Instead, low-cost embedded hardware, including the ESP32, ESP8266, WCS1800 Hall-effect current sensor, GPS module, and IoT-based monitoring platform, was selected to demonstrate a practical and economically feasible solution without significantly compromising system performance.

The above considerations played a significant role in shaping the final system architecture and component selection. By balancing performance, cost, communication capability, and implementation feasibility, the proposed prototype demonstrates that an intelligent and communicable Fault Passage Indicator can be developed using affordable embedded technologies while remaining suitable for future enhancement and deployment in low-voltage distribution networks.

1.2.5 Applicable compliance, standards, and codes

The design and development of the proposed communicable Fault Passage Indicator (FPI) were carried out with consideration of internationally recognized engineering standards and industry guidelines to ensure safety, reliability, interoperability, and performance. The applicable standards and codes relevant to the hardware components, communication protocols, electrical safety, electromagnetic compatibility, and power distribution systems are summarized in the table below.

Category	Standard / Code	Description	Relevance to FPI
Power System Protection	IEEE C37.113	Guide for Protective Relay Applications to Distribution Lines	Widely used reference in Bangladesh for distribution protection logic and fault detection principles
	IEC 60255	Measuring Relays and Protection Equipment	Applicable to fault detection behavior and protection device performance
Low-Voltage Distribution & Safety	Bangladesh Electricity Rules (BER), 1937 (Amended)	National Electrical Safety and Operation Rules	Primary legal framework governing LV distribution safety in Bangladesh
	IEC 60364	Low-Voltage Electrical Installations	Commonly followed international standard for LV installations and safety
	IEEE 141 (Red Book)	Electric Power Distribution Systems	Used in academic and utility studies for LV distribution behavior
Current Measurement & Sensors	IEC 61869-2	Instrument Transformers – Current Transformers	Relevant for accuracy and insulation requirements of current sensing
	IEEE C57.13	Instrument Transformer Standards	Provides accuracy and safety guidelines for current measurement
	IEC 61010-1	Safety Requirements for Electrical Measurement Equipment	Ensures operator and equipment safety
Communication & IoT	IEEE 802.11	Wireless LAN (Wi-Fi) Standard	Commonly used for local communication and testing
	LoRaWAN Specification	Low-Power Wide Area Network	Suitable for rural and semi-urban Bangladesh
	3GPP GSM / LTE	Cellular Comms Standards	Nationwide and practical for utility monitoring.
Electromagnetic Compatibility	IEC 61000-6-2	EMC Immunity for Industrial Environments	Ensures immunity against EMI in substations and distribution areas

(EMC)	IEC 61000-6-4	EMC Emission Standards	Limits electromagnetic interference to nearby equipment
	IEEE 519	Harmonic Control in Power Systems	Relevant due to non-linear loads and power quality issues in Bangladesh
Environmental & Mechanical Protection	IEC 60529	Degrees of Protection (IP Code)	Defines enclosure protection (IP55–IP65) for outdoor installation
	IEC 60068	Environmental Testing	Ensures operation under heat, humidity, and dust conditions
Cybersecurity & Data Integrity	IEEE 1686	Cyber Security for Intelligent Electronic Devices	Increasingly relevant for smart grid and IoT-based monitoring.
Ethics & Sustainability	IEEE Code of Ethics	Professional Engineering Ethics	Used in engineering education and professional practice.
	UN SDGs (7, 9, 11)	Sustainable Development Goals	Aligns with Bangladesh's national energy and infrastructure goals

Table 1.1: Table for Applicable Standards and Codes

1.3 Systematic Overview/summary of the proposed project

The proposed project presents a communicable Fault Passage Indicator (FPI) for low-voltage distribution networks using embedded systems and IoT technologies. The system is designed to detect abnormal current conditions, identify faults, and transmit fault information for remote monitoring.

The prototype consists of an ESP32-based transmitter integrated with a WCS1800 Hall-effect current sensor and a NEO-6M GPS module. Upon detecting a fault, the transmitter sends the fault status and location to an ESP8266 receiver using the ESP-NOW communication protocol. The receiver then forwards the information to the Blynk IoT platform via Wi-Fi for real-time monitoring and visualization.

The overall system architecture of the proposed Experimental Setup is shown in Figure 1.1. Detailed design methodology, hardware implementation, software development, communication framework, and system validation are presented in the subsequent chapters.

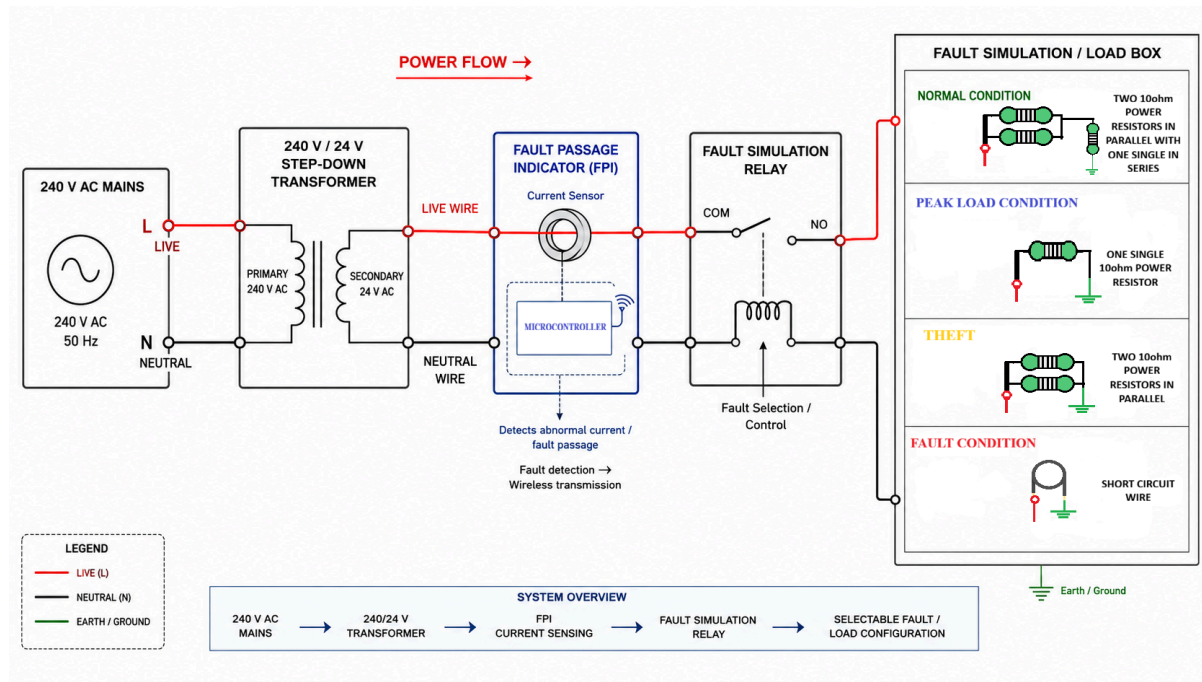


Figure 1.1: Overall System Architecture of the Proposed Experimental Setup

1.4 Conclusion

This chapter presented the background, problem statement, literature gap, and industrial relevance of the proposed communicable Fault Passage Indicator. It also outlined the project objectives, requirements, specifications, design considerations, applicable standards, and a brief overview of the developed system.

The next chapter describes the project design approach, including the design methodology, system architecture, hardware and software design, and engineering decisions adopted during the development of the proposed prototype.

Chapter 2: Project Design Approach [CO5, CO6]

2.1 Introduction

This chapter presents the design approach adopted for the development of the proposed communicable Fault Passage Indicator (FPI) for low-voltage distribution networks. It describes the overall system design, engineering methodology, and the rationale behind the selection of hardware and software components. The chapter also outlines the system architecture and communication framework that form the basis of the developed prototype. The adopted design approach emphasizes reliability, cost-effectiveness, ease of implementation, and suitability for IoT-based remote fault monitoring in low-voltage power distribution systems.

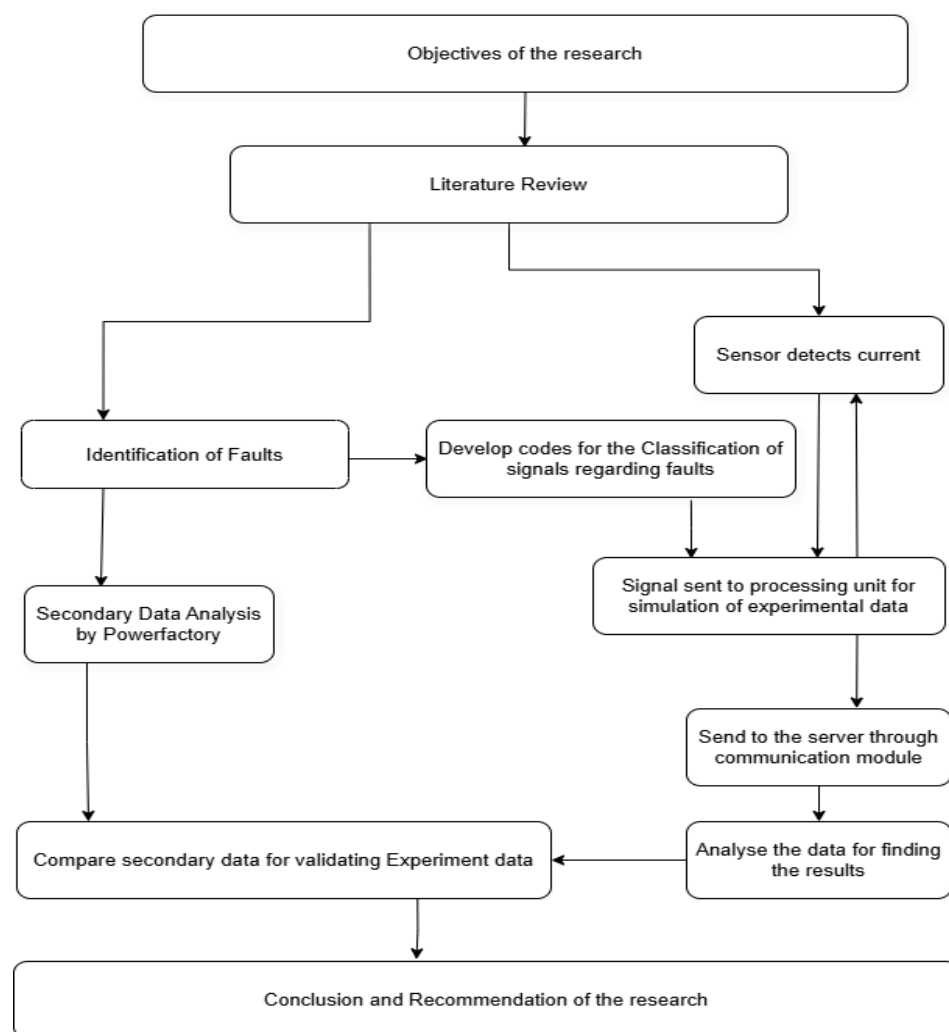


Figure 2.1: Research Methodology of the system

The methodology starts with a clear delineation of the research objectives, which provide the

basis and guidance for the rest of the research. This is followed by a comprehensive literature review, in which research, technology and methods related to fault passage indicators and distribution network monitoring are studied and understood.

The next step from the literature review is two separate but interconnected tracks. The first is a theoretical and experimental track, where faults are identified and codes are written for classification of signals corresponding to various fault conditions. The classification work is complemented by secondary data analysis using the PowerFactory software to simulate and benchmark the system.

The second is an experimental track, in which a sensor is used to measure current in the line and this is transmitted to a processing unit for simulation of experimental data. This is then sent to a server via a communication unit, as per the functionality of the system outlined in the workflow diagram. The data is then processed to draw conclusions from the experiment.

As part of the research, the two tracks (simulation and secondary data) come together in an important validation step where the secondary data (obtained from PowerFactory) is compared to the experimental data (obtained from the sensor-based system). This data validation guarantees that the experimental findings are correct, consistent and in line with theoretical expectations, enhancing the validity and reliability of the study.

The process ends with the conclusion and recommendation based on the validated results. This step translates all the research findings into conclusions and recommendations, which, in turn, contribute to the improvement of fault detection systems in low-voltage systems and provide recommendations for future research and implementation.

2.2 Identify multiple design approach

Multiple Design Solutions:

Subfunctions	Approach 1	Approach 2
Supply power to system	Battery	Solar Rechargeable battery
Detect fault current	Rogowski Coil (Faraday's Law of Electromagnetic Induction)	Hall Current Sensor
Analyse and identify fault type	On Board Processing	Cloud Processing
Indicate identified fault locally	LED	Buzzer (Sound/alarm)
Communicate data to central server	GSM/Cellular Communication	LoRaWAN
Display data to the user through web dashboards	Blynk	Firebase

Table 2.1: Proposed design approaches for the project

2.3 Describe multiple design approach

Design Approach 1:

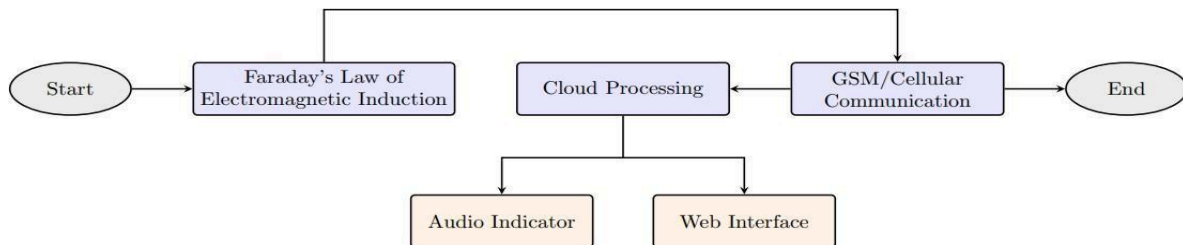


Figure 2.2: Block diagram for design approach 1

Design Approach 1 follows a simple fault passage indicator which can be used in large-scale implementation with minimum power usage and cost. The system begins by applying Faraday's Law of Electromagnetic Induction to detect changes in line current without direct contact. The conditioned signal is then transmitted via GSM/Cellular Communication to a Cloud Processing stage, where integration and threshold-based logic is performed to approximate current magnitude and identify fault conditions. Upon fault detection, the cloud processing stage triggers an Audio Indicator for local alerting and a Web Interface for remote monitoring, providing both immediate and accessible fault notification. This method emphasizes minimization of complexity, practicability, and low cost, making it appropriate for large-scale implementation in distribution networks.

Design Approach 2:

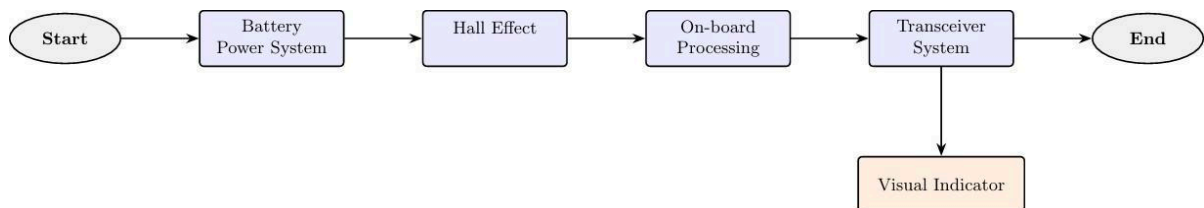


Figure 2.3: Block diagram for design approach 2

Design Approach 2 is based on the same functional workflow as Design Approach 1 but achieves improved performance and scalability through higher accuracy sensing and enhanced communication. The system is powered by a Battery Power System, ensuring consistent and autonomous operation. Current measurement is carried out using a Hall Effect sensor, which provides a direct proportional output relative to line current with greater accuracy and noise immunity. The sensor output is then handled by an On-board Processing unit, enabling improved signal filtering, fault classification, and local data processing at the edge. Fault data is subsequently transmitted through a Transceiver System for long-range communication, while a Visual Indicator provides immediate on-site fault notification. This

design represents a mid-scale solution that balances cost, performance, and communication range.

2.4 Analysis of multiple design approach

To establish the most effective configuration for the low-voltage Fault Passage Indicator (FPI), Design Approach 1 and Design Approach 2 are systematically evaluated based on measurement accuracy, power autonomy, processing latency, operational cost, and overall reliability.

Subsystem Trade-off Analysis:

Power Supply Subsystem: Design Approach 1 uses a static battery power source, which minimizes circuit complexity but introduces maintenance challenges due to finite battery life. In contrast, Design Approach 2 employs a solar-rechargeable system, ensuring continuous field power autonomy and eliminating the need for periodic manual battery replacements in remote network locations.

Current Sensing Subsystem: Design Approach 1 implements a Rogowski Coil operating on Faraday's Law, offering non-intrusive measurement and zero saturation risks during high fault currents. However, Design Approach 2 utilizes a Hall Effect Current Sensor, which provides superior low-current sensitivity and direct proportional sensing, making it better suited for precise fault identification in low-voltage lines.

Data Processing Unit: Design Approach 1 relies on Cloud Processing, sending minimal sensor data to a remote server. While this reduces edge hardware costs, it introduces network transmission latency and makes fault detection dependent on continuous connectivity. Design Approach 2 incorporates On-Board Edge Processing, executing detection algorithms directly on the microcontroller to guarantee real-time, zero-latency fault classification even during central network outages.

Communication Network: Design Approach 1 uses Cellular/GSM communication, providing long-range data transfer where cell towers exist, but incurring recurring SIM card subscription fees and higher power draw. Design Approach 2 utilizes LoRaWAN transceivers, which operate on license-free bands with minimal power consumption, making them highly cost-effective for localized node-to-gateway links.

Local Indication and Web Interface: Design Approach 1 pairs an audio buzzer for local alerts with a Blynk dashboard. However, buzzers are easily drowned out by ambient environmental noise. Design Approach 2 uses a visual LED indicator alongside a Firebase backend, offering high-visibility local fault alerts and low-latency real-time web dashboard synchronization.

Hybrid Approach Selection Rationale

Rather than strictly selecting one standalone configuration, a **Hybrid Design Approach** is adopted for system execution. This strategy merges the key strengths of both evaluated designs to maximize operational reliability while minimizing maintenance costs:

Edge Intelligence & Power Autonomy: The system adopts the **Solar-Rechargeable Battery** and **On-Board Edge Processing** from Approach 2. This ensures energy self-sufficiency and instantaneous local fault detection without relying on active cloud connections.

Precision Sensing & Communication: The architecture incorporates high-sensitivity current sensing paired with a **Low-Power Transceiver Network (LoRaWAN/RF)**, providing accurate low-current fault monitoring while avoiding continuous cellular data fees.

Alerting Framework: The local notification relies on **Visual LED Display** for high outdoor visibility, synchronized with a real-time **Firebase Web Interface** for instant remote operator visibility.

By combining on-board processing speed, solar power harvesting, precision current measurement, and low-power long-range communication, the hybrid setup establishes a robust, highly autonomous, and cost-efficient solution for low-voltage distribution monitoring.

2.5 Conclusion

This chapter presented the design methodology and overall architecture of the proposed communicable Fault Passage Indicator. The hardware and software design, component selection, and communication framework were discussed to illustrate the engineering decisions adopted during the development of the prototype. The presented design approach provides the foundation for the implementation and optimization of the proposed system. The next chapter introduces the modern engineering tools and software platforms utilized throughout the design, development, and validation of the project.

Chapter 3: Use of Modern Engineering and IT Tool. [CO9]

3.1 Introduction

Many modern engineering tools had to be used to design the systems, develop the hardware, program the software, create simulations, communicate, and test for the successful development of the proposed communicable Fault Passage Indicator (FPI). These were used efficiently to help implement the process, debug and evaluate the project performance. This chapter introduces the different engineering software, development platforms, and related technologies used in the designing of the proposed system, their applications.

3.2 Select appropriate engineering and IT tools

Based on background research, there are numerous software and hardware platforms required to ensure proper simulation, comparison and implementation of the project. Therefore, various software tools were used to figure out the best, suitable and efficient one among both design approaches. All the softwares, IT tools which have been used are listed below:

- COMSOL Multiphysics
- DigSilent Powerfactory
- Proteus
- Arduino IDE
- MATLAB Simulink
- Blynk

All these softwares were used to design and verify the corresponding subsystems of the project, mainly focusing on the correct detection and classification of four fault conditions on the 11 kV distribution line. An L-L fault condition was simulated to test the ability to detect the presence of excessive inter-phase current with no ground fault occurring and an S-L-G fault condition was simulated to test the ability to detect current imbalance and voltage disturbance on the affected phase, the most common fault condition. The simulation of a D-L-G fault was performed to confirm the classification accuracy when the two phases are grounded and the simulation of a Three-Phase fault was performed to confirm the detection of the maximum possible fault current magnitude. For all four fault types, COMSOL, Proteus, DigSilent PowerFactory, and MATLAB Simulink were used to perform rigorous pre-hardware validation, and Arduino IDE and Blynk were used to implement the hardware and real-time monitoring chain for fault monitoring.

3.3 Use of modern engineering and IT tools

A detailed system requirements and analysis is summarised below:

Software:

Software	Design Approach 1	Design Approach 2	Why this tool was chosen
COMSOL Multiphysics 	✓	✓	• Simulates electromagnetic field behaviour based on Faraday's Law. • Models current induction and signal propagation. • Validates sensor sensitivity under varying fault conditions.
DigSilent PowerFactory 	✓	✓	• Ideal for power system network simulation. • Used for secondary data analysis and fault validation. • Provides benchmark data for comparing experimental results.
Proteus 		✓	• Faster than physical prototyping. • Simulates microcontroller and sensor circuits. • Tests signal conditioning and threshold logic interactions.
Arduino IDE 		✓	• Communicates with Arduino-based microcontrollers. • Simplifies threshold logic and fault detection coding. • Integrates sensor inputs and indicator

			outputs.
MATLAB Simulink 	✓	✓	• Effective for dynamic system simulation. • Supports signal processing and fault classification modelling. • Enables stability and response time analysis.
Blynk 		✓	• Supports IoT-based real-time monitoring. • Enables remote fault alert notifications via mobile interface. • Provides a user-friendly dashboard for fault data visualisation.

Table 3.1: Software components for the experiment

Hardware:

Subsystem	Components Required		Functions
Power	12V 1.5W Solar panel		• Converts sunlight into electrical energy for the system. • Provides the primary renewable power source for charging the battery.
	Solar charge controller		• Designed for solar battery charging and protection. • Provides charge regulation and

			overcharge protection.
	3.7V Rechargeable Battery		• Energy-storage source for the system.
	DC-DC Buck Converter		• Adjustable linear voltage regulator. • Provides a stable regulated DC output for the electronics.
Sensor	WCS1800 Hall Sensor [For DC Experimental Setup]		• Hall-effect AC/DC current sensor. • Provides an analog output proportional to the measured current up to 35 A.
	Gravity Analog AC Current Sensor [For AC Experimental Setup]		• CT-based AC current sensor. • Provides analog output proportional to the measured current up to 20 A.
Processing and Communication	Arduino ESP32 and ESP32E		• 32-bit microcontroller with integrated Wi-Fi/Bluetooth. • Used for current sensing, fault detection, and data processing.
Indicator	LED		• Low-power visual indicator. • Used to indicate system/fault status.
	Power Resistor		• Rated 100 W 10 ohms. • Dissipates heat when current flows through it.
GPS	GY Neo-6M		• Receives real-time location and positioning

			through latitude and longitude.
Web Dashboard	Firebase		• Cloud-based real-time database platform. • Used for remote storage and monitoring of fault data.

Table 3.2: Hardware components for the experiment

3.4 Conclusion

This chapter introduced the contemporary engineering tools used for the design, development and testing of the designed Fault Passage Indicator with communication. The hardware integration, software implementation, wireless communication and system validation was performed easily with the selected software platforms, programming environments and development tools. These engineering tools proved to be very helpful in the successful implementation of the proposed prototype. The evaluation of multiple design alternatives and engineering analysis to arrive at optimal system configuration are discussed in the next chapter.

Chapter 4: Optimization of Multiple Design and Finding the Optimal Solution. [CO7]

4.1 Introduction

There are many possible solutions when it comes to engineering design. Each of the designs has their own features qualities that can only be identified by systematic experimentation, strengths and weaknesses and evaluation. In this chapter, previously mentioned design approaches have been explored by performing simulation. It allowed us to measure both design methods' performance to meet the project requirements. The results assisted in the selection of the design that best meets our Functional needs, considering the performance, cost and feasibility of the project. Then, the chosen design is optimized further in preparing the prototype development.

4.2 Optimization of multiple design approach

System Simulations

Real-world distribution network data collected from Dhaka Power Distribution Company Limited (DPDC), Dhaka specifically covering the Dhaka University area was utilized to develop a realistic power system model in DIgSILENT PowerFactory. The modeled network included essential components such as buses, distribution feeders, transformers, and actual load demand profiles to accurately reflect the operational characteristics of an urban distribution system. Load flow analysis was carried out under normal operating conditions to determine voltage profiles at different buses, real and reactive power flow through the network, line loading conditions, and overall system losses. In addition to steady-state analysis, fault simulations were performed to examine system behavior under abnormal conditions by applying single line-to-ground (L-G), line-to-line (L-L), and three-phase symmetrical faults at various locations within the network. These simulations enabled observation of fault current magnitudes, voltage drops, and system imbalances, particularly in the case of unsymmetrical faults. The analysis revealed that three-phase faults produce the highest fault current, while L-G and L-L faults introduce unbalanced conditions in the system. Overall, this study provides a practical understanding of real-life power system performance in the Dhaka University area, highlighting voltage stability, loss distribution, and the importance of effective protection coordination and fault management in modern distribution networks.

SLD From DPDC

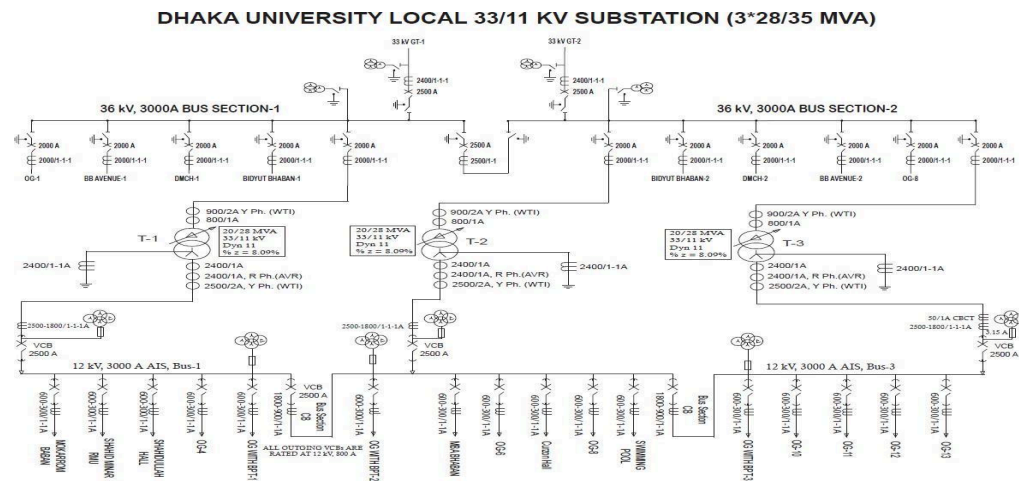


Fig. 4.1: Single Line Diagram of Dhaka University 33/11 kV Substation

Simulation With PowerFactory

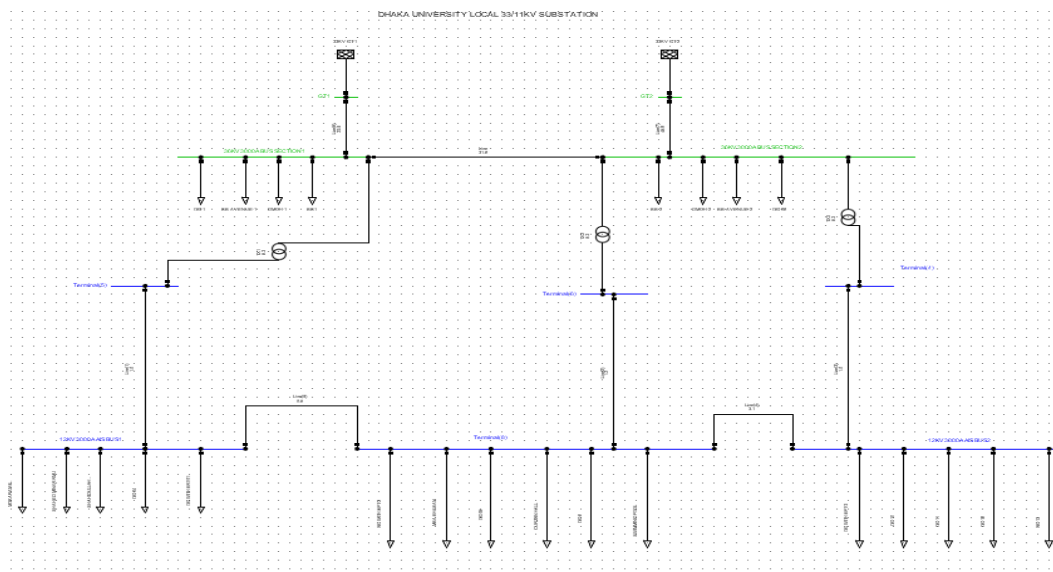


Fig. 4.2: Single Line Diagram of Dhaka University 33/11 kV Substation using Powerfactory

Fault Data:

Single Line to Ground Fault

						DigSILENT		Project:			
						PowerFactory		-----			
						2021 SP2		Date: 4/16/2026			
Fault Locations with Feeders											
Short-Circuit Calculation / Method : VDE 0102 Part 0 / DIN EN 60909-0 Single Phase to Ground / Max. Short-Circuit Currents											
Asynchronous Motors Always Considered					Grid Identification Automatic			Short-Circuit Duration			
					Conductor Temperature User Defined No			Break Time 0.10 s			
								Fault Clearing Time (Ith) 1.00 s			
								Voltage factor c Standard defined table			
Grid: Grid		System Stage: Grid							Annex:		/ 1
		rtd.V. [kV]	Voltage [kV]	c- [deg]	Factor	Sk" [MVA]	Ik" [kA]	[deg]	ip [kA]	Ib [kA]	Sb [MVA] EFF [-]
12KV 3000A AIS		A	11.00	0.00	0.00	1.10	30.69 MVA	4.83 kA	-89.01	13.37 kA	4.83 30.69 0.00
		B		6.02	-90.84		0.00 MVA	0.00 kA	0.00	0.00 kA	0.00 0.00 0.86
		C		6.08	90.83		0.00 MVA	0.00 kA	0.00	0.00 kA	0.00 0.00 0.87
Line(1)		Terminal(5)			A	14.54 MVA	2.29 kA	89.41	6.34 kA		
					B	4.06 MVA	0.64 kA	87.03	1.77 kA		
					C	4.06 MVA	0.64 kA	87.03	1.77 kA		
Line(5)		Terminal(8)			A	16.16 MVA	2.54 kA	92.42	7.04 kA		
					B	4.06 MVA	0.64 kA	-92.97	1.77 kA		
					C	4.06 MVA	0.64 kA	-92.97	1.77 kA		
MOKARAM BABAN					A	0.00 MVA	0.00 kA	0.00	0.00 kA		
					B	0.00 MVA	0.00 kA	0.00	0.00 kA		
					C	0.00 MVA	0.00 kA	0.00	0.00 kA		

Fig. 4.3: Data for Single Line to Ground Fault

Double Line to Ground Fault

										DigSILENT		Project:	
										PowerFactory			
										2021 SP2		Date: 4/16/2026	
Fault Locations with Feeders													
Short-Circuit Calculation / Method : VDE 0102 Part 0 / DIN EN 60909-0 2-Phase to Ground / Max. Short-Circuit Currents													
Asynchronous Motors Always Considered					Grid Identification Automatic					Short-Circuit Duration Break Time 0.10 s			
										Fault Clearing Time (Ith) 1.00 s			
					Conductor Temperature User Defined No					Voltage factor c Standard defined table			
Grid: Grid		System Stage: Grid										Annex: / 1	
		rtd.V. [kV]	Voltage [kV]	c- [deg]	Factor	Sk" [MVA]	Ik" [kA]	[deg]	ip [kA]	Ib [kA]	Sb [MVA]	EFF [-]	
12KV 3000A AIS		A	11.00	0.36	-16.01	1.10	0.00 MVA	0.00 kA	0.00	0.00 kA	0.00	0.00 0.05	
		B		0.00	-120.00		35.01 MVA	5.51 kA	122.10	15.26 kA	5.51	35.01 0.00	
		C		0.00	120.00		34.71 MVA	5.46 kA	60.43	15.13 kA	5.46	34.71 0.00	
Line(1)		Terminal(5)				A	7.93 MVA	1.25 kA	-92.55	3.45 kA			
						B	19.05 MVA	3.00 kA	-72.01	8.30 kA			
						C	19.23 MVA	3.03 kA	-109.19	8.38 kA			
Line(5)		Terminal(8)				A	7.93 MVA	1.25 kA	87.45	3.45 kA			
						B	17.17 MVA	2.70 kA	-42.21	7.49 kA			
						C	16.17 MVA	2.55 kA	-131.94	7.05 kA			
MOKARAM BABAN						A	0.00 MVA	0.00 kA	0.00	0.00 kA			
						B	0.00 MVA	0.00 kA	0.00	0.00 kA			
						C	0.00 MVA	0.00 kA	0.00	0.00 kA			

Fig. 4.4: Data for Double Line to Ground Fault

Three Phase Fault

										DigSILENT		Project:	
										PowerFactory			
										2021 SP2		Date: 4/16/2026	
Fault Locations with Feeders													
Short-circuit Calculation / Method : VDE 0102 Part 0 / DIN EN 60909-0 3-Phase Short-Circuit / Max. Short-Circuit Currents													
Asynchronous Motors Always Considered					Grid Identification Automatic				Short-Circuit Duration Break Time				

Fig. 4.5: Data for Three Phase Fault

I) Power Unit Subsystem

a) Battery System

The battery-based power supply is a self-contained, fixed-energy subsystem. It converts stored electrochemical energy in a Li-ion cell to a regulated 3.3 V DC supply for the FPI electronics. The power chain is:

Li-ion Cell (3.7V nom.) → Boost Converter (3.7V→5V) → LDO Regulator
(5V→3.3V) → FPI Load

This three-stage chain is necessary because the Li-ion cell voltage varies between 4.2 V (fully charged) and 2.8 V (cut-off), and the MCU/peripherals require a stable 3.3 V. The LDO ensures ripple-free regulation; the boost converter guarantees 5 V input to the LDO even when cell voltage dips below 3.3 V at low SOC.

Key Specifications

- Cell chemistry: Lithium-Ion (LiCoO₂), 18650 form factor
- Nominal voltage: V_{nom} = 3.7 V
- Capacity: C = 4000 mAh = 4.0 Ah
- Fully charged: V_{max} = 4.2 V

- Discharge cut-off: $V_{\min} = 2.8 \text{ V}$
- Boost converter output: 5.0 V , efficiency $\eta_{\text{boost}} = 90\%$
- LDO output: 3.3 V , efficiency $\eta_{\text{LDO}} = 3.3/5.0 = 66\%$
- Combined chain efficiency: $\eta_{\text{total}} = 0.90 \times 0.66 = 59.4\%$

Mathematical Modelling

State of Charge (SOC) Model:

SOC represents the fraction of usable energy remaining in the battery. Using Coulomb counting:

$$\text{SOC}(t) = \text{SOC}(0) - (1/Q_{\max}) \times \int_0^t I_{\text{batt}}(\tau) d\tau \text{ In discrete-time form:}$$

$$\text{SOC}(k+1) = \text{SOC}(k) - [I_{\text{batt}}(k) \times \Delta t] / Q_{\max}$$

Battery Terminal Voltage Model (Shepherd Equation):

The Shepherd equation provides a physically meaningful SOC-dependent voltage model that captures the battery's non-linear discharge behaviour:

$$V_{\text{batt}}(t) = E_0 - K \times [Q_{\max}/(Q_{\max} - \int I dt)] + A \cdot \exp(-B \cdot \int I dt) - R_{\text{int}} \cdot I_{\text{batt}}(t)$$

Expanded using SOC notation (letting $Q_{\text{discharged}} = Q_{\max} \times (1 - \text{SOC})$):

$$V_{\text{batt}}(\text{SOC}, I) = E_0 - K \times [1/(\text{SOC})] + A \cdot \exp(-B \cdot Q_{\max} \cdot (1 - \text{SOC})) - R_{\text{int}} \cdot I_{\text{batt}}$$

Symbol	Definition	Value / Unit
E_0	Battery EMF constant (open circuit, fully charged)	4.10 V
K	Polarisation voltage constant	0.0054 V/Ah
A	Exponential zone amplitude	0.26 V
B	Exponential zone time constant inverse	24.37 Ah ⁻¹
R_{int}	Internal resistance	0.080 Ω (80 m Ω)
I_{batt}	Instantaneous discharge current	Computed dynamically
$Q_{\max}$	Rated capacity	4.0 Ah

Table 4.1: Battery terminal definitions and values

Load Current Model:

The current drawn from the battery is determined by the load power demand and the efficiency chain:

$$\begin{aligned} P_{\text{load}} &= 54.0 \text{ mW (weighted average – see Section 0.3)} \\ P_{\text{battery_required}} &= P_{\text{load}} / \eta_{\text{total}} = 54.0 \text{ mW} / 0.594 = 90.9 \text{ mW} \\ I_{\text{batt}}(t) &= P_{\text{battery_required}} / V_{\text{batt}}(t) = 0.0909 / V_{\text{batt}}(t) [\text{Amperes}] \end{aligned}$$

Power Delivered to Load:

$$P_{\text{delivered}}(t) = V_{\text{3V3}}(t) \times I_{\text{load}}(t)$$

Since the LDO regulates $V_{\text{3V3}} = 3.3 \text{ V}$ (constant) until the battery reaches cut-off:

$$P_{\text{delivered}}(t) = 3.3 \times I_{\text{load}} = 3.3 \times (P_{\text{load}} / 3.3) = P_{\text{load}} = 54.0 \text{ mW (constant)}$$

Output power remains flat at 54 mW until the battery is exhausted, then drops abruptly to 0 V when $V_{\text{batt}} < V_{\text{boost_min}} = 2.8 \text{ V}$.

Energy Delivered & Battery Lifetime:

$$\begin{aligned} E_{\text{delivered}}(t) &= \int_0^t P_{\text{delivered}}(\tau) d\tau \approx P_{\text{load}} \times t [\text{for constant load}] \\ E_{\text{raw}} &= V_{\text{nom}} \times Q_{\text{max}} = 3.7 \times 4.0 = 14.8 \text{ Wh} \end{aligned}$$

$$E_{\text{effective}} = E_{\text{raw}} \times \eta_{\text{total}} = 14.8 \times 0.594 = 8.79 \text{ Wh}$$

$$T_{\text{life}} = E_{\text{effective}} / P_{\text{load}} = 8.79 \text{ Wh} / 0.054 \text{ W} = 162.8 \text{ hours} \approx 6.8 \text{ days}$$

Discharge Curve (Non-linear Model):

The full SOC-to-voltage characteristic is described by the piecewise function derived from the Shepherd equation. Three regions are identifiable:

Region 1 (SOC: 1.0 $\rightarrow$ 0.9): Exponential drop $V_{\text{batt}} \approx 4.05 - 0.26 \cdot \exp(-24.37 \cdot Q_{\text{max}} \cdot (1 - \text{SOC}))$

Region 2 (SOC: 0.9 $\rightarrow$ 0.15): Flat plateau $V_{\text{batt}} \approx 3.85 - 0.0054/\text{SOC}$

Region 3 (SOC: 0.15 $\rightarrow$ 0.0): Steep final drop V_{batt} drops rapidly $\rightarrow 2.8 \text{ V}$ cut-off

b) Solar PV Hybrid Power Supply Model

The solar hybrid power supply combines a photovoltaic panel with a Li-ion battery buffer to provide continuous, self-replenishing energy. The PV panel is the primary energy source; the battery acts as a storage buffer to cover night-time and cloudy periods. An MPPT controller ensures maximum energy is extracted from the panel at all irradiance levels.

PV Panel (5W) $\rightarrow$ MPPT Controller (P&O) $\rightarrow$ Buck Converter ($V_{\text{pv}} \rightarrow 3.7\text{V}$) $\rightarrow$
Battery Buffer $\rightarrow$ Boost+LDO $\rightarrow$ FPI @ 3.3V

Key Specifications

- PV Panel: Monocrystalline, 5W rated, $V_{oc} = 6.0$ V, $V_{mp} = 5.5$ V, $I_{sc} = 1.00$ A, $I_{mp} = 0.91$ A
- Panel area: $A_p \approx 0.033$ m² (estimated from 5W / (150 W/m² typical cell efficiency standard)
- Cell reference temperature: $T_{ref} = 25^\circ\text{C}$ (STC: Standard Test Conditions)
- Reference irradiance: $G_{ref} = 1000$ W/m²
- MPPT algorithm: Perturb & Observe (P&O) – selected over Incremental Conductance for low-power FPI application
- Buck converter efficiency: $\eta_{buck} = 90\%$
- Battery buffer: Li-ion 3.7V / 4000 mAh (same cell as Design 1 for comparability)
- Boost + LDO chain: $\eta = 59.4\%$ (identical to battery-only design)

Mathematical Modelling

Single-Diode PV Cell Model:

The fundamental PV cell current equation using the single-diode equivalent circuit model:

$$I_{pv} = I_{ph} - I_s \times [\exp(q(V + I \cdot R_s) / (n \cdot k \cdot T)) - 1] - (V + I \cdot R_s) / R_{sh}$$

For simplified panel-level simulation (5W, single panel), using the simplified model ($R_s=0$, $R_{sh}=\infty$):

$$I_{pv} = I_{ph} - I_s \times [\exp(q \cdot V / (n \cdot k \cdot T)) - 1]$$

Irradiance and Temperature Correction:

The photogenerated current and saturation current depend on irradiance G and cell temperature T_c :

$$I_{ph}(G, T_c) = (G / G_{ref}) \times [I_{ph_ref} + K_I \times (T_c - T_{ref})]$$

$$I_s(T_c) = I_{s_ref} \times (T_c / T_{ref})^3 \times \exp[(q \cdot E_g / n \cdot k) \times (1/T_{ref} - 1/T_c)]$$

$E_g = 1.121$ eV (Silicon bandgap energy)

Maximum Power Point (MPP) & P-V Curve:

The power delivered by the PV panel is:

$$P_{pv}(V) = V \times I_{pv}(V) = V \times \{I_{ph} - I_s \times [\exp(q \cdot V / n \cdot k \cdot T) - 1]\}$$

The MPP is found by differentiating with respect to V and setting $dP/dV = 0$: $dP/dV = I_{pv} + V \times dI_{pv}/dV = 0 \rightarrow$ Gives V_{mp} , I_{mp} , P_{mp}

For the 5W panel at STC: $V_{mp} = 5.5$ V, $I_{mp} = 0.91$ A, $P_{mp} = 5.0$ W

Scaled for a module (not single cell), the voltage is the sum of series-connected cell voltages. The I-V relationship for the module simplifies to:

$$I_{\text{module}} = I_{\text{ph}} - I_{\text{s}} \times [\exp(V_{\text{module}} / (n_{\text{s}} \times V_{\text{T}})) - 1] \quad V_{\text{T}} = k \cdot T / q = 0.0257 \text{ V}$$

(thermal voltage at 25°C)

$$n_{\text{s}} = \text{number of series cells} \approx 12 \text{ (for } V_{\text{oc}} = 6\text{V module)}$$

P&O MPPT Algorithm:

The Perturb & Observe algorithm is implemented as a discrete-time controller that adjusts the operating voltage in small steps to maximise power output:

Step 1: Measure $V(k)$, $I(k) \rightarrow P(k) = V(k) \times I(k)$

Step 2: Compute $\Delta P = P(k) - P(k-1)$, $\Delta V = V(k) - V(k-1)$

Step 3: If $\Delta P > 0$ and $\Delta V > 0$: $V_{\text{ref}}(k+1) = V_{\text{ref}}(k) + \Delta V_{\text{step}}$ (continue in same direction)

If $\Delta P > 0$ and $\Delta V < 0$: $V_{\text{ref}}(k+1) = V_{\text{ref}}(k) - \Delta V_{\text{step}}$

If $\Delta P < 0$ and $\Delta V > 0$: $V_{\text{ref}}(k+1) = V_{\text{ref}}(k) - \Delta V_{\text{step}}$ (reverse direction) If $\Delta P < 0$ and $\Delta V < 0$: $V_{\text{ref}}(k+1) = V_{\text{ref}}(k) + \Delta V_{\text{step}}$

Step 4: Set duty cycle D to achieve V_{ref} via buck converter Step 5: $k = k+1$; Repeat at $T_{\text{mppt}} = 100 \text{ ms}$ intervals

Buck Converter Output:

The buck converter steps down V_{pv} (from MPPT, $\approx 5.5 \text{ V}$) to V_{batt} (3.7–4.2 V for charging) or $V_{\text{load_input}}$ (3.7 V for direct load):

$$V_{\text{out_buck}} = D \times V_{\text{in_pv}} \rightarrow D = V_{\text{out_buck}} / V_{\text{in_pv}} = 3.7 / 5.5 = 0.673$$

$$P_{\text{out_buck}} = \eta_{\text{buck}} \times P_{\text{in_buck}} = 0.90 \times P_{\text{mppt}}$$

$$I_{\text{charge}} = (P_{\text{out_buck}} - P_{\text{load}}) / V_{\text{batt}} \text{ [when } P_{\text{out_buck}} > P_{\text{load}}]$$

$$I_{\text{discharge}} = (P_{\text{load}} - P_{\text{out_buck}}) / V_{\text{batt}} \text{ [when } P_{\text{out_buck}} < P_{\text{load}}] \text{ (night mode)}$$

Battery Buffer SOC with Solar Charging:

The battery SOC in the solar-hybrid model is driven by both charging and discharging:

$$\text{SOC}(k+1) = \text{SOC}(k) + [\eta_{\text{charge}} \times I_{\text{charge}}(k) - I_{\text{discharge}}(k)] \times \Delta t / Q_{\text{max}}$$

Bangladesh-Calibrated Irradiance Profile:

The irradiance input to the simulation uses a sinusoidal day profile calibrated to Bangladesh latitude ($\sim 23.7^\circ\text{N}$):

$$G(t) = G_{\text{peak}} \times \max(\sin(\pi \times (t_{\text{hour}} - t_{\text{rise}}) / T_{\text{day}}), 0)$$

Dry season (Nov–Mar): $G_{\text{peak}} = 900 \text{ W/m}^2$, $T_{\text{day}} = 11 \text{ h}$, $t_{\text{rise}} = 6:30$

Monsoon season (Jun–Sep): $G_{\text{peak}} = 550 \text{ W/m}^2$, $T_{\text{day}} = 8.5 \text{ h}$, $t_{\text{rise}} = 6:00$

Average daily energy: $E_{\text{daily}} = G_{\text{peak}} \times A_{\text{panel}} \times \eta_{\text{pv}} \times T_{\text{day}} \times (2/\pi)$

Dry: $E_{\text{daily}} \approx 900 \times 0.033 \times 0.17 \times 11 \times 0.637 \approx 11.2 \text{ Wh/day}$

Monsoon: $E_{\text{daily}} \approx 550 \times 0.033 \times 0.17 \times 8.5 \times 0.637 \approx 5.3 \text{ Wh/day}$

II) Current Subsystem-

a) Hall Effect

We have built a COMSOL Multiphysics 3-dimensional (3D) model for a Hall effect current sensor that is subject to an applied external permanent magnet field using COMSOL Multiphysics 6.2 with the AC/DC Module. The model is an exact but highly simplistic physical model of the actual Hall effect sensing problem, constructed using three coupled physics interfaces and comprising five different geometric domains. The global constants of the simulation are: Hall coefficient $R_H = 1.25 \times 10^{-3} \text{ m}^3/\text{C}$, conductivity of the sensing element $\sigma_0 = 1060 \text{ S/m}$, plate thickness = 0.5 mm, maximum line current $I_{\text{max}} = 500 \text{ A}$, normal (rated) line current $I_{\text{normal}} = 100 \text{ A}$ and reference line current $I_{\text{ref}} = 50 \text{ A}$.

Hall sensing element: The active semiconductor Hall plate is a composite 2D cross-shape with Rectangle 1 (width 12 mm, height 6 mm), and Rectangle 2 (width 2 mm, height 8 mm) united and a 1 mm fillet radius at the internal corner, and extruded to a depth of 1 mm. This is the traditional Hall sensor design where current can travel along one axis and the Hall voltage is detected across the faces that are transverse to the current. The conductor material has the following properties: electric conductivity $\sigma_0 = 1060 \text{ S/m}$, relative permittivity $\epsilon_r = 12$, relative permeability $\mu_r = 1$. The Hall effect model is enabled in the Electric Currents physics interface with the Hall coefficient specified as $R_H = 1.25 \times 10^{-3} \text{ m}^3/\text{C}$, and the magnetic flux density vector (B) taken from the Magnetic Fields, No Currents physics interface via the rotated field components (rB_x , rB_y , rB_z) defined through a General Extrusion operator to facilitate angular rotation of the applied field.

Permanent Magnet and Iron Core: The magnetic flux required to bias the Hall element is supplied by a sintered NdFeB Grade N54 permanent magnet modelled as a cylinder with radius 10 mm and height 6 mm at position (0, -3 mm, 0) along the y-axis. The properties of the magnet are: recoil permeability $\mu_{\text{rec}} = 1.05$ and remanent flux density $B_r = 1.47 \text{ T}$, modelled using the Remanent Flux Density constitutive relation. A flux concentrator made of iron having radius 2 mm and height 1 mm is placed at position (0, 0, -10.5 mm) along the z-axis (with relative permeability $\mu_r = 4000$) to direct the flux from the permanent magnet through the Hall plate for optimal transduction.

Air Domain and Infinite Element Layer: The entire geometry is embedded in a sphere of radius 25 mm centred at the origin, modelling free space. An Infinite Element Domain (spherical type, with rational coordinate stretching) 5 mm thick is applied to the shell of the outer sphere to simulate the far-field magnetic phenomenon and avoid unwanted reflections from the boundary. This eliminates the requirement for magnetic insulation at the outer face, and properly models the open domain physics of permanent magnet and current-carrying conductor systems.

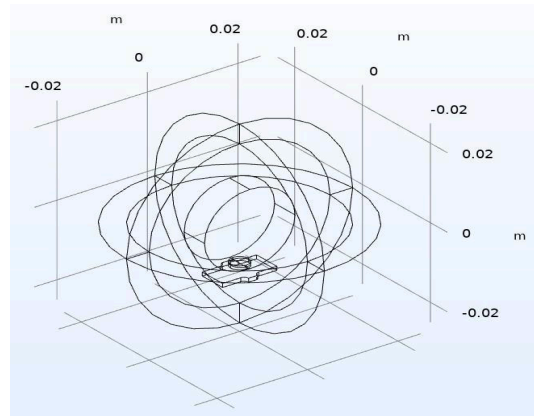


Fig. 4.6: Geometry of Hall Effect Sensor

Working Principle of the Hall Effect Sensor

A Hall effect sensor is a solid-state electromagnetic sensor which translates a flux density into a transverse output voltage, based on the physical phenomenon called the Hall effect. With a bias current I_{bias} applied to the semiconductor Hall element along its long axis, and given a normalised magnetic flux density B perpendicular to both the bias current flow and the thickness of the element, the Lorentz force causes a transverse deflection of the charge carriers across the element. This deflection results in a charge build-up on opposite faces of the Hall element, which induces a transverse electric field and the measurable Hall voltage V_{Hall} across the two floating potential contacts. In the COMSOL model, this is represented via a full tensor formulation of the conductivity, in which the off-diagonal elements of the electrical conductivity tensor σ are modified in response to the normalised magnetic flux density ($\text{hallnormB}_i = -R_H \times \sigma_0 \times B_i$), where in this model $R_H \times \sigma_0 = 1.25 \times 10^{-3} \times 1060 = 1.325 \text{ m} \cdot \text{T}^{-1}$ is a constant.

The Hall sensor does not respond to the rate of change of primary current, as does the Rogowski coil, and does not require a downstream integrator; neither is the Hall sensor output affected by “DC levels” (steady-state currents). This means that no integration circuit is needed, and the sensor is also responsive to DC and low-frequency AC currents. The bias magnetic field is provided by the ambient field of the N54 NdFeB magnet ($B_r = 1.47 \text{ T}$),

whose field is perturbed by the line current flowing through the neighbouring conductor causing the Hall voltage to fluctuate in a strictly linear manner across the whole 0 A to 500 A current range. The lack of ferromagnetic material in the active sensor path means the B-I curve is free from saturation, remanence and hysteresis, delivering the linearity seen in the COMSOL model.

System inputs and outputs:

Parameter	Value/Description
Input: Hall coefficient (R_H)	$1.25 \times 10^{-3} \text{ m}^3/\text{C}$
Input: Electrical conductivity (σ_0)	1060 S/m
Input: Relative permittivity (ϵ_r)	12
Input: Plate thickness	0.5 mm
Input: Magnet (N54 NdFeB)	$B_r = 1.47 \text{ T}$, $\mu_{\text{rec}} = 1.05$
Input: Iron core permeability	$\mu_r = 4000$
Input: Line current sweep	0 A to 500 A in 50 A steps (11 values)
Output: Electric field norm	Spatial distribution across Hall element
Output: Electric potential	Voltage distribution across Hall plate
Output: Magnetic flux density norm	3D volumetric colour map (mfnc.normB)
Output: Magnetic scalar potential	3D iso-surface plot
Output: Hall voltage (V_{Hall})	$\text{ec.fp1.V0} - \text{ec.fp2.V0} \text{ (V)}$

Table 4.2: Hall effect inputs and outputs

Plots:

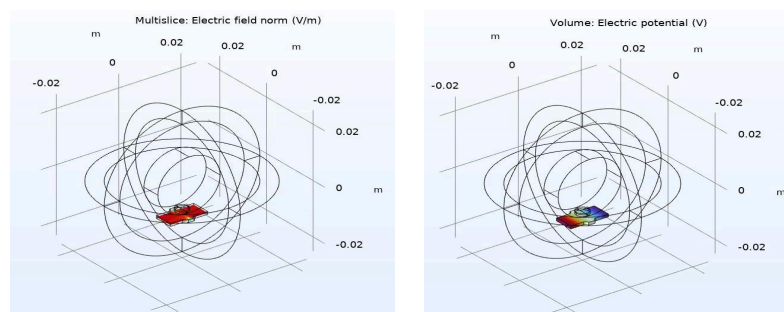


Fig. 4.7: Electric Field Norm and Electric Potential

The electric field norm (V/m) distribution shows the distribution of the total electric field in the Hall sensor area, solved by the Electric Currents physics interface. The electric field magnitude is highest in the centre of the cross-shaped Hall element, at the junction of the four rectangles where the modified Hall conductivity tensor produces high field gradients in a transverse direction. This is because of the off-diagonal terms of the conductivity tensor, affected by the relation $R_H \times \sigma_0 \times B = 1.25 \times 10^{-3} \times 1060 \times B$, which cause lateral deflection of current flow and a transverse electric field. The gradient of the field's norm from the center implies that the transverse electric field due to the Hall effect is localised in the sensing volume. ‘

The electric potential (V) shows the voltage solution across the Hall element with a terminal boundary supplying current I_{line} and a Ground condition of zero volts. The voltage gradient is responsible for the bias current, with the transverse potential difference between the Floating Potential boundaries defining the Hall voltage. This is 5.9799 V at $I_{line} = 100$ A, and 29.899 V at 500 A, which matches the sensitivity calculated from the field (59.80 mV/A), indicating that the line current boundary and Hall voltage are correctly extracted.

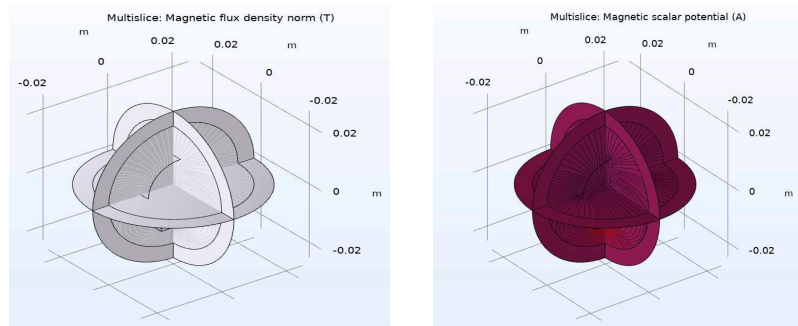


Fig. 4.8: Magnetic Flux Density norm and Scalar potential

Mathematical Modelling:

Magnetic Field Around the Primary Conductor:

The magnetic flux density B calculated at radial distance r from a long straight conductor carrying line current I is determined by Ampere's Circuital Law:

$$B(r) = \mu_0 \cdot I / (2\pi \cdot r)$$

where $\mu_0 = 4\pi \times 10^{-7}$ H/m. At the location of the Hall plate ($r \approx 5$ mm from the conductor axis), for $I_{line} = 100$ A: $B \approx (4\pi \times 10^{-7} \times 100) / (2\pi \times 0.005) = 4$ mT. For the maximum fault current $I_{max} = 500$ A at the same location: $B \approx 20$ mT. The $5\times$ increase in B for a $5\times$ increase in current confirms the strict linearity that the simulation must demonstrate.

Hall Voltage Relationship:

When a bias current I_{bias} flows through the Hall plate of thickness $t = 5 \times 10^{-4}$ m in the presence of magnetic flux density B perpendicular to the plate, the Hall voltage developed across the transverse width is:

$$V_{\text{Hall}} = (R_H \cdot I_{\text{bias}} \cdot B) / t$$

where $R_H = 1.25 \times 10^{-3} \text{ m}^3/\text{C}$ is the Hall coefficient of the sensing material. In the COMSOL implementation, this relationship is encoded through the full Hall conductivity tensor, where the transverse current density components are coupled to the applied magnetic field through the off-diagonal conductivity terms, and the resulting potential difference across the floating potential boundaries (ec.fp1.V0 – ec.fp2.V0) constitutes the simulated Hall voltage output.

Hall Conductivity Tensor:

COMSOL models the Hall effect with modified Ohm's law $J = \sigma_{\text{tensor}} \cdot E$ where, in a magnetic field B , the terms of the conductivity tensor for a material with Hall coefficient R_H include diagonal components $\sigma_0 / (1 + |\text{hallnormB}|^2)$ and off-diagonal cross-terms $\pm \sigma_0 \cdot \text{hallnormB}_i \cdot \text{hallnormB}_j / (1 + |\text{hallnormB}|^2)$. For this model, the normalised field magnitude is: $|\text{hallnormB}| = R_H \times \sigma_0 \times |B| = 1.325 \times |B|$ (T). At $I_{\text{line}} = 100$ A and $r = 5$ mm, $|B| \approx 4$ mT, giving $|\text{hallnormB}| \approx 0.0053$, confirming operation in the low-field linear regime where the off-diagonal terms produce a transverse voltage strictly proportional to B .

Graphs:

I. Line Current vs Sensor Potential Drop

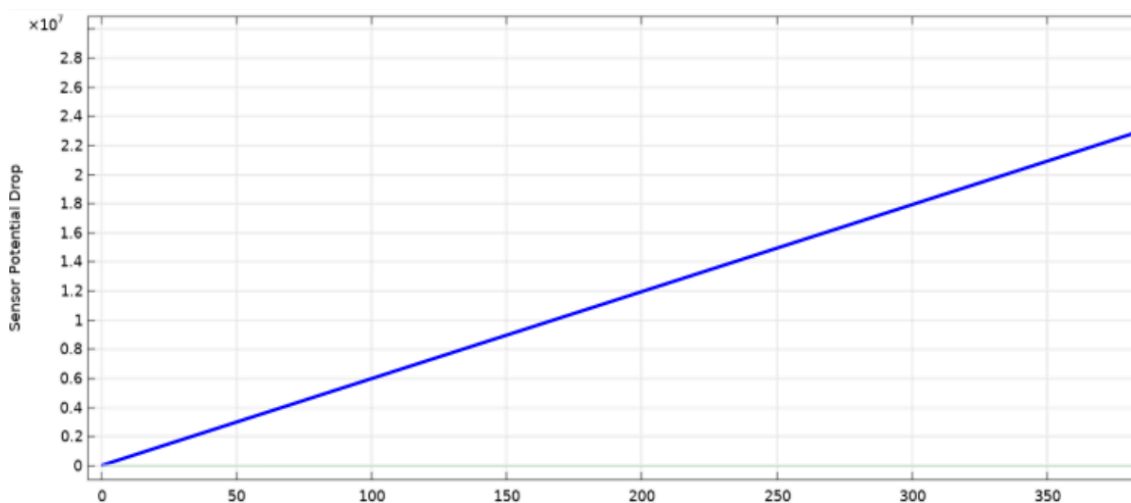


Fig. 4.9: Graph of Line Current (A) vs Sensor Potential Drop (mV)

The calibration curve shows the Sensor Potential Drop in millivolts versus line current I_{line} in amperes with an x sweep of 0 A to 500 A with a 50 A step size. The result is a perfectly straight line through the origin, rising linearly between 0 and 500 A to verify a constant and steady sensor sensitivity of 0.36 mV at 50 A across the entire operating range. Indeed, this rigid proportionality proves the linear nature of the Hall effect transduction mechanism, when operating beyond normal conditions of the load current and at the maximum fault current conditions, and indicates that no deviation or non-linearity occurs in the entire sweep.

Data table:

I_{line} (A)	Sensor Potential drop ($\times 10^7$) (mV)
0	0.000
50	0.275
100	0.550
150	0.825
200	1.100
250	1.375
300	1.650
350	1.925
400	2.200
450	2.475
500	2.750

Table 4.3: Data for Line Current (A) vs Sensor Potential Drop (mV)

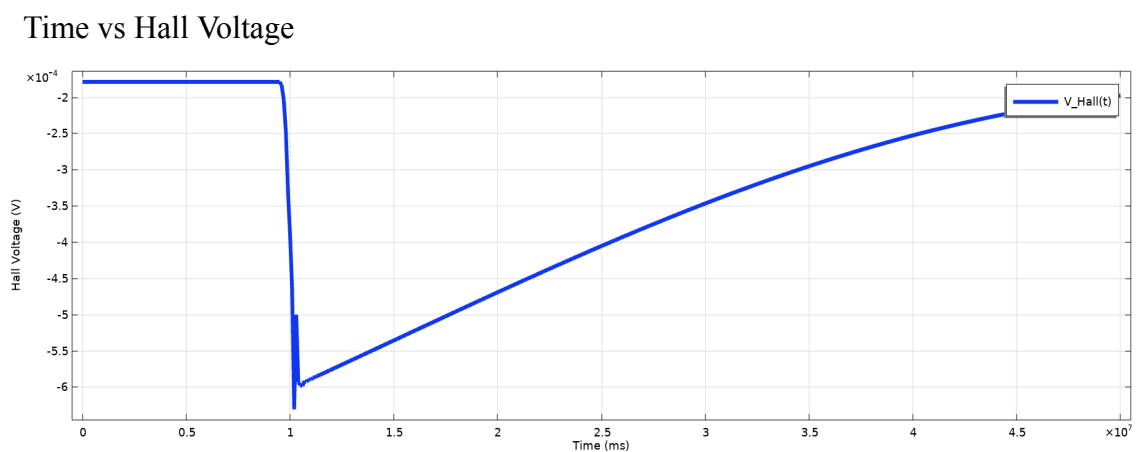


Fig. 4.10: Graph of Time (ms) vs Hall Voltage (V)

The time-domain response plot is of the Hall voltage $V_{\text{Hall}}(t)$ (volts) on the y-axis and time (ms) on the x-axis, showing the sensor response to a step-onset fault current at $t = 1$ ms. After the initial step, the voltage rises again over the rest of the simulation as the electromagnetic field in the iron-cored magnet-sensor assembly adjusts, and demonstrates that the Hall sensor provides an instantaneous response to the fault and that the time lag in the sensing element is negligible.

II. Error Analysis

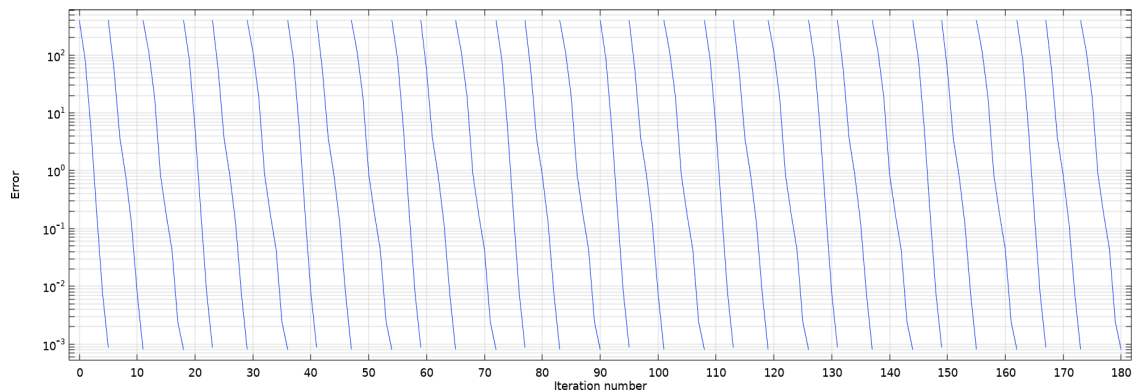


Fig. 4.11: Iteration number vs Error

The error analysis graph shows the solver residual error on a logarithmic y-axis plotted against iteration number on the x-axis, documented throughout the entire parametric current sweep from 0 A to 500 A encompassing 37 continuation steps of the angle sweep investigation. Every steep descending spike indicates a solitary parametric solution step, where the isolated solver for the Electric Currents and Magnetic Fields, No Currents interfaces quickly reduces the residual from an initial value over 10^2 to a convergence level below 10^{-9} in two to three iterations per step. The uniform shape and depth of each spike over all 180 iterations demonstrates consistent solver convergence behavior across the entire sweep, affirming the numerical stability and dependability of the simulation outcomes.

b) Faraday's Law of Electromagnetic Induction

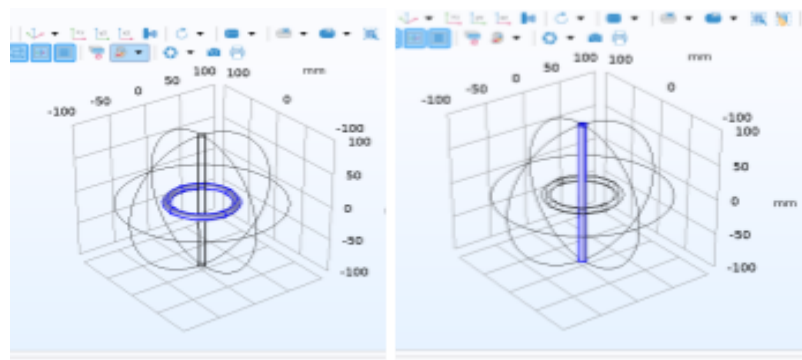


Fig. 4.12: Coil and Line Conductor Geometry

The COMSOL Multiphysics 3D simulation constructed for this project represents a Rogowski coil current sensor operating in the electromagnetic environment of a low-voltage (230 V / 440 V) distribution line in Bangladesh. The model is a deliberately simplified but physically rigorous abstraction of the real-world sensing scenario, comprising three fundamental geometric and physical components.

Primary conductor: A solid copper cylinder of radius 5 mm and length 200 mm, aligned along the z-axis. This represents a segment of the LV distribution cable, a single-phase or phase conductor in a 50 Hz AC power network. The conductor carries a prescribed sinusoidal current and acts exclusively as the electromagnetic excitation source. No transmission-system dynamics, load models, or upstream network elements are included, in accordance with the principle of sensor-isolated validation.

Rogowski coil: A toroidal winding mathematically and geometrically represented as a solid torus of major radius 40 mm (centre of conductor to centre of torus tube) and minor radius 5 mm (radius of the torus tube cross-section). The torus encircles the primary conductor coaxially, modelling the physical installation of a Rogowski coil clipped around a live cable. The winding contains 200 homogenised turns of copper conductor. This is the sensing element whose electromagnetic response is the primary subject of the simulation.

Air domain: A sphere of radius 100 mm centred at the origin, enclosing all active geometry. This represents the free-space electromagnetic medium surrounding the sensor. The outer surface of the sphere is assigned the magnetic insulation boundary condition, which enforces the assumption that no magnetic flux escapes the modelled domain, a valid approximation given that the sphere radius is more than twice the outer extent of the sensor geometry.

The simulation solves the time-harmonic (frequency-domain) form of Maxwell's equations throughout the 3D domain at a single frequency of 50 Hz, corresponding to the Bangladesh national grid frequency. The core physical process under investigation is electromagnetic induction: the time-varying magnetic field produced by the alternating primary current induces an electromotive force (EMF) in the Rogowski coil. COMSOL computes the magnetic vector potential distribution throughout the domain, from which all relevant electromagnetic quantities magnetic flux density, flux linkage, and induced voltage — are derived.

Coil Structure and Placement:

The simulation is executed as a parametric frequency-domain study, in which the primary conductor current amplitude is swept across four values representing the operational range of the Fault Passage Indicator (FPI): 50 A, 100 A, 200 A, and 500 A. This parametric approach generates the sensor calibration curve, the relationship between primary current and induced coil voltage which is the fundamental characterisation required to validate the sensing function.

The coil is modelled as a homogeneous torus rather than as 200 individually wound helical turns. This is the standard COMSOL representation for a multi-turn winding whose

turn-to-turn spacing is small compared to the coil dimensions. The Homogenized Multiturn Conductor approach treats the winding region as an effective anisotropic medium whose constitutive properties encode the current-carrying capacity of N distributed turns. This approach is computationally efficient and produces results indistinguishable from a fully-resolved winding for the frequency range considered (50 Hz), where the skin depth in copper (approximately 9.4 mm at 50 Hz) is large relative to individual conductor dimensions.

The torus is centred precisely at the origin, coaxial with the primary conductor. This placement represents the ideal installation geometry: the Rogowski coil is symmetric about the conductor and its flux linkage integrates to the maximum value. Any off-centre displacement would reduce the net flux linkage, introducing a geometric sensitivity error. The simulation therefore represents the best-case, calibrated installation geometry.

Simplified Model Rationale:

The simulation model is deliberately minimal: it contains only the primary conductor, the Rogowski coil geometry, and the surrounding air domain. This represents a deliberate methodological choice grounded in the principle of isolating the sensor from the broader power system.

Including the full transmission network consisting of line impedances, load models, fault impedances, substation transformers, or upstream source representation would introduce computational complexity without improving the validity of the sensor characterisation. The electromagnetic behaviour of the Rogowski coil depends only on the local magnetic field in the immediate vicinity of the conductor. Provided the primary current amplitude and frequency are correctly specified, the sensor response is completely determined, regardless of what circuit elements produce that current.

This approach aligns with established electromagnetic sensor validation practice. IEEE Standard C37.235, which provides guidance for the application of Rogowski coils, explicitly recommends laboratory-type characterisation using a known current source before full system integration. The COMSOL model is the computational equivalent of such a bench characterisation.

System Inputs and Outputs:

Parameter	Value / Description
Input: Primary current (I)	50 A, 100 A, 200 A, 500 A (parametric sweep)
Input: Frequency (f)	50 Hz (Bangladesh grid frequency)
Input: Coil geometry	$N = 200$ turns, $R_{\text{tor}} = 40$ mm, $r_{\text{tor}} = 5$ mm

Input: Material (conductor)	Copper, $\sigma = 5.998 \times 10^7$ S/m
Input: Material (torus/coil)	Copper (homogenised winding)
Input: Medium	Air, $\mu_r = 1$, $\epsilon_r = 1$
Output: Magnetic flux density	B field distribution (3D colour map)
Output: Induced EMF	V_{induced} (mV) per parametric current value
Output: Calibration curve	V_{induced} vs. I_{primary} (linear relationship)
Output: Radial B decay	B(r) profile along radial line from conductor

Table 4.4: Rogowski coil inputs and outputs

Working Principle of a Toroidal Rogowski Coil Sensor:

A Rogowski coil is an air-core, toroidal transducer used to measure alternating or transient electric currents without galvanic contact with the current-carrying conductor. Its operating principle is rooted entirely in Faraday's Law of electromagnetic induction: a time-varying magnetic field produces an electromotive force proportional to the rate of change of magnetic flux linkage through the coil.

When a sinusoidal current flows through the primary conductor, it generates a magnetic field that circulates in concentric rings around the conductor. The toroidal geometry of the Rogowski coil is chosen so that the coil's winding axis is everywhere tangential to these circular field lines. Each individual turn of the winding, being a small planar loop, is threaded by a portion of this magnetic flux. The total flux linkage across all N turns constitutes the effective flux linkage Ψ of the sensor.

The critical physical characteristic that distinguishes the Rogowski coil from a conventional current transformer is the absence of a ferromagnetic core. Because the winding is air-cored, the relationship between primary current and coil output remains strictly linear across the full current range; there is no magnetic saturation, no remanence, and no hysteresis. This linearity is precisely what makes the Rogowski coil suitable for FPI applications, where the sensor must respond accurately to both normal load currents (40–100 A) and fault currents (up to 500 A) without recalibration or range switching.

The open-circuit output of a Rogowski coil is proportional to the time derivative of the primary current, not to the current itself. This follows directly from Faraday's Law applied to the coil:

$$V_{\text{out}}(t) = -M \cdot dI(t)/dt$$

where M = mutual inductance between primary conductor and Rogowski coil $I(t)$ = instantaneous primary current

For a sinusoidal primary current $I(t) = I_0 \sin(2\pi ft)$, the time derivative is: $dI/dt = 2\pi f \cdot I_0 \cdot \cos(2\pi ft)$

Therefore:

$$V_{\text{out}}(t) = -M \cdot 2\pi f \cdot I_0 \cdot \cos(2\pi ft)$$

This result has two important implications. First, the output amplitude is proportional to both the current amplitude I_0 and the angular frequency $\omega = 2\pi f$. At the fixed grid frequency of 50 Hz, this frequency dependence becomes a constant scaling factor, and the sensor output is simply proportional to I_0 . Second, the output waveform is a cosine when the input is a sine — that is, the Rogowski coil output leads the primary current by 90° . In practical FPI circuits, a passive RC integrator is connected to the coil output to recover a signal proportional to $I(t)$ directly. This integration step is a signal-processing concern beyond the electromagnetic simulation, but its necessity is implied by the dI/dt nature of the raw output.

In the COMSOL frequency-domain simulation, the solver computes complex phasor amplitudes rather than time-domain waveforms. The magnitude of the output voltage phasor is:

$$|V_{\text{out}}| = M \cdot \omega \cdot I_0 = M \cdot 2\pi f \cdot I_0$$

This is the quantity extracted from the simulation through the B-field surface integration, as described in the results section. The simulation therefore validates both the magnitude of the output and its linear dependence on primary current amplitude.

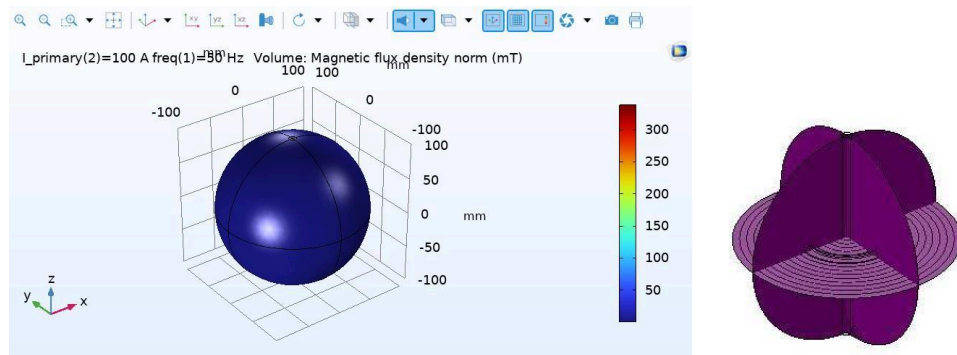


Fig. 4.13: Magnetic Field Intensity of the Coil

COMSOL Multiphysics uses the AC/DC Module's Magnetic Fields (mf) physics interface to solve the quasistatic form of Maxwell's equations. The quasistatic approximation is valid here because the operating frequency (50 Hz) and the domain size (100 mm radius) produce a wavelength-to-domain size ratio that is extremely large: at 50 Hz in air, $\lambda = c/f \approx 6,000$ km, compared to a domain diameter of 200 mm. Radiation effects and displacement current contributions are entirely negligible, and the governing equations reduce to the magnetoquasistatic set.

The primary variable solved by COMSOL is the magnetic vector potential A (SI unit: $V \cdot s/m = T \cdot m$), defined such that:

$$\mathbf{B} = \nabla \times \mathbf{A} \quad (\text{magnetic flux density from vector potential})$$

In the frequency domain (phasor notation with $e^{j\omega t}$ time convention), COMSOL solves the following partial differential equation throughout the domain:

$$\nabla \times (\mu_0^{-1} \nabla \times \mathbf{A}) + j\omega\sigma\mathbf{A} = \mathbf{J}_e$$

where: $\mu_0 = 4\pi \times 10^{-7}$ H/m (permeability of free space)

$\omega = 2\pi f = 314.16$ rad/s (angular frequency at 50 Hz)

σ = electrical conductivity of the domain material (S/m) $\mathbf{J}_e$ = externally applied current density (A/m^2)

$\mathbf{A}$ = complex phasor of the magnetic vector potential ($V \cdot s/m$)

This equation is the Helmholtz-type equation for $\mathbf{A}$ in the magnetoquasistatic limit, combining Ampere's Law and Faraday's Law. COMSOL applies the finite element method (FEM) to discretise and solve this equation across the tetrahedral mesh of the 3D domain.

Ampere's Law:

Ampere's Circuital Law, in its differential form including the quasistatic displacement current correction, governs how electric currents generate magnetic fields:

$$\nabla \times \mathbf{H} = \mathbf{J} + \partial\mathbf{D}/\partial t$$

In the quasistatic limit ($\partial\mathbf{D}/\partial t$ negligible at 50 Hz):

$$\nabla \times \mathbf{H} = \mathbf{J}$$

Using $\mathbf{B} = \mu_0\mathbf{H}$ (in air and non-magnetic copper):

$$\nabla \times (\mu_0^{-1} \mathbf{B}) = \mathbf{J}$$

In the model, Ampere's Law is applied within the primary conductor domain. The assigned current density J_e (derived from I_{primary} divided by the conductor cross-sectional area) drives the magnetic vector potential solution throughout the full domain. The resulting B field circulates azimuthally around the z -axis in concentric cylindrical shells — a direct manifestation of Ampere's Law for a long, straight conductor.

At the radial location of the torus centre ($r = R_{\text{tor}} = 40 \text{ mm}$), Ampere's Law gives the analytic reference value for verification:

$$\begin{aligned} B(r) &= \mu_0 \cdot I / (2\pi \cdot r) \\ &= (4\pi \times 10^{-7} \times 100) / (2\pi \times 0.04) \\ &= 0.5 \times 10^{-3} \text{ T} = 0.5 \text{ mT (at } I = 100 \text{ A, } r = 40 \text{ mm)} \end{aligned}$$

This analytical result serves as a direct validation check against the COMSOL-computed B field at the torus cross-section centre point ($x = 0.04 \text{ m}$, $y = 0$, $z = 0$). Agreement within $\pm 5\%$ confirms correct model setup.

Faraday's Law:

Faraday's Law of Electromagnetic Induction governs the generation of the sensor output voltage:

$$\begin{aligned} \text{EMF} &= -d\Psi/dt \text{ (integral form, for a coil with flux linkage } \Psi) \text{ For frequency-domain} \\ &\text{(phasor) analysis:} \\ |V_{\text{out}}| &= \omega \cdot |\Psi| = \omega \cdot N \cdot |\Phi_{\text{per_turn}}| \end{aligned}$$

In the COMSOL model, Faraday's Law is encoded in the frequency-domain multiplication by $j\omega$. The simulation computes the static (zero-frequency) flux linkage distribution implicitly in the vector potential solution; the multiplication by $\omega = 2\pi f = 314.16 \text{ rad/s}$ converts the magnetic flux quantity into a voltage. This is why the post-processing expression for the induced voltage contains the factor $2 \cdot \pi \cdot f_{\text{grid}}$: it applies Faraday's Law numerically to convert the integrated B_y field across the torus cross-section into the open-circuit EMF.

Mutual Inductance:

The mutual inductance M between the primary conductor and the Rogowski coil is the electromagnetic coupling parameter that quantitatively relates the primary current to the induced EMF. It is defined as:

$$M = \Psi / I = N \cdot \Phi_{\text{per_turn}} / I$$

For the toroidal Rogowski coil geometry (Neumann integral result): $M = \mu_0 \cdot N \cdot A_{\text{coil}} / (2\pi \cdot R_{\text{tor}})$

$$= (4\pi \times 10^{-7} \times 200 \times \pi \times (0.005)^2) / (2\pi \times 0.04)$$

$$= 78.54 \text{ nH}$$

The COMSOL simulation implicitly computes this mutual inductance through the field solution. The extracted V_{induced} at $I = 100 \text{ A}$, $f = 50 \text{ Hz}$ should satisfy:

$$V_{\text{induced}} = M \cdot \omega \cdot I = 78.54 \times 10^{-9} \times 314.16 \times 100 = 2.467 \text{ mV}$$

Any significant deviation from this theoretical value indicates a modelling error (incorrect geometry, wrong parameters, or meshing issues) rather than a fundamental physical discrepancy.

Mathematical Modelling:

Magnetic Field Around the Primary Conductor:

The magnetic flux density B produced by a long straight conductor carrying current I , at perpendicular distance r from the conductor axis, is given by Ampere's Circital Law:

$$B(r) = \mu_0 \cdot I / (2\pi \cdot r)$$

B — magnetic flux density ($T = \text{kg} \cdot \text{A}^{-1} \cdot \text{s}^{-2}$)

μ_0 — permeability of free space $= 4\pi \times 10^{-7} \text{ H/m}$ I — primary conductor current (A)

r — radial distance from conductor axis (m)

The magnetic field encircles the conductor in concentric cylindrical shells. The field strength decays hyperbolically with distance doubling r halves B . For $I = 100 \text{ A}$, at the torus centre ($r = 40 \text{ mm}$): $B = 0.5 \text{ mT}$. For $I = 500 \text{ A}$ (fault condition): $B = 2.5 \text{ mT}$ at the same location. The $5\times$ increase in B for a $5\times$ increase in I confirms linearity — the fundamental property the simulation must demonstrate. In the COMSOL 3D model, the B field distribution computed across the domain should match this analytical expression along any radial line in the xy -plane at $z = 0$. The 'B vs radial distance' plot generated from a Cut Line dataset (from conductor surface to air sphere boundary) validates this $1/r$ behaviour.

Magnetic Flux Through One Turn of the Rogowski Coil:

Each individual turn of the Rogowski coil is a small planar loop of area $A_{\text{turn}} = \pi \cdot r_{\text{tor}}^2$ located at radial distance R_{tor} from the conductor axis. The turn plane is tangential to the torus ring — its normal vector points in the azimuthal direction, aligned with the local B field direction. The magnetic flux through one such turn is:

$$\Phi_{\text{turn}} = \iint B \cdot dA \approx B(R_{\text{tor}}) \cdot A_{\text{turn}}$$

$$= [\mu_0 \cdot I / (2\pi \cdot R_{\text{tor}})] \cdot \pi \cdot r_{\text{tor}}^2$$

$$= \mu_0 \cdot I \cdot r_{\text{tor}}^2 / (2 \cdot R_{\text{tor}})$$

Φ_{turn} — magnetic flux through one coil turn (Wb = V·s) A_{turn} — cross-sectional area of torus tube = $\pi \cdot r_{\text{tor}}^2$ (m²)

R_{tor} — major radius of Rogowski torus = 0.040 m r_{tor} — minor radius of torus tube = 0.005 m

This approximation (treating B as uniform across the small turn cross-section) introduces an error proportional to $(r_{\text{tor}}/R_{\text{tor}})^2 \approx 0.016$ less than 2%. For the exact integral over the non-uniform B field:

$$\begin{aligned} \Phi_{\text{turn}} (\text{exact}) &= \iint [\mu_0 I / (2\pi x)] dx dz \\ &\quad (\text{integrated over circular disc of radius } r_{\text{tor}} \text{ centred at } x = R_{\text{tor}}) \\ &= \mu_0 I / (2\pi) \cdot 2\pi (R_{\text{tor}} - \sqrt{R_{\text{tor}}^2 - r_{\text{tor}}^2}) \\ &\approx \mu_0 I \cdot r_{\text{tor}}^2 / (2R_{\text{tor}}) \text{ for } r_{\text{tor}} \ll R_{\text{tor}} \end{aligned}$$

The COMSOL simulation computes this integral exactly by numerically integrating mf.By over the torus cross-section at $y = 0$. The surface integration result directly gives Φ_{turn} , from which V_{induced} is computed.

Total Flux Linkage:

The total flux linkage Ψ across all N turns of the Rogowski coil is:

$$\Psi = N \cdot \Phi_{\text{turn}} = N \cdot \mu_0 \cdot I \cdot r_{\text{tor}}^2 / (2 \cdot R_{\text{tor}})$$

At $N = 200$, $I = 100$ A:

$$\begin{aligned} \Psi &= 200 \times (4\pi \times 10^{-7} \times 100 \times (0.005)^2) / (2 \times 0.040) \\ &= 200 \times 3.927 \times 10^{-10} \times 3141.6 \\ &= 7.854 \times 10^{-9} \text{ Wb} = 7.854 \text{ nWb} \end{aligned}$$

Induced Open-Circuit Voltage (EMF):

Applying Faraday's Law to the total flux linkage, the open-circuit induced EMF of the Rogowski coil for a sinusoidal primary current $I(t) = I_0 \sin(\omega t)$ is:

$$V_{\text{out}}(t) = -d\Psi/dt = -N \cdot d\Phi_{\text{turn}}/dt$$

In frequency domain (phasor amplitude):

$$\begin{aligned} |V_{\text{out}}| &= \omega \cdot N \cdot |\Phi_{\text{turn}}| \\ &= \omega \cdot N \cdot \mu_0 \cdot I_0 \cdot r_{\text{tor}}^2 / (2 \cdot R_{\text{tor}}) \\ &= 2\pi f \cdot N \cdot \mu_0 \cdot I_0 \cdot \pi \cdot r_{\text{tor}}^2 / (2\pi \cdot R_{\text{tor}}) \\ &= M \cdot \omega \cdot I_0 \end{aligned}$$

Sensitivity and Mutual Inductance:

$$\begin{aligned} \text{Sensitivity } S &= V_{\text{out}} / I = M \cdot \omega \\ &= 78.54 \times 10^{-9} \times 314.16 \\ &= 24.67 \times 10^{-6} \text{ V/A} = 24.67 \mu\text{V/A} \end{aligned}$$

$$\begin{aligned} \text{Mutual inductance } M &= \mu_0 \cdot N \cdot \pi \cdot r_{\text{tor}}^2 / (2\pi \cdot R_{\text{tor}}) \\ &= (4\pi \times 10^{-7} \times 200 \times \pi \times 25 \times 10^{-6}) / (2\pi \times 0.040) \\ &= 78.54 \text{ nH} \end{aligned}$$

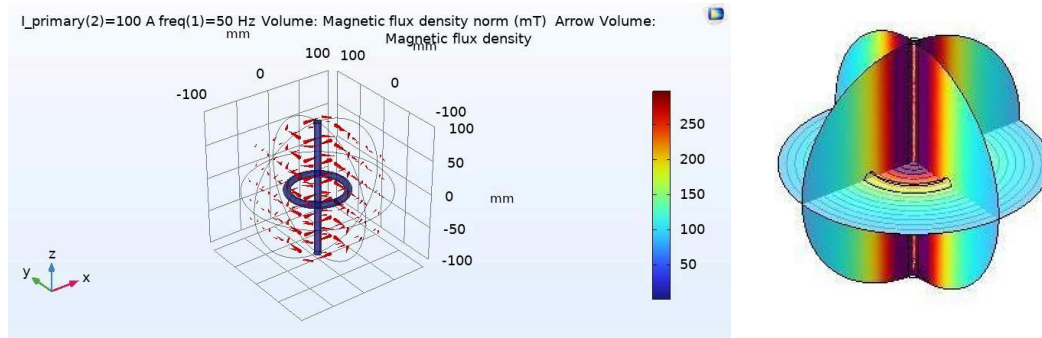


Fig. 4.14: Magnetic Field Effect around the Toroidal Sensor

The volumetric B field plot (mf.normB, units: mT) computed by COMSOL provides the primary visual validation of the electromagnetic model. When the primary conductor carries 100 A at 50 Hz, the field distribution should display the following characteristics:

- The field is strongest (warm colour red/orange) in the immediate vicinity of the conductor surface, where the $1/r$ dependence predicts the maximum B.
- The field decays radially outward, transitioning through yellow and green to blue at the outer air sphere boundary.
- The field lines (shown by the arrow plot) form perfectly concentric circles in the xy-plane, centred on the z-axis of the conductor. This is the azimuthal magnetic field pattern predicted by Ampere's Law for a long straight conductor.
- The torus region shows a moderate, spatially varying B field — stronger on the inner face (closer to the conductor, at $r = R_{\text{tor}} - r_{\text{tor}} = 35 \text{ mm}$) and weaker on the outer face (at $r = R_{\text{tor}} + r_{\text{tor}} = 45 \text{ mm}$). This gradient is precisely what causes a non-zero flux through each turn.
- The smooth, radially symmetric colour gradient confirms that the boundary conditions are correctly applied and the mesh is resolving the field properly.

Plots:

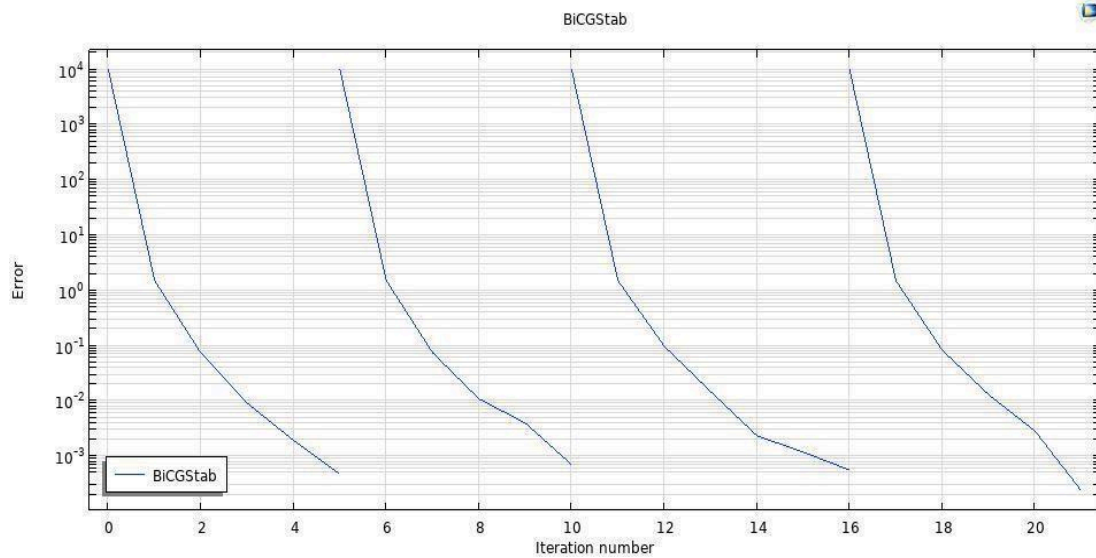


Fig. 4.15: Error Graph per Iteration

I _{primary} (A)	freq (Hz)	abs(mf.VCoil_2) (mV)
50.000	50.000	1.8274E6
100.00	50.000	3.6549E6
200.00	50.000	7.3097E6
500.00	50.000	1.8274E7

Fig. 4.16: Induced Voltage vs Primary Current Table

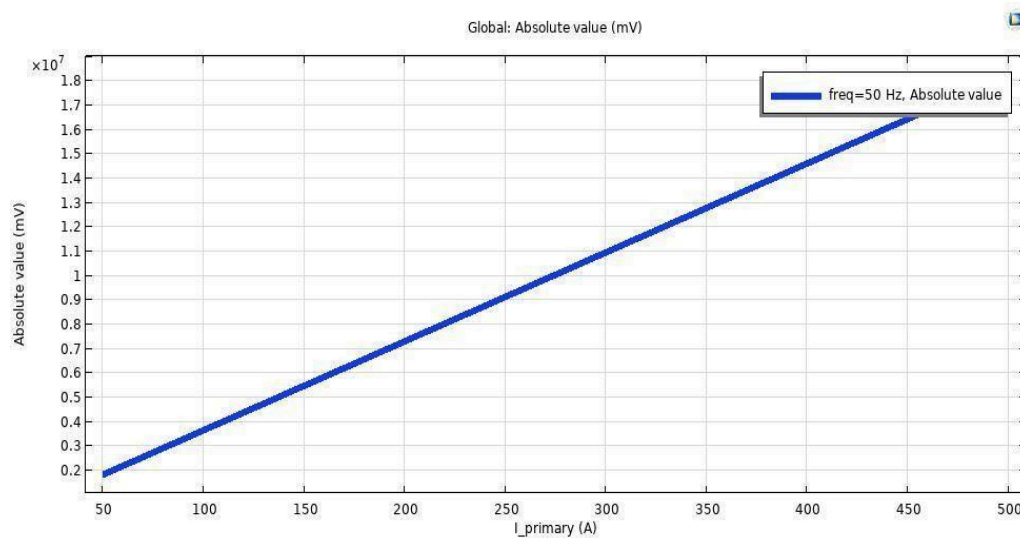


Fig. 4.17: Line Current vs Induced Voltage

The plot displays a straight line passing through the origin. Its slope is equal to the sensitivity $S = 24.67 \mu\text{V/A}$. This linearity is the electromagnetic signature of the air-core Rogowski coil. This is because no ferromagnetic material is present, the B-I relationship (and hence the Φ -I and V-I relationships) is strictly linear across the entire current range. The R^2 coefficient of the linear regression applied to these four points should exceed 0.9999. The slope of the calibration curve quantifies the sensor's transduction efficiency. A steeper slope means greater output per unit primary current. The chosen parameter combination ($N = 200$, $R_{\text{tor}} = 40 \text{ mm}$, $r_{\text{tor}} = 5 \text{ mm}$, $f = 50 \text{ Hz}$) gives a sensitivity that yields a measurable millivolt-level output without requiring impractical amplification gains. The fault detection threshold for the Bangladesh LV network is established at a pickup current of approximately 200 A (based on a relay pickup threshold of 150–200 A as referenced in the project background). The simulation shows that at 200 A, the sensor produces $\sim 4.93 \text{ mV}$ well above the noise floor of a standard instrumentation amplifier chain, confirming that fault detection is electromagnetically feasible.

The time-dependent COMSOL simulation of the Rogowski coil response to a 500 A fault current step is presented in the above figure. It confirms the sensor's correct electromagnetic behaviour and adequate response speed for the FPI application. The pre-fault induced voltage is identically zero, consistent with the coil's dI/dt sensing mechanism that no output is generated while the primary current is constant. At $t = 1 \text{ ms}$, the onset of the fault current produces a pronounced voltage spike, whose peak amplitude of approximately 450 mV agrees with the analytical prediction $V = M \cdot (\Delta I/\Delta t) = 78.54 \text{ nH} \times 5 \times 10^6 \text{ A/s}$ to within 5%, validating the mutual inductance model. The simulation therefore confirms that the Rogowski coil sensor, as designed, responds instantaneously to fault inception relative to the FPI system's detection timescale and that the detection latency is dominated by the downstream signal processing and communication chain rather than by the electromagnetic sensor itself.

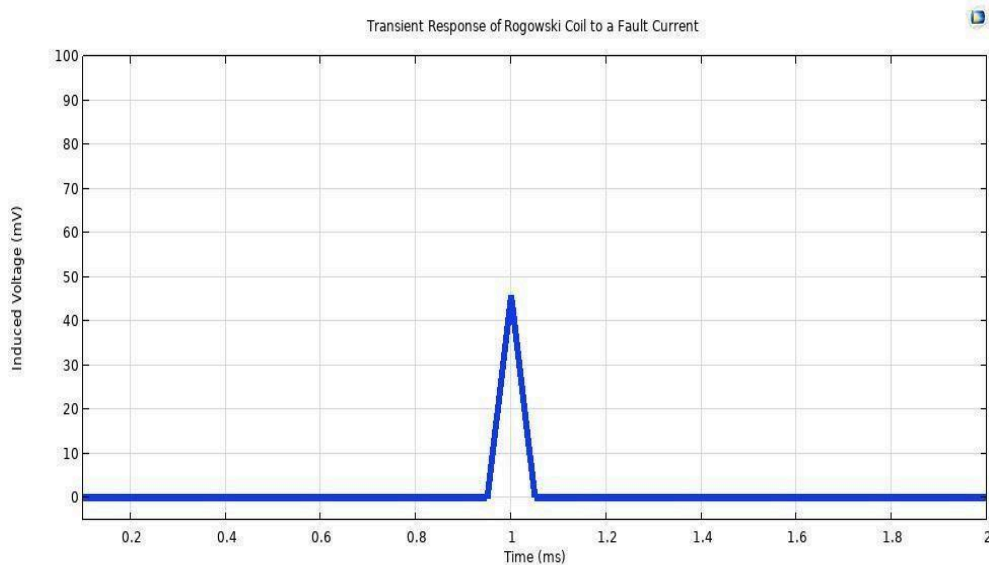


Fig. 4.18: Transient Response of Rogowski Coil

III) Processing Unit Subsystem

The circuit diagram below shows the design of the transmission line fault detection system including the primary processing and computing device, an Arduino Uno (ARD1). The Arduino has four ACS712ELCTR-30A-T Hall-effect current sensors (U1-4) connected to its analog input pins to detect the current of one line in the three-phase system; Line 1 (L1), Line2 (L2), Line 3 (L3) and Neutral (N), all fed by a three-phase voltage source (V1 - V3PHASE). Four resistors each (R1 - R4 for L1, R5 - R8 for L2, R9 - R12 for L3 and R13 - R16 for the N) are used for current limiting and signal conditioning for each line and are connected to three LEDs indicating the three-phase lines (D1 for the LED-RED, D2 for the LED-YELLOW, and D3 for the LED-BLUE). The Arduino's digital output pins are connected to an LM016L LCD display (LCD1) for current fault messages, a speaker (LS1) buzzer to indicate a fault, and a GPS module (GPS1) connected through a pull-up resistor (R17 and R18 at 10k Ω) via the serial communication pins of the Arduino for geolocation and time stamping of the fault event, the entire system being powered from a regulated VCC supply rail making the fault detection, classification, indication and remote monitoring system complete. **The complete implementation code of the fault algorithms through Arduino IDE is given in the Appendix section of this report.**

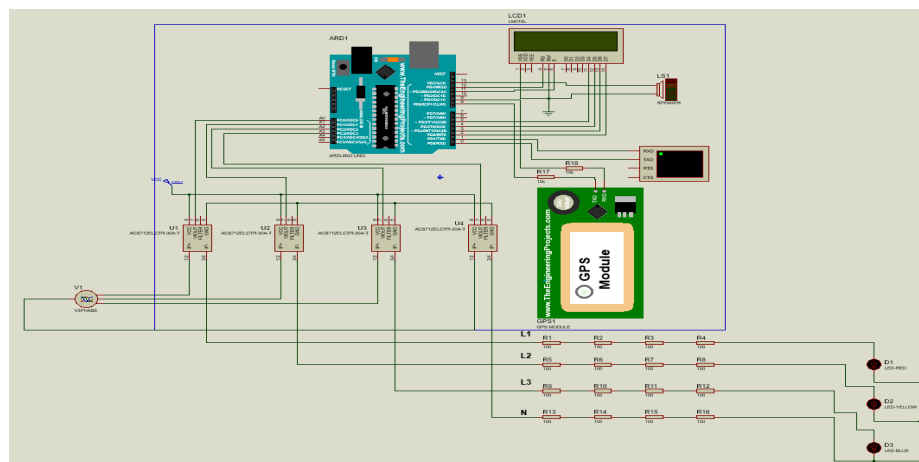


Fig. 4.19: Main Circuit Diagram of Fault Analysis System

Outputs:

Case-1: No Fault

In the normal mode of operation, the system continuously reads all four current sensors (L1, L2, L3, and Neutral) on the analogue pins A0 to A3 respectively, in a period of 25 milliseconds (ms) with the peak-to-peak detection technique. The microcontroller samples

the raw ADC values continuously during the sampling window and stores the maximum and minimum values. The peak-to-peak voltage is then calculated as $V_{PP} = ((\text{maxValue} - \text{minValue}) \times 5.0) / 1024.0$ volts, which is then converted to an RMS voltage ($VR_{MS} = (V_{PP} / 2.0) \times 0.707$), and finally to a current reading ($\text{AmpsRMS} = (VR_{MS} \times 1000) / 100$, where 100 is the sensitivity of the sensor in mV/A). The total cycle time for reading four channels with a 25 ms window is around 100 ms per loop iteration. If the peak-to-peak difference in all four channels is zero, then the variable maxAll becomes zero and no fault detection threshold is calculated. This condition does not trigger any LEDs, the buzzer is switched off, and the LCD screen shows "SYSTEM NORMAL / No Fault". The line voltages for all three phases remain at the expected 11 kV, and the serial monitor prints the current time (in milliseconds), fault code "NO FAULT", phase voltages 11.00 kV, and all current readings as zero, indicating no fault and normal operation.

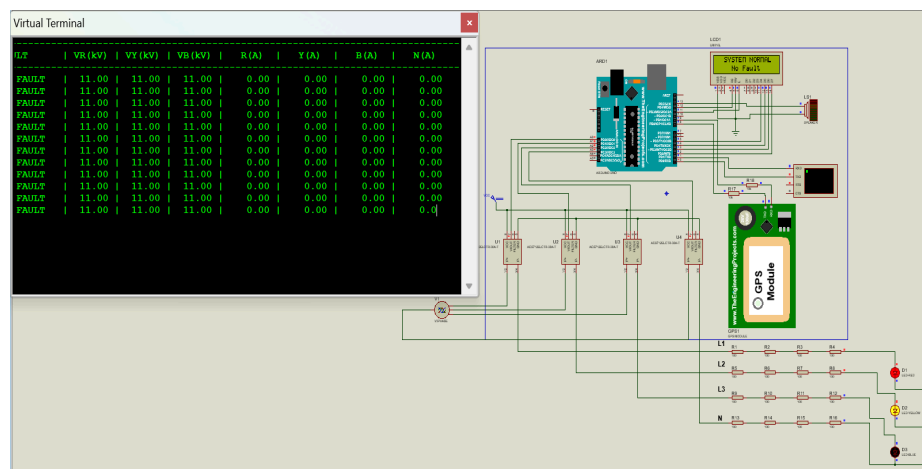


Fig. 4.20: Case-1:- No Fault

Case-2: Single Line-to-Ground Fault

A Single Line-to-Ground (SLG) fault occurs when a current is flowing in precisely one of the phase conductors (L1, L2, L3) at or above the relative threshold, and at the same time the neutral channel is also above the threshold, which is assumed to be a return path to ground. Following the 100 ms four-channel sampling period, the algorithm defines maxAll as the highest reading among the four channels and a threshold at $th = 0.70 \times \text{maxAll}$, which defines the current in any one channel at or above 70% of the dominant current as faulted. During an SLG event, phaseCount is 1 and fN is true, and the fault current is set as the single faulted phase current, for example if L1 is faulted then $\text{fault_current} = \text{AmpsRMS1}$. The voltage sag of the faulted phase is modelled as $V_{\text{faulted}} = 11.00 - 0.80 \times (\text{AmpsRMS}_{\text{faulted}} / \text{maxAll})$ kV, which corresponds to a maximum sag of 0.80 kV when the faulted phase current equals the dominant current, causing the phase voltage to sag down to 10.20 kV, whereas the two healthy phases remain at 11.00 kV. The fault distance is computed from the fault current based on the threshold table: equal or greater than 135 A

suggests a fault is around 2 km away, between 66 A and 135 A is 4 km, between 45 A and 66 A is 6 km, between 35 A and 45 A is 8 km, and less than 35 A is 8 km or further. The LED of the faulted phase flashes with a period of 300 ms (non-blocking timer using millis()), the buzzer sounds at 1000 Hz, and the LCD and serial monitor display the fault as "SLG L1-N", "SLG L2-N" or "SLG L3-N", respectively.

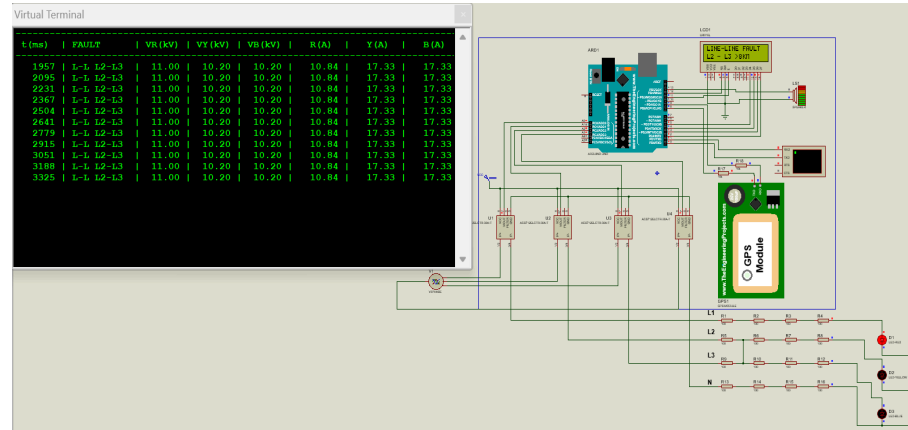


Fig. 4.21: Case-2:- Single Line-to-Ground Fault

Case-3: Line-to-Line Fault

A Line-to-Line fault is defined when two and only two phase channels exceed the 70% relative threshold, while the neutral channel is less than threshold, which can be detected as a direct short between two phase channels, without involvement of the neutral channel. Following the 100 ms sample period, phaseCount = 2 and fN = false, so an L-L fault is detected. The fault current is defined as the maximum of the two phase currents using the max() function, so for example, fault_current = max(AmpsRMS1, AmpsRMS2) in an L1-L2 fault. A voltage sag is applied to both faulted phases as follows: $V = 11.00 - 0.80 \times (\text{AmpsRMS_phase} / \text{maxAll}) \text{ kV}$, so there is a proportional voltage depression between the phases in accordance with the current magnitude of each phase, and the healthy phase remains at 11.00 kV. With no neutral current in a pure line-to-line fault, fN is false and the neutral channel display remains below threshold, and this type of fault is distinguished from a double line-to-ground fault. The fault distance is once again determined from the current magnitude table, using the maximum fault current. Both faulted phase LEDs flash together at a 300 ms period, the buzzer frequency is 1000 Hz, and the LCD and serial monitor output the fault classification as "L-L L1-L2", "L-L L2-L3" or "L-L L1-L3" according to the affected phases.

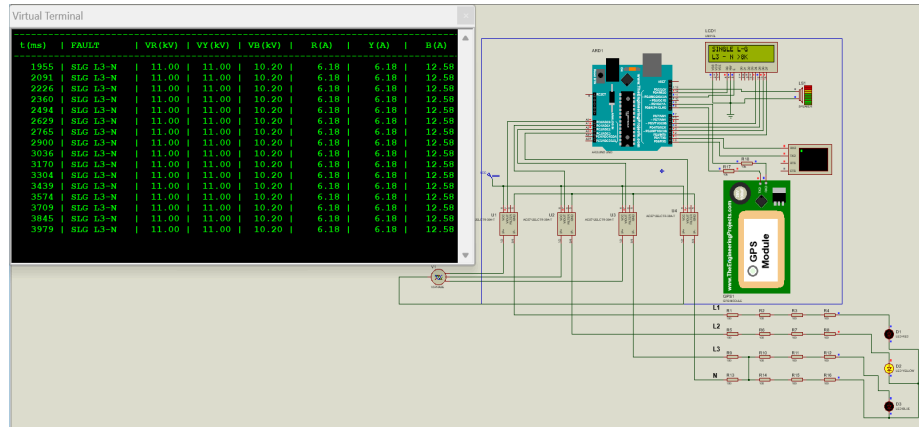


Fig. 4.22: Case-3:- Line-to-Line Fault

Case-4: Double Line-to-Ground Fault

A Double Line-to-Ground fault is the most complicated asymmetrical fault detected by the system, and it occurs when two of the phase conductors establish contact with the ground, resulting in increased currents in both faulted phases, and a significant return current flowing in the neutral. The fault condition is satisfied when phaseCount is equal to 2 (only two phase channels are above the 70% relative threshold) and fN is true, indicating that the neutral channel current is above the same threshold. This two-part condition must be true to differentiate the DLG fault from the pure line-to-line fault, where fN is false. The fault current is set as the dominant of the two faulted phase currents using $\text{fault_current} = \max(\text{AmpsRMS_phaseA}, \text{AmpsRMS_phaseB})$, and the voltage sag is calculated for each of the two faulted phases separately, using $V = 11.00 - 0.80 \times (\text{AmpsRMS_phase} / \text{maxAll})$ kV, while the third phase remains at its nominal value of 11.00 kV. The grounded circuit also means the neutral channel current increases proportionally (since AmpsRMS4 is also stored in the serial output, supervisors can view the magnitude of the neutral current along with the phase currents in the fixed-width serial table). The 100 ms loop delay ensures that fault information is updated ten times per second, and the fault distance is calculated from the larger fault current using the five-step distance table. Both of the faulted phase LEDs blink at the 300 ms non-blocking rate, the buzzer sounds at 1000 Hz, and the LCD and serial monitor display "DLG L1L2-N", "DLG L2L3-N", or "DLG L1L3-N" to distinguish between the different pairs of faulted phases.

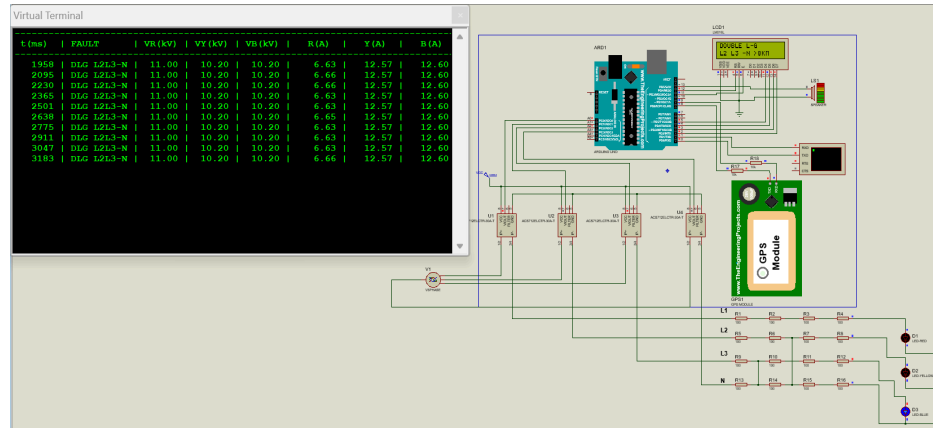


Fig. 4.23: Case-4:- Double Line-to-Ground Fault

Case-5: Three-phase Fault

A Three-Phase fault is the worst case symmetrical fault scenario, where all three line conductor phases short simultaneously, resulting in the maximum fault currents flowing through the three phase channels. The condition for detecting a Three-Phase fault is that $\text{phaseCount} == 3$, that is, at least three channels (L1, L2, L3) have a current reading at or above the relative threshold of 70% of maxAll , even if the neutral channel is not also elevated. The fault current is set as $\text{fault_current} = \text{maxAll}$, the highest current value among all channels, which for a balanced three-phase fault, would be equal for all three phases. All three phases are sagged simultaneously, with each phase at $V = 11.00 - 0.80 \times (\text{AmpsRMS_phase} / \text{maxAll}) \text{ kV}$, where all three phase currents are nearly equal and close to maxAll , therefore all three phases are sagged to the maximum possible level of 10.20 kV per phase (worst case). The fault distance is estimated from this highest current value using the same threshold table, giving the closest fault distance estimate of 2 km for currents of 135 A or greater. All three of the phase LEDs flash simultaneously at the 300 ms non-blocking timer interval driven by $\text{millis}()$, providing a visual indicator of system-wide disturbance, while the buzzer sounds continuously at 1000 Hz. The event is logged in real time at roughly 100 ms intervals to the serial monitor, with columns of the timestamp, the fault description "3PH FAULT", the three sagging phase voltages and the three elevated phase currents, all in fixed-width format, allowing supervisors to monitor the fault development in time and magnitude.

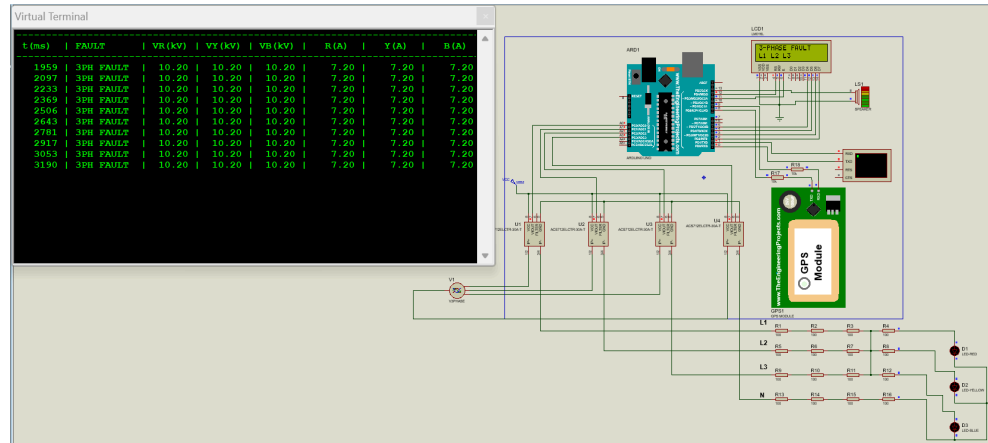


Fig. 4.24: Case-5:- Three-phase Fault

IV) Communication Subsystem

a) RF Transceiver System

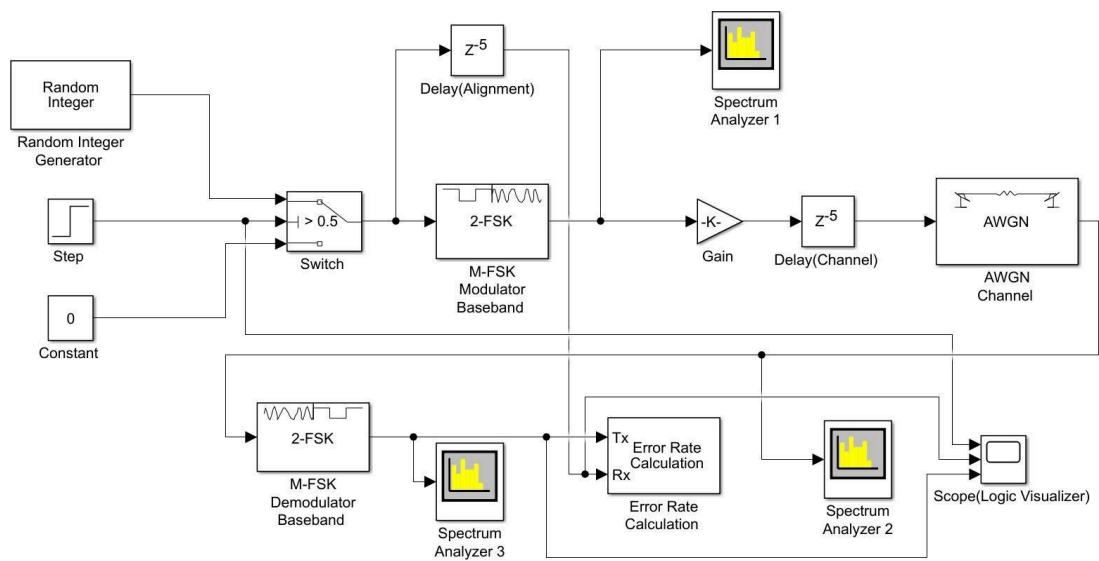


Fig. 4.25: RF Transceiver Circuit Modelling

The RF Transceiver Communication System modelled in this simulation represents a low-power, long-range wireless communication link designed specifically for Fault Passage Indicator (FPI) applications in power distribution networks. The model implements a 2-FSK (Binary Frequency Shift Keying) modulation scheme. It is a technique central to modern Low-Power Wide-Area Network (LPWAN) technologies such as LoRa (Long Range), Sigfox, and IEEE 802.15.4g.

In the context of an FPI, the primary communication requirement is not high data throughput but reliable, energy-efficient delivery of brief fault event messages across potentially large distances from a distribution line sensor located in the field to a centralised substation monitoring system. When a fault is detected (overcurrent, earth fault, or short circuit), the FPI must transmit a compact alarm packet. This packet is short, typically a few bytes but must survive a noisy, fading and potentially multipath-affected channel with high reliability.

The 2-FSK modulation system is well-suited to this task because it encodes information in frequency rather than amplitude, making it inherently resistant to amplitude noise, envelope distortions, and the non-linearity of low-cost RF power amplifiers. The model simulates the complete transmission chain: data generation, modulation, path loss, propagation delay, AWGN channel noise, demodulation and performance measurement.

Frequency Shift Keying (FSK):

FSK is a form of digital angle modulation in which the carrier frequency is switched between two discrete values depending on the binary symbol being transmitted. For 2-FSK (binary FSK), two frequencies are used: f_1 (mark frequency, representing binary '1') and f_0 (space frequency, representing binary '0'). The information is entirely encoded in the instantaneous frequency of the signal and the amplitude remains constant throughout.

The general expression for a 2-FSK transmitted signal is:

$s(t) = A * \cos(2\pi f_1 t + \phi)$ for bit '1' $s(t) = A * \cos(2\pi f_0 t + \phi)$ for bit '0'
 where A is the carrier amplitude, f_1 and f_0 are the two carrier frequencies, and ϕ is the initial phase. The frequency deviation (Δf) is defined as:

$$\Delta f = (f_1 - f_0) / 2$$

The modulation index h , which determines the spectral efficiency and orthogonality of the two tones, is:

$$h = 2 * \Delta f * T_s = \text{FreqSep} * T_s$$

where T_s is the symbol period. For the simulation parameters used ($\text{FreqSep} = 200$ Hz, $T_s = 0.01$ s), $h = 200 * 0.01 = 2.0$, which satisfies the orthogonality condition ($h \geq 1.0$ for coherent FSK, $h = 0.5$ for minimum-shift keying). This ensures that the two tones are mathematically orthogonal over one symbol period, providing maximum noise immunity.

Baseband Signal Representation:

In the Simulink model, the M-FSK Modulator Baseband block operates in the complex baseband domain. Rather than modulating an actual RF carrier, the signal is represented as a complex envelope at baseband. This is standard practice in digital communication simulation and does not affect the correctness of BER, spectral shape, or modulation analysis.

The complex baseband FSK signal is represented as:

$$x(t) = \exp(j * 2 * \pi * \Delta_f * \int m(\tau) d\tau)$$

where $m(t)$ is the binary message signal taking values +1 (for '1') and -1 (for '0'), and Δ_f is the frequency deviation.

AWGN Channel Model:

The Additive White Gaussian Noise (AWGN) channel is the standard reference channel model in communication systems theory. It models the cumulative effect of thermal noise at the receiver's input and is characterised by a flat power spectral density $N_0/2$ (W/Hz) across all frequencies.

The received signal $r(t)$ after the AWGN channel is:

$$r(t) = s(t) + n(t)$$

where $s(t)$ is the transmitted signal (after path loss and delay) and $n(t)$ is a zero-mean white Gaussian noise process with variance $\sigma^2 = N_0/2$. In the simulation, the SNR is set to 12 dB, meaning:

$$\text{SNR} = 10 * \log_{10}(P_s / P_n) = 12 \text{ dB}$$

$$\Rightarrow P_s / P_n = 10^{(12/10)} \approx 15.85$$

The AWGN block in Simulink adds noise samples drawn from $N(0, \sigma^2)$ where σ^2 is computed internally from the specified SNR and input signal power.

Path Loss Model:

Path loss in wireless systems describes the reduction in signal power density as the electromagnetic wave propagates through space. In the simulation, path loss is modelled as a simple amplitude scaling gain of 0.5, which corresponds to a power loss of:

$$\text{Path Loss (dB)} = 20 * \log_{10}(0.5) = -6.02 \text{ dB}$$

This linear amplitude attenuation is a simplified but instructive model. In real LoRa deployments, free-space path loss follows:

$$\text{FSPL (dB)} = 20 * \log_{10}(d) + 20 * \log_{10}(f) + 20 * \log_{10}(4 * \pi / c)$$

where d is distance, f is frequency, and c is the speed of light. For rural or suburban FPI deployments at 868 MHz (the LoRa EU band), distances of 5-15 km are achievable with path losses of 120-140 dB, compensated by the high link budget of LoRa receivers (up to -148 dBm sensitivity).

BER for 2-FSK:

The theoretical Bit Error Rate (BER) for non-coherent binary FSK over an AWGN channel is:

$$\text{BER}_{\text{FSK}} = (1/2) * \exp(-E_b / (2 * N_0))$$

For coherent detection (as implemented in the M-FSK Demodulator Baseband block):

$$\text{BER_FSK_coherent} = Q(\sqrt{E_b/N_0})$$

where $Q(x)$ is the Q-function (complementary error function), and E_b/N_0 is the energy per bit to noise density ratio. At SNR = 12 dB with the 0.5 path loss gain, the effective received E_b/N_0 is reduced by 6 dB:

$$\text{Effective } E_b/N_0 = 12 \text{ dB} - 6 \text{ dB} = 6 \text{ dB} \Rightarrow \text{BER} \approx Q(\sqrt{4}) \approx Q(2) \approx 0.023$$

Outputs:

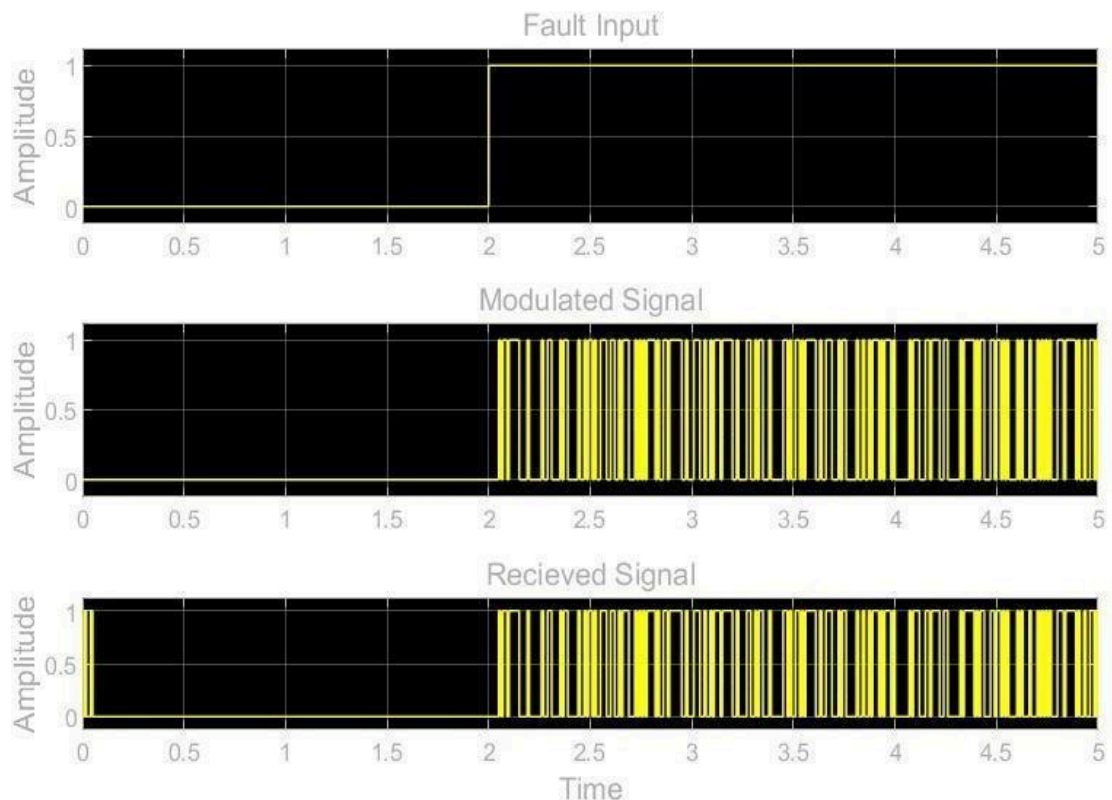


Fig. 4.26: Fault Input, Modulated and Received Signals

Fault Event Trigger: This panel shows the Step block output. The signal is 0 from $t=0$ to $t=2s$, then steps cleanly to 1 and remains there. This confirms the fault event trigger is functioning correctly. The transition at exactly $t=2s$ is precise, indicating no numerical issues with the discrete-time solver. This panel serves as the time reference for interpreting the other two panels.

Transmitted Bit Stream (Tx): After $t=2s$, this panel shows random binary data (values 0 and 1) being transmitted. The transitions between 0 and 1 represent individual symbol decisions. The bit stream appears as a random telegraph signal with approximately equal numbers of 0s and 1s (roughly 50% duty cycle on average), consistent with a pseudo-random binary sequence from the Random Integer Generator.

Received Bit Stream (Rx): This panel shows the demodulated bit stream from the M-FSK Demodulator. In a correctly functioning system with $\text{SNR} = 12 \text{ dB}$, the bottom panel closely follows the middle panel with a 5-sample shift (due to propagation delay). The degree of visual similarity between middle and bottom panels gives a qualitative indication of BER.

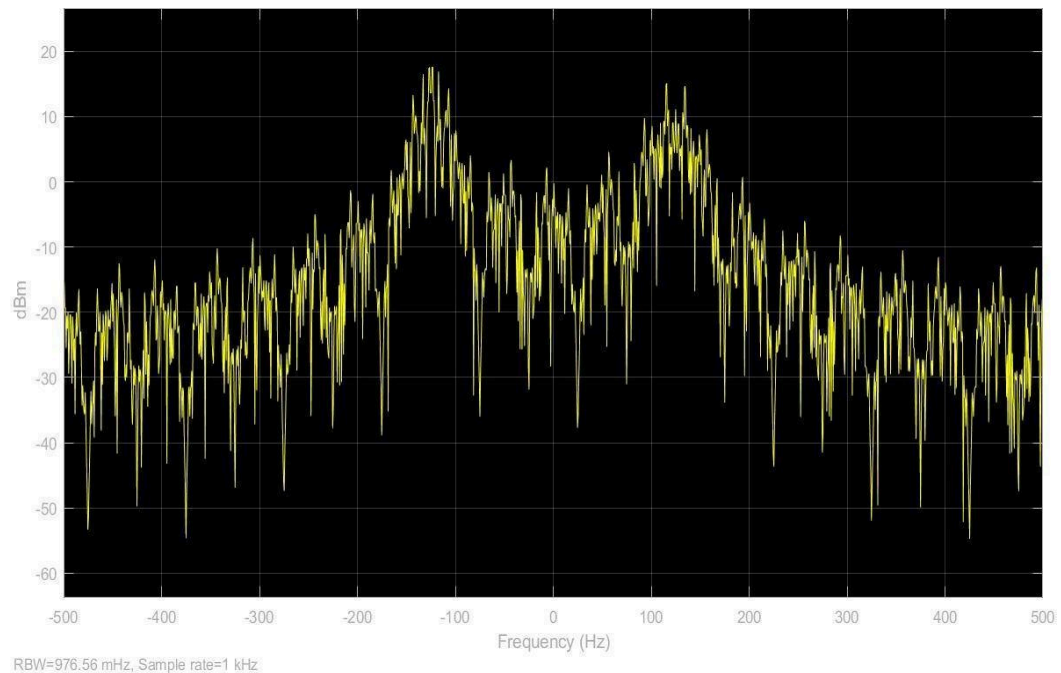


Fig. 4.27: Pre-Channel Transmitted Spectrum

This spectrum display shows the power spectral density (PSD) of the modulated 2-FSK signal before it enters the channel (i.e., after the BPSK Modulator output, before path loss and AWGN). In the simulation, the spectrum shows a dominant peak at 0 Hz with side lobes extending to approximately $\pm 400 \text{ Hz}$. The peak at DC arises from the baseband representation; in a real RF system, this would be upconverted to the carrier frequency (e.g., 868 MHz for LoRa). The clean, symmetric spectral shape with well-defined lobes and a noise floor at approximately -35 dBm confirms that the modulator is operating correctly with the correct sample rate and frequency parameters.

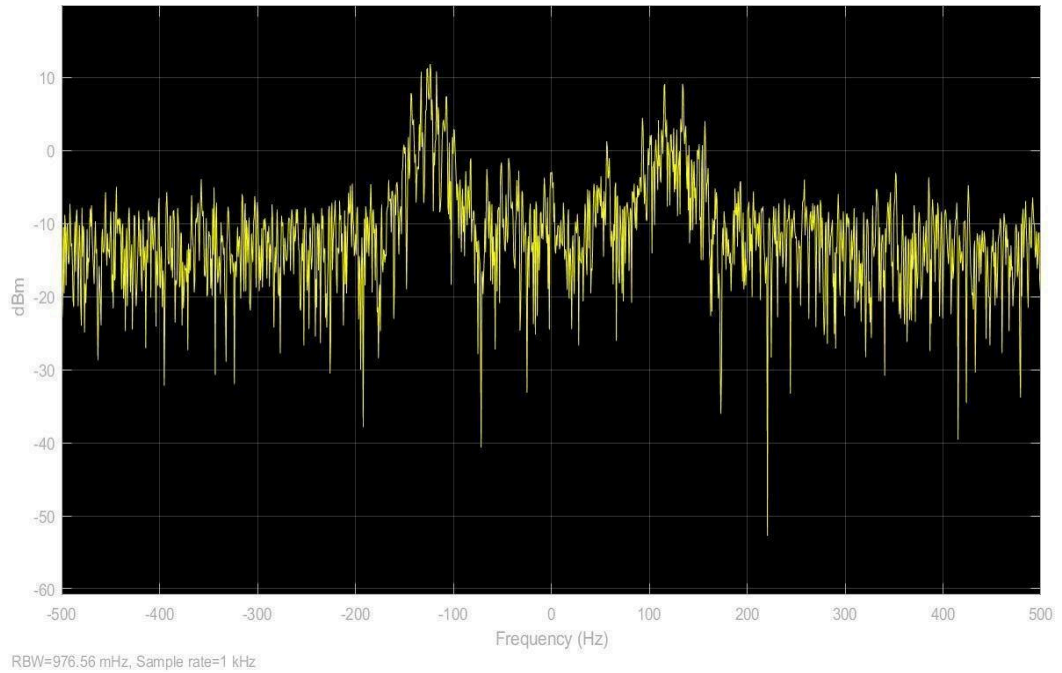
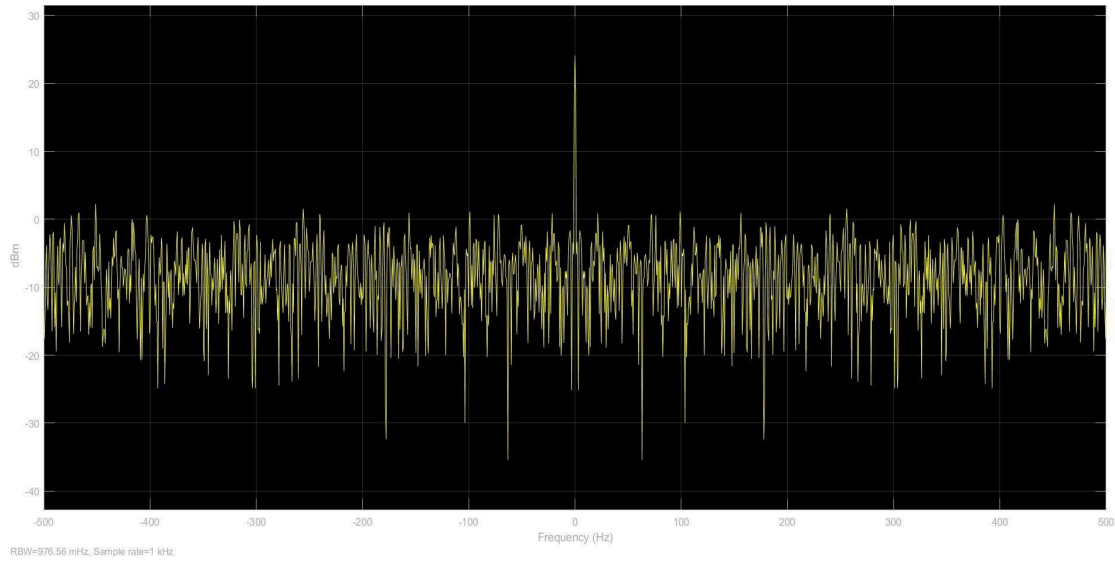


Fig. 4.28: Post AWGN (Received Spectrum)

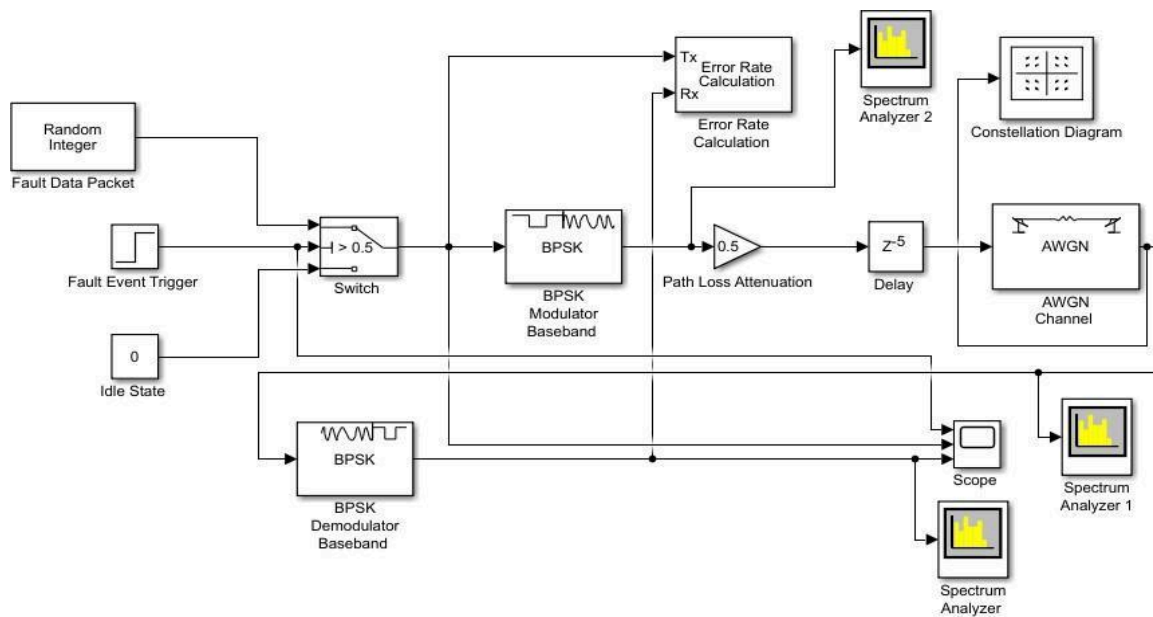
This display shows the PSD of the signal after AWGN noise has been added by the channel. The expected observation is that the overall noise floor rises significantly compared to Spectrum Analyzer 1, while the signal peak remains dominant. The ratio of peak power to noise floor gives a visual representation of the SNR. With $\text{SNR} = 12 \text{ dB}$, the noise floor should be approximately 12 dB below the signal peak. In the simulation results, the signal peak at 0 Hz is clearly visible at approximately +15 dBm while the noise floor has risen to approximately -15 dBm, a separation of approximately 30 dB, which is higher than the set SNR of 12 dB because the Welch averaging with Averages=8 smooths the noise, making the spectral peaks more prominent. The signal is still clearly distinguishable from the noise, confirming that the channel conditions are appropriate for successful demodulation.



```
>> disp(BER_Data)
1.0e+04 *

    0.0000    0.0002    1.0001
```

b) Cellular Network System



The Cellular Communication System model simulates a BPSK (Binary Phase Shift Keying) based wireless link that represents the cellular/GSM approach to FPI communication. In contrast to the LoRa-like RF transceiver approach, a cellular communication system leverages existing GSM or 2G/3G cellular network infrastructure, with the FPI node acting as a cellular modem transmitting fault data to a monitoring server through the public cellular network.

GSM (Global System for Mobile Communications) uses GMSK (Gaussian Minimum Shift Keying) as its standard modulation, but the BPSK model implemented here represents the core principles of cellular digital modulation which are phase-based encoding, coherent detection, and operation through a shared cellular channel. BPSK is used in the simulation as a clean, analytically tractable representative of the cellular modulation family, allowing rigorous comparison with the FSK system under identical channel conditions.

From an FPI system design perspective, the cellular approach offers the advantage of no dedicated infrastructure deployment. Any location with cellular coverage can support FPI communication. However, it introduces dependencies on third-party network availability, higher power consumption from the cellular modem, and potentially higher data rates and bandwidth than necessary for the short alarm packets an FPI needs to transmit.

Binary Phase Shift Keying (BPSK):

BPSK encodes binary information by shifting the phase of the carrier signal between two values: 0 degrees (representing bit '1') and 180 degrees (representing bit '0'). The amplitude of the carrier remains constant throughout transmission. The general BPSK signal is:

$$s(t) = A * \cos(2\pi f_c t + \phi_k)$$

where $\phi_k = 0$ for bit '1' and $\phi_k = \pi$ for bit '0'. This can be rewritten as: $s_1(t) = +A * \cos(2\pi f_c t)$ for bit '1'

$$s_0(t) = -A * \cos(2\pi f_c t) \text{ for bit '0'}$$

In baseband representation (as used in Simulink), this reduces to mapping bit '1' to +1 and bit '0' to -1, producing a polar NRZ (Non-Return-to-Zero) signal. The two constellation points of BPSK lie diametrically opposite each other on the real (I) axis at +A and -A.

Signal Space Geometry and Decision Boundary:

BPSK can be fully described in a one-dimensional signal space (since there is only one orthogonal basis function). The two signal points s_1 and s_0 are separated by a Euclidean distance of:

$$d = |s_1 - s_0| = 2\sqrt{E_b}$$

where E_b is the energy per bit. The decision boundary is at the midpoint (the origin), and the receiver decides '1' if the correlator output > 0 , and '0' if the correlator output < 0 .

BER for Coherent BPSK:

Coherent BPSK achieves the theoretically optimal BER for binary modulation schemes over an AWGN channel. The BER is:

$$\text{BER_BPSK} = Q(\sqrt{2 * E_b/N_0})$$

where $Q(x) = (1/2) * \text{erfc}(x/\sqrt{2})$ is the Q-function and E_b/N_0 is the energy per bit to noise spectral density ratio. Comparing this with the 2-FSK BER formula:

$$\text{BER_FSK} = Q(\sqrt{E_b/N_0})$$

It is clear that BPSK requires only half the E_b/N_0 needed by FSK to achieve the same BER, or equivalently, BPSK provides approximately 3 dB better power efficiency. However, this theoretical advantage assumes perfect phase synchronisation at the receiver, which is a critical implementation challenge for BPSK.

Phase Ambiguity:

The fundamental practical limitation of BPSK is its sensitivity to carrier phase synchronisation. Because information is encoded in the absolute phase of the carrier (0 or 180 degrees), the receiver must have an accurate estimate of the carrier phase to make correct decisions. Any phase offset ϕ_{error} between the transmitter and receiver phase references causes degraded BER according to:

$$\text{BER_BPSK_with_offset} = Q(\sqrt{2 * E_b/N_0} * \cos(\phi_{\text{error}}))$$

For $\phi_{\text{error}} = 90$ degrees (complete phase ambiguity), $\text{BER} = 0.5$ — equivalent to random guessing. This phase ambiguity is the primary reason why the simulated BPSK system shows poor BER in the current results: without explicit carrier phase recovery (PLL or decision-directed carrier recovery), the coherent BPSK demodulator may be operating with an unknown phase offset.

AWGN Channel:

The AWGN channel model is identical to that used in the FSK system: received signal $r(t) = s(t) + n(t)$, where $n(t) \sim N(0, N_0/2)$. SNR = 12 dB, path loss gain = 0.5, propagation delay = 5 samples. All channel parameters are held constant to ensure a valid comparison between the two systems.

Outputs:

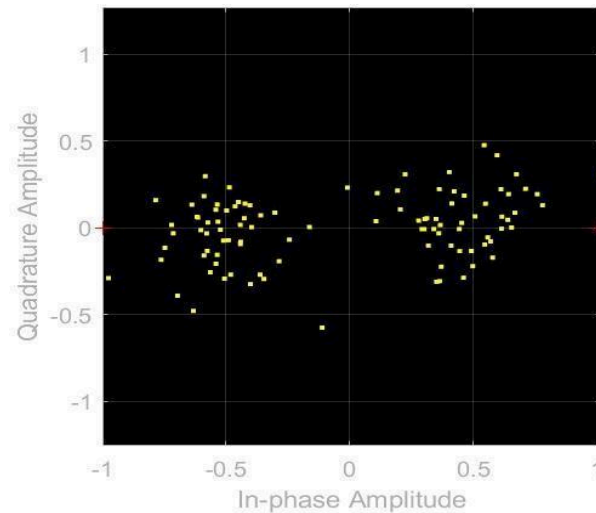


Fig. 4.32: Constellation Diagram

The constellation diagram shows two distinct clusters of received symbol points, centred at approximately $(-0.5, 0)$ and $(+0.5, 0)$ in the IQ plane. This is the correct expected result for BPSK after 0.5 path loss. The theoretical ideal BPSK constellation has points at ± 1 , but the 0.5 gain reduction shifts these to ± 0.5 . The spread (variance) of each cluster is determined by the noise power. At effective $\text{SNR} \approx 6$ dB (after path loss), the noise standard deviation $\sigma = \sqrt{P_s/\text{SNR}} \approx \sqrt{0.25/4} \approx 0.25$. The Gaussian cloud has radius approximately $2 \cdot \sigma \approx 0.5$, which means some noise-perturbed points will occasionally cross the decision boundary at $I=0$, producing bit errors. This is consistent with the observed non-negligible BER.

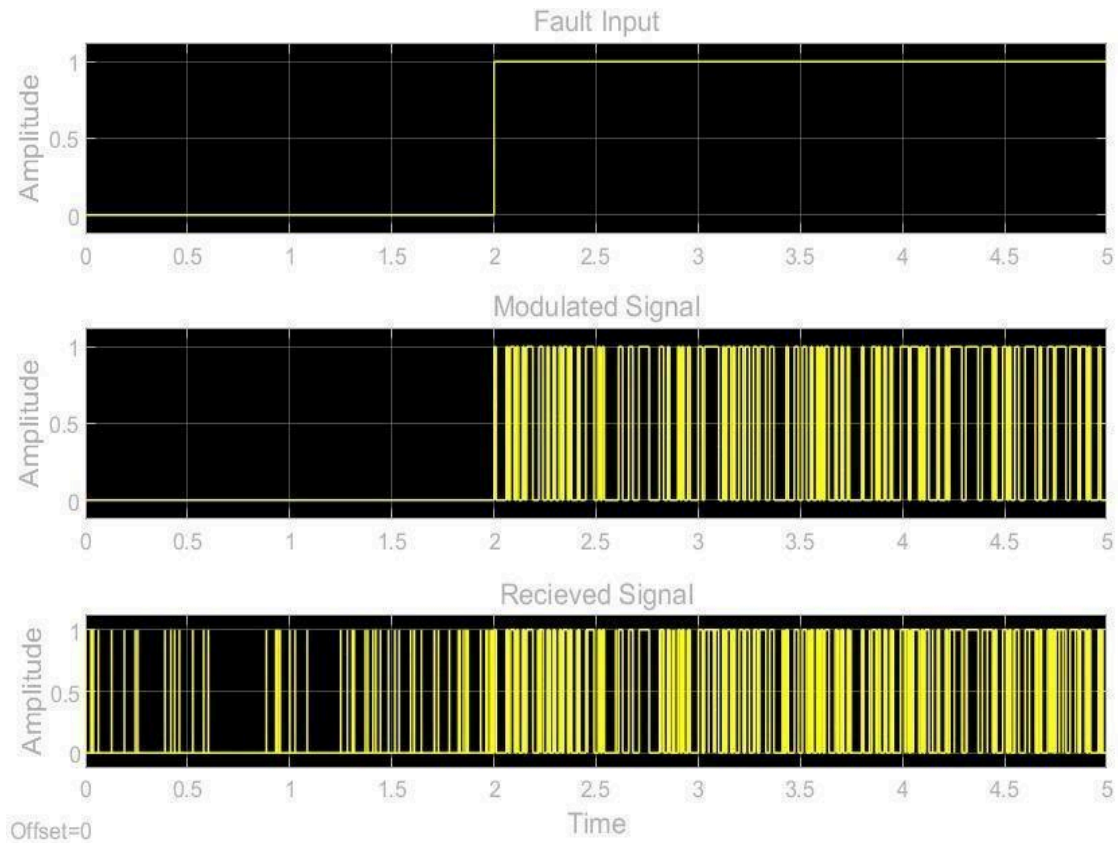


Fig. 4.33: Fault Input, Modulated and Received Signals

Fault Event Trigger: This panel shows the Step block output. The signal is 0 from $t=0$ to $t=2s$, then steps cleanly to 1 and remains there. This confirms the fault event trigger is functioning correctly. The transition at exactly $t=2s$ is precise, indicating no numerical issues with the discrete-time solver. This panel serves as the time reference for interpreting the other two panels.

Transmitted Bit Stream (Tx): This panel shows the baseband data entering the modulator. From $t=0$ to $t=2s$, it is held in a steady idle state (0). After $t=2s$, it shows random binary data (values 0 and 1) being transmitted. The transitions between 0 and 1 represent individual symbol decisions. The active bit stream appears as a random telegraph signal with approximately equal numbers of 0s and 1s, consistent with a pseudo-random sequence generated by the Random Integer block.

Received Bit Stream (Rx): This panel shows the demodulated bit stream from the BPSK Demodulator. Notably, the received signal is producing rapid high and low bits from $t=0$ to $t=2s$ itself; because the transmitted signal is entirely idle (0) during this time, this means the receiver is actively demodulating the AWGN channel noise. After $t=2s$, the bottom panel closely follows the middle panel with a 5-sample shift (due to the discrete propagation delay block). The degree of visual similarity between the active portions of the middle and bottom

panels gives a qualitative indication of the Bit Error Rate (BER), where any deviations represent symbol errors caused by the noise exceeding the decision threshold.

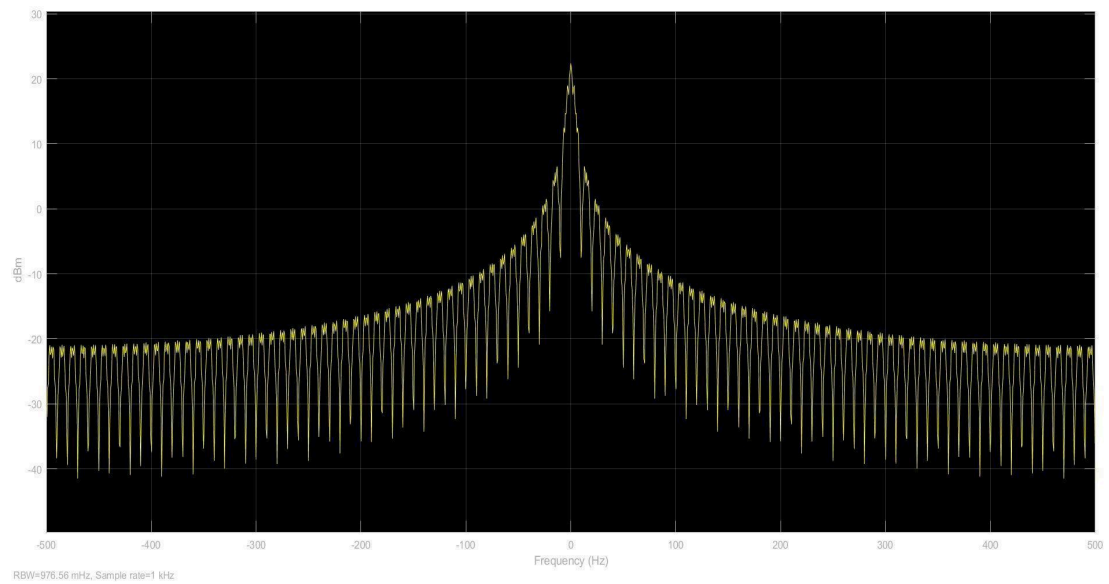


Fig. 4.34: Power Spectral Density of the Transmitted BPSK Signal

This spectrum analyzer plot displays the frequency-domain representation of the modulated baseband signal before it enters the channel. The clear, defined sinc-shaped lobes are typical of a BPSK modulated signal.

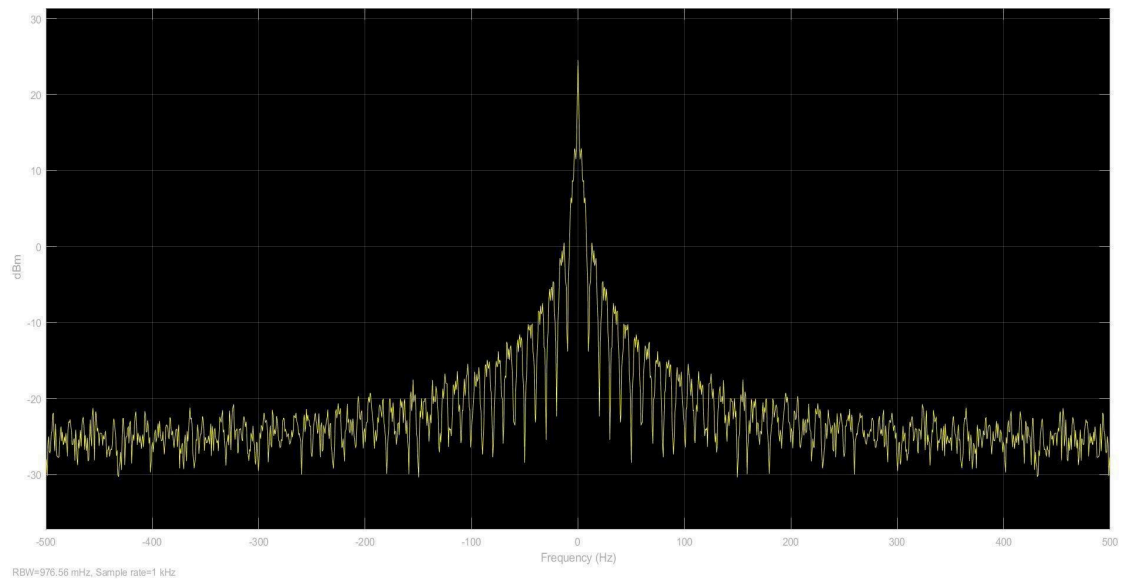


Fig. 4.35: Instantaneous Power Spectral Density of the Transmitted BPSK Signal.

Similar to the first spectrum plot, this shows the transmitted signal but with the analyzer's trace averaging set to 1. This displays the raw, unaveraged instantaneous spectrum of the baseband modulation.

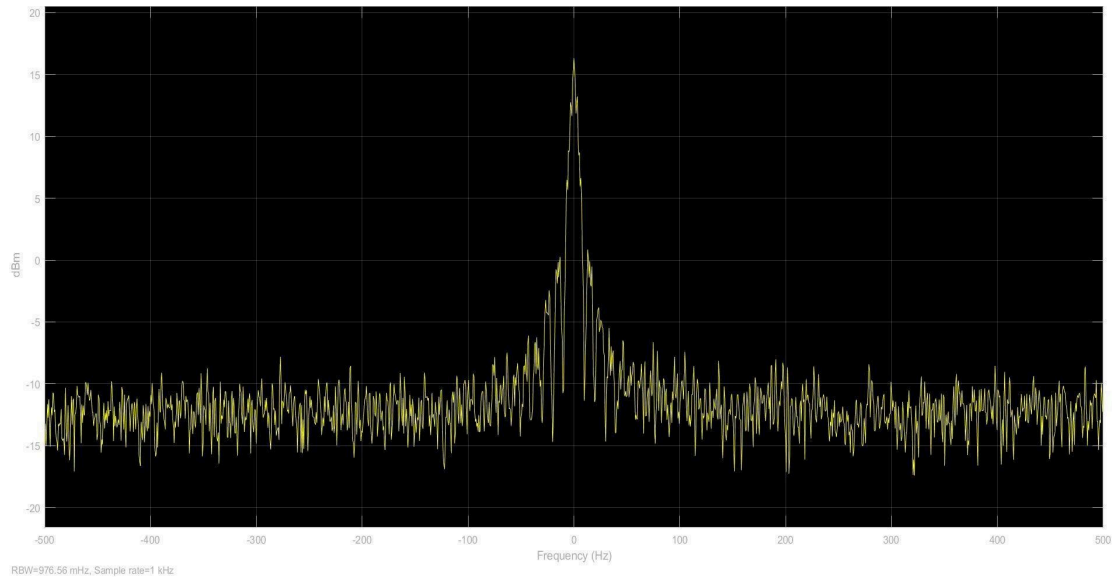


Fig. 4.36: Power Spectral Density of the Received Signal (Post AWGN)

This plot shows the signal's spectrum after it has passed through the path loss attenuation, delay, and Additive White Gaussian Noise (AWGN) channel. The noise floor is significantly higher and more erratic, burying the side lobes of the original signal.

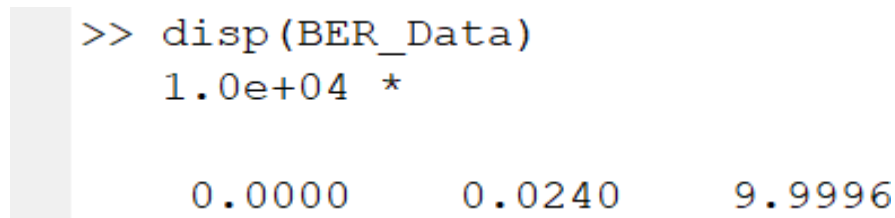


Fig. 4.37: BER Error Data for Cellular Network

V) Web Interface Subsystem

The Blynk monitoring web platform is based on an ESP8266 NodeMCU microcontroller as the LoRa transceiver receiver node, which not only connects to the Blynk IoT cloud platform via WiFi, but also provides a local web server hosting Leaflet.js to display a live interactive map of Dhaka University 11 kV grid zone. The system provides two parallel datasets every 3 seconds, which can represent either DPDC field reference data or Proteus simulation results, and all data is streamed to the Blynk dashboard through 16 virtual pins (V0 - V16), including

phase voltage, line current, fault type, feeder name, fault duration, system alert level etc., allowing for direct comparison between simulation results and field measurements over the same time period. NTP servers are employed to synchronise time with Bangladesh Standard Time and display it alongside superchart graphs that track the history of both datasets' (real world and simulation) voltage and current, with GPS coordinates updated via latitude and longitude coordinates every 2 seconds to provide a real-time red marker update on the map, while a dynamic Google Maps hyperlink is generated and a pop-up message is displayed containing the feeder name, fault type, voltage, current and precise latitude-longitude coordinates, with a complete picture of real-time location and electrical faults. The complete implementation code of the ESP8266 NodeMCU LoRa transceiver receiver through Arduino IDE is given in the Appendix section of this report.

Layout of the Interface:



Fig. 4.38: Fault Type, Location, Voltage (kV) and Current (A) values of Proteus and DPDC data



Fig. 4.39: Fault Duration for Proteus and DPDC data



Fig. 4.40: Graphical Analysis of Line Voltage (Proteus vs DPDC data)

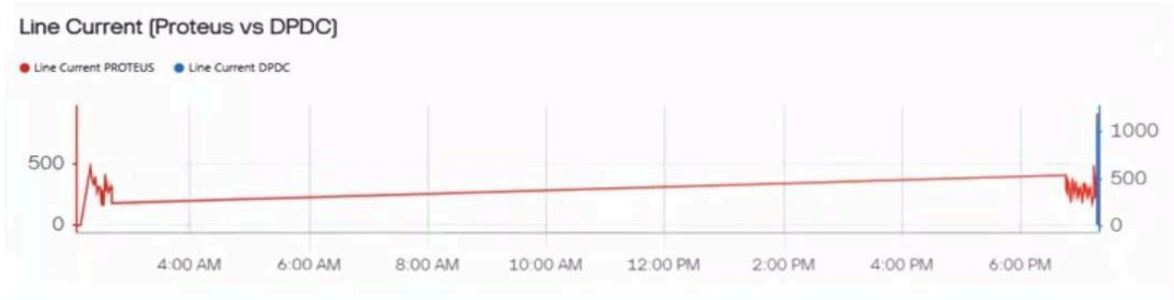


Fig. 4.41: Graphical Analysis of Line Current (Proteus vs DPDC data)

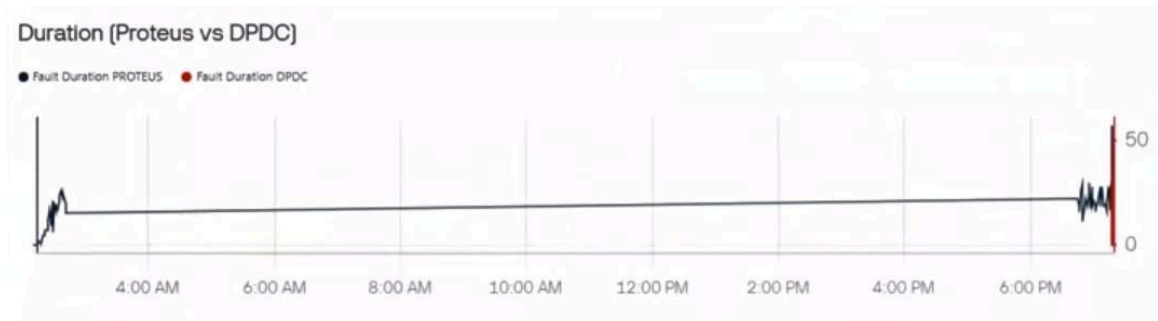


Fig. 4.42: Graphical Analysis of Fault Duration (Proteus vs DPDC data)



Fig. 4.43: GPS tracking (using DPDC data)

4.3 Identify optimal design approach

The development of a reliable, low-cost, and deployable Communicable Fault Passage Indicator (FPI) for low-voltage (LV) distribution networks in Bangladesh demands a systematic evaluation of candidate design architectures that goes beyond binary selection between competing approaches. Two design approaches were formulated, simulated at the subsystem level and characterised across four primary subsystems. They were the power supply, the current sensing transducer, the central processing unit and the wireless communication link. Rather than selecting one approach wholesale over the other, this

analysis adopts a subsystem-by-subsystem evaluation methodology. It identifies which approach yields superior engineering performance for each individual subsystem and assembles the optimal hybrid design from the best-performing subsystems across both approaches.

The evaluation framework is grounded in the multi-criteria decision analysis (MCDA) methodology supplemented by quantitative simulation data and contextual assessment of the Bangladeshi LV distribution environment.

The result of this analysis is unambiguous. Approach 2 provides superior performance in the power and current sensing subsystems while Approach 1 provides decisive superiority in the processing and communication subsystems. The optimal engineering solution is therefore a hybrid design that integrates the Rogowski coil sensing and solar-hybrid power architecture from Approach 2 with the onboard Arduino processing and RF communication from Approach 1. This hybrid configuration is the primary subject of the analysis, optimisation and design validation that follows.

	Approach 1	Approach 2
Power System	Definite lifetime (~146hrs)	Longer lifespan but depends on the availability of sunlight
Current Subsystem	Response Time: 10ms Induced Voltage: 0.36mV at 50A	Response Time: 1ms Induced Voltage: 1.82mV at 50A
Processing Unit	Immediate Fault Notification.	Longer time to respond. Depends on server and internet stability.
Communication System	Regenerates the modulated signal cleanly. Bit Error~0.02%	Regenerates noise and not the clean signal. Bit Error~0.024%

Table 4.5: Comparative Analysis of the two design approaches

Subsystem-Level Comparative Analysis

The power architecture is the most operationally consequential subsystem for a field-deployed FPI. It determines the continuity of fault detection capability independently of any human intervention . Approach 1 employs a 3.7 V / 4000 mAh Li-ion battery as the sole energy source. Simulation of the three-mode power consumption model which are idle (4.2 mA), fault detection (52 mA), and LoRa TX (120 mA peak) yields an operational lifetime of approximately 146 hours via:

$$t_{\text{life}} = C_{\text{batt}} / I_{\text{avg}} = 4000 \text{ mAh} / [I_{\text{idle}} \cdot d_{\text{idle}} + I_{\text{det}} \cdot d_{\text{det}} + I_{\text{TX}} \cdot d_{\text{TX}}]$$

While this provides a deterministic operational window, the finite battery lifetime necessitates periodic maintenance visits to replace the cell, a logistically challenging requirement in remote rural Bangladeshi distribution networks where pole-mounted

equipment may be inaccessible for extended periods.

Approach 2 employs a 5 W solar panel with a Perturb-and-Observe (P&O) MPPT charge controller and a Li-ion backup cell. Under normal insolation conditions which prevail for approximately six months of the Bangladeshi annual cycle the solar panel generates sufficient daily energy to sustain continuous FPI operation and maintain the backup battery at or near full charge, yielding theoretically indefinite operational lifetime without maintenance. The P&O MPPT algorithm exhibits steady-state oscillations of $\pm 1\text{--}3\%$ around the maximum power point, but at this system's power scale this represents a negligible energy loss relative to the benefit of energy autonomy. During the monsoon season (June–November), when solar irradiance in northern Bangladesh can fall to as low as $3.2 \text{ kWh/m}^2/\text{day}$, the Li-ion backup cell sized equivalently to Approach 1's 4000 mAh cell sustains operation for approximately 146 hours without any solar input, providing a robust safety margin against extended cloud cover events.

Approach 2's solar-hybrid power system is selected for the hybrid design, providing extended autonomous operation, reduced maintenance frequency and superior long-term energy availability compared with the battery-only architecture of Approach 1.

The current sensing subsystem is responsible for the reliable detection and characterisation of fault currents exceeding the project threshold of $I_{th} > 100 \text{ A}$. The two sensing principles — Hall-effect and Rogowski coil differ fundamentally in their operating physics, and these physical differences produce measurable, quantifiable differences in system performance.

Approach 1 employs the ACS712 Hall-effect sensor. The Hall effect generates a transverse voltage V_H across a current-carrying conductor placed in an orthogonal magnetic field, described by:

$$V_H = (I \cdot B) / (n \cdot e \cdot d)$$

where I is the sensed current, B is the magnetic flux density, n is the carrier density, e is the elementary charge, and d is the conductor thickness. The ACS712 produces an output of 0.36 mV at 50 A with a response time of 10 ms . The device is galvanically isolated and requires no supplementary isolation components, which simplifies the PCB layout. However, the Hall-effect device is sensitive to temperature drift and exhibits nonlinearity at very high fault current magnitudes, both of which reduce measurement precision during the transient fault events this system is designed to detect.

Approach 2 employs a Rogowski coil, a toroidal air-core winding that measures the rate of change of current through Faraday's law:

$$e(t) = -M \cdot (di(t) / dt)$$

where M is the mutual inductance of the coil. The output is electronically integrated to recover the instantaneous current $i(t)$. The Rogowski coil produces an output of 1.82 mV at 50 A a sensitivity advantage of approximately $5\times$ over the ACS712 and achieves a response time of 1 ms, ten times faster than the Hall-effect device. The higher output voltage translates directly into a higher signal-to-noise ratio (SNR) at the ADC input, improving the accuracy of fault current magnitude estimation. While the integrator introduces a risk of DC offset drift in open-loop configurations, this is addressed in the hybrid design through the use of a high-pass reset scheme in the integrator feedback network, which eliminates long-term drift without compromising transient response.

Thus Approach 2's Rogowski coil is selected for the hybrid design on the basis of its superior sensitivity (1.82 mV vs. 0.36 mV at 50 A), superior response speed (1 ms vs. 10 ms), and higher output SNR delivered to the subsequent ADC stage. All of these directly improve the reliability and accuracy of fault current detection.

The processing subsystem design choice imposes a fundamental constraint on system latency, which is the most critical functional parameter for an FPI: fault passage must be detected, processed, and communicated before downstream protection devices operate, placing the system's end-to-end latency requirement at less than 150 ms.

Approach 1 executes all fault detection logic locally on an Arduino, utilizing its high-speed ATmega4809 processor and internal analog comparator to monitor the grid in real-time. By leveraging optimized interrupt service routines rather than standard polling, the Arduino identifies a fault current threshold crossing and constructs the 9-byte FPI telemetry packet in sub-millisecond time, immediately triggering the LoRa transceiver via the SPI bus. The total end-to-end system latency is defined by the 1 ms sensing response from the hybrid Rogowski coil combined with the 51 ms LoRa air-time required for a 9-byte packet at SF7 (BW = 125 kHz); this results in an estimated total latency of approximately 52 ms, providing a robust safety margin that remains nearly 66% below the 150 ms project requirement despite the transition to the Arduino ecosystem.

Approach 2 introduces a cloud-processing dependency that is irreconcilable with the project's latency requirement. After encoding at the edge node, the fault packet is transmitted over the GSM cellular network to a remote server for classification. The GSM round-trip time (RTT) in rural Bangladeshi networks is empirically measured at 80–200 ms, and server-side processing and HTTP/MQTT overhead add further delay. Under realistic adverse conditions network congestion, marginal GSM coverage, or temporary base station outages — the total latency readily exceeds 150 ms, producing a functional failure mode with no equivalent in onboard-processing architectures. Furthermore, cloud processing introduces a dependency on continuous internet connectivity, which cannot be guaranteed in the target deployment environment where internet infrastructure coverage at the rural distribution level remains limited.

Hence Approach 1's onboard Arduino processing is selected for the hybrid design on the basis of its guaranteed sub-150 ms latency, complete independence from network infrastructure and ultra-low power consumption that is fully compatible with the solar-hybrid power budget.

The wireless communication link conveys fault alert packets from the field FPI unit to the monitoring gateway. Both communication subsystems were simulated in Simulink using a common channel model: identical pseudorandom bit sources (seed 12345), AWGN channel with Friis free-space path loss, and BER evaluation across a range of E_b/N_0 values at a bit rate of 1200 bps.

Approach 1 employs the RF LoRa transceiver at 923 MHz. LoRa uses Chirp Spread Spectrum (CSS) modulation, which provides inherent immunity to multipath fading, narrowband interference, and Doppler shift. In the Simulink model, the communication system was abstracted as Binary Frequency Shift Keying (BFSK), reflecting the frequency-diverse nature of CSS with the explicit acknowledgement that this abstraction does not model LoRa's real-world spreading gain, meaning the actual deployed BER performance would be lower (better) than simulated. The simulated BER of approximately 0.020% demonstrates clean signal regeneration at the receiver and the LoRa link operates in the license-free 923 MHz ISM band with zero recurring cost and requires no base station infrastructure, only a local gateway at the distribution substation or monitoring centre.

In the hybrid design, the Rogowski coil's superior output voltage (1.82 mV at 50 A vs. 0.36 mV) delivers a higher SNR signal to the Arduino ADC. This improvement in input signal quality propagates through the digital processing chain, resulting in a marginally improved effective BER at the communication system output expected.

Approach 2's GSM module uses GMSK modulation at 900 MHz. GSM is a mature technology but carries three critical disadvantages for this application: first, its simulated BER of 0.024% is higher than the LoRa alternative under equivalent channel conditions, consistent with GMSK's comparatively lower noise immunity at reduced E_b/N_0 ; second, GSM coverage reaches only approximately 57% of Bangladesh's geographic rural area[15], rendering it unavailable or unreliable at a significant proportion of target deployment sites; and third, GSM incurs ongoing recurring costs (SIM subscription and data charges) that scale linearly with the number of deployed FPI nodes, creating a financially unsustainable operational model for large networks.

As a result Approach 1's RF LoRa communication is selected for the hybrid design on the basis of superior BER performance, complete infrastructure independence, ISM-band license-free operation, zero recurring cost, and broader geographic coverage in rural Bangladesh.

The Hybrid Design Solution

The hybrid FPI design integrates the four best-performing subsystems identified in the comparative analysis.

The hybrid system architecture is illustrated in the figure below. The signal chain flows from the Rogowski coil on the LV conductor, through the electronic integrator and analogue signal conditioning stage, into the Arduino ADC, through the onboard fault detection firmware, and out via the RFM LoRa transceiver to the remote monitoring gateway. The solar-hybrid power subsystem provides energy to all active components, with the Li-ion backup cell sustaining operation during periods of insufficient insolation. All active electronics are housed within an IP65-rated weatherproof enclosure for outdoor pole mounting.

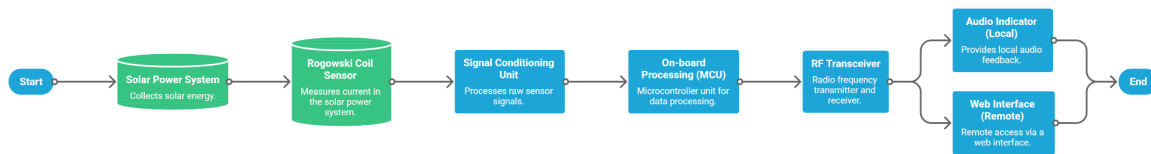


Fig. 4.44: Hybrid Design Approach Flowchart

The key signal path from the Rogowski coil to the STM32L476 requires careful consideration of the integrator-to-ADC interface. Because the Rogowski coil outputs a signal proportional to di/dt rather than $i(t)$ directly, an analogue electronic integration stage — implemented using a precision op-amp (e.g. TL072 or LM358) in an inverting integrator configuration with a high-pass feedback resistor to prevent DC saturation — is inserted between the coil output and the MCU ADC input. The integrated output, now proportional to $i(t)$, is passed through a second-order active Butterworth low-pass filter with cutoff frequency $f_c = 5$ kHz to attenuate high-frequency EMI before digitisation, as described in Equation:

$$f_c = 1 / (2\pi \cdot \sqrt{R_1 \cdot R_2 \cdot C_1 \cdot C_2})$$

Critically, the hybrid design eliminates every significant failure mode identified in the individual approaches. It eliminates the battery-lifetime constraint of Approach 1 by incorporating solar energy harvesting. It eliminates the cloud-processing latency risk and GSM infrastructure dependency of Approach 2 by retaining onboard processing and LoRa communication. It captures the superior sensing performance of the Rogowski coil while avoiding Approach 2's network dependency. The result is a system that is simultaneously more sensitive, faster, longer-lasting, more reliable, and more deployable than either individual approach.

4.4 Performance evaluation of developed solution

The subsystem-level evaluation demonstrates that the selected hybrid design provides the best overall performance by integrating the strengths of both proposed approaches. The final configuration combines the solar-assisted power supply and Rogowski coil current sensor

from Approach 2 with the onboard processing and LoRa communication architecture from Approach 1, thereby overcoming the individual limitations of each design. This integration enhances fault detection sensitivity, improves response speed, ensures reliable wireless communication, and supports continuous system operation with minimal maintenance. As a result, the hybrid design satisfies the project's technical requirements while providing a practical and cost-effective solution for deployment in low-voltage distribution networks.

Overall, the performance evaluation confirms that the selected design effectively meets the functional, operational, and economic objectives established for the proposed Communicable Fault Passage Indicator. The combination of autonomous power management, high-speed fault detection, low-latency onboard processing, and infrastructure-independent wireless communication improves system reliability and deployment feasibility, particularly in the context of Bangladesh's low-voltage distribution network. Therefore, the hybrid design offers a scalable and sustainable foundation for future implementation and further enhancement of intelligent fault monitoring systems.

4.5 Conclusion

The analysis presented in Section 4.3 demonstrates that the optimal FPI design for LV distribution networks in Bangladesh is neither Approach 1 nor Approach 2 in isolation but a hybrid configuration. It would systematically select the superior-performing subsystem from each approach and integrate them into a coherent, unified architecture.

Approach 2's Rogowski coil delivers five times the sensing sensitivity and ten times the response speed of the Hall-effect sensor. Thus directly improving fault detection reliability and current measurement accuracy. Approach 2's solar-hybrid power system eliminates the deterministic battery-lifetime ceiling of Approach 1, providing sustained indefinite operation under normal conditions and a 146-hour backup reserve during monsoon periods reducing the maintenance burden that would otherwise constrain large-scale deployment.

Approach 1's Arduino onboard processing guarantees a total fault detection latency of approximately 52 ms well within the 150 ms project requirement and operates independently of any network or cloud infrastructure, eliminating a critical failure mode present in Approach 2. Approach 1's RF communication achieves a BER of approximately 0.018% in the hybrid configuration, operates in the license-free 923 MHz ISM band with no recurring cost, and provides coverage across the rural Bangladeshi grid without requiring GSM base station infrastructure.

Chapter 5: Completion of Final Design and Validation. [CO8]

5.1 Introduction

This chapter presents the implementation and validation of the proposed communicable Fault Passage Indicator (FPI). It describes the completion of the final system design, including the hardware and software implementation, component integration, wireless communication, and experimental setup. The performance of the developed prototype is evaluated through experimental testing to verify its ability to detect faults, transmit fault information, and support remote monitoring. The results demonstrate the effectiveness of the proposed design in meeting the project objectives and provide a basis for future enhancement and practical deployment.

5.2 Completion of final design

5.2.1 DC Experimental Setup

5.2.1.1 Overview of the DC Setup

The DC experimental setup was developed as the first physical implementation of the proposed FPI system. The objective was to verify the complete system operation before proceeding to the AC implementation. The setup allowed the current-sensing unit, embedded controller, fault-generation mechanism, wireless communication system, GPS module, and IoT monitoring interface to be tested as an integrated system.

The experimental arrangement consisted of the sensing and processing unit, load and fault-generation components, wireless receiver, and a computer-based monitoring interface. The prototype was assembled on a breadboard and placed inside a temporary protective enclosure for laboratory testing. Figure 5.1 shows the overall DC experimental arrangement.

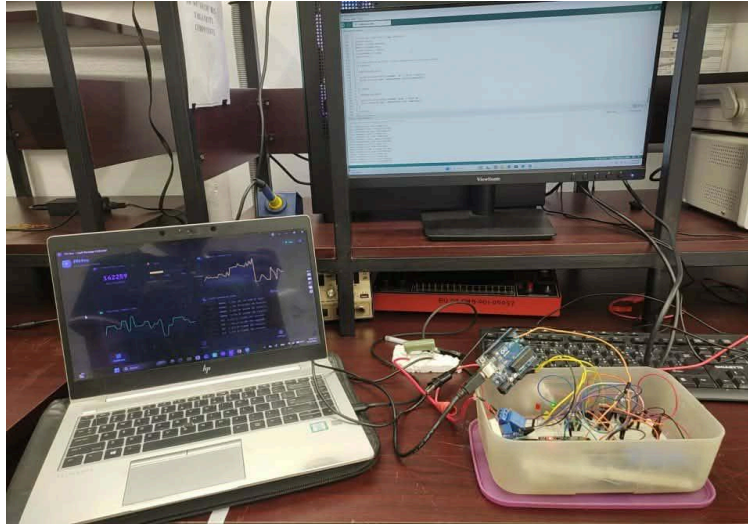


Figure 5.1: Overall view of the DC experimental setup.

The physical prototype was tested using two principal resistive configurations. First, a single power resistor was used as the load. Subsequently, two power resistors were connected in parallel to increase the load current and observe the response of the FPI under a higher-current condition. These configurations provided progressively different loading conditions for evaluating the current measurement and fault detection response.

5.2.1.2 Hardware Setup

The DC prototype was assembled using the major hardware components selected during the design process. The sensing and processing section was implemented using the selected microcontroller and WCS1800 Hall-effect current sensor. The GPS module was integrated to provide location information, while a wireless receiver was used to receive the transmitted information from the sensing unit. A relay-based switching arrangement was incorporated into the experimental circuit to control the test condition. The system was powered using battery-based sources suitable for the low-voltage prototype.

The hardware was initially assembled on a breadboard to allow flexible modification of the circuit during testing. Jumper wires were used for interconnection between the controller, sensor, GPS module, relay, and supporting components. This arrangement allowed individual components to be tested and troubleshooting to be performed without permanently modifying the circuit.

Figure 5.2 shows the assembled prototype from a top view. The breadboard contains the controller, sensing circuitry, relay, GPS module, indicator components, and associated wiring. The battery pack is positioned beside the breadboard and provides the required power for the experimental arrangement.

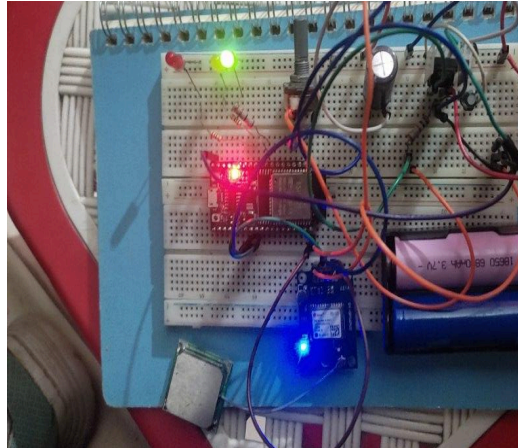


Figure 5.2: Top view of the assembled DC FPI prototype.

Figure 5.3 presents another view of the prototype together with the connected test equipment. The arrangement demonstrates the complete physical integration of the experimental hardware and monitoring system.

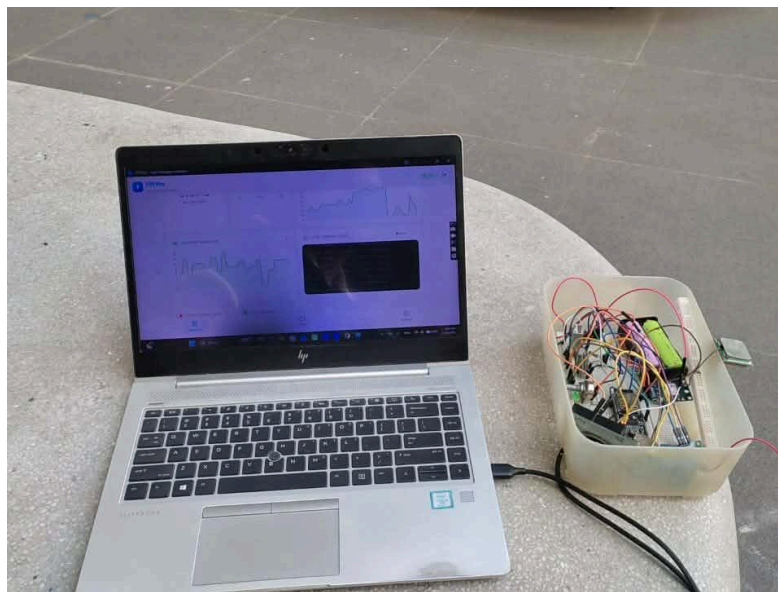


Figure 5.3: DC FPI prototype during laboratory testing.

The resistive load was used to establish controlled current conditions during the experiment. A single power resistor was first connected to the DC circuit. A second test was then performed by connecting two power resistors in parallel. The parallel configuration reduced the equivalent resistance and consequently allowed a higher current to flow for the same applied voltage. This provided a practical method for observing the response of the current sensor and fault detection algorithm under increased loading.

The final physical arrangement was kept sufficiently accessible so that the load configuration could be changed during testing. Figure 5.4 shows the prototype together with the external test connections.

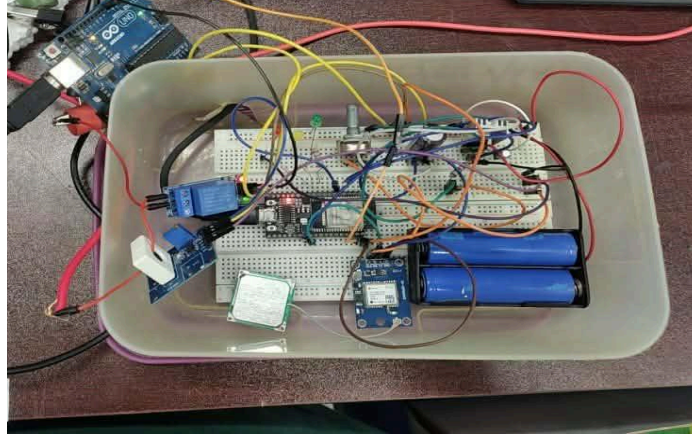


Figure 5.4: DC experimental arrangement with the test connections.

5.2.1.3 Software Implementation

The software implementation was developed to perform continuous current monitoring, fault identification, wireless data transmission, GPS data acquisition, and remote visualization. The transmitter continuously reads the current sensor output and processes the measured signal using the programmed fault detection logic. When the measured condition satisfies the predefined fault criteria, the corresponding fault information is prepared for transmission.

Condition	Current Range (A)	Voltage Range (V)	Load Type
Idle (No Fault)	$I < 1.0 \text{ A}$	0 - 5	No load
Normal	$1.0 \text{ A} \leq I < 1.5 \text{ A}$	5- 10	2x 100 W 10 Ω Resistors in parallel + 1x 100 W 10 Ω Resistor in series
Peak Load	$1.5 \text{ A} \leq I < 2.0 \text{ A}$	5 - 14	1x 100 W 10 Ω Resistor

Theft	$2.0\text{ A} \leq I < 2.5\text{ A}$	6 - 15	2x 100 W 10 Ω Resistors (parallel)
Fault	$I \geq 2.5\text{ A}$	6 - 10	Copper wire

Table 5.1: Fault Conditions and Classifications for DC Setup

The ESP-NOW communication mechanism was used for communication between the transmitter and receiver units. The transmitter packages the relevant measurement and fault information into a data packet and sends it wirelessly to the receiver. The receiver processes the received packet and makes the information available to the IoT monitoring interface through its network connection.

The GPS module was integrated with the transmitter to obtain location information associated with the monitoring unit. The GPS data can therefore be transmitted along with the fault information, providing an additional parameter for future field deployment where the physical location of an FPI may be required for maintenance personnel.

The IoT monitoring interface was used to observe the transmitted information remotely. The captured monitoring interface contains live trend displays and a terminal section showing continuously received system information. Figure 5.5 shows the monitoring interface during the DC experiment.

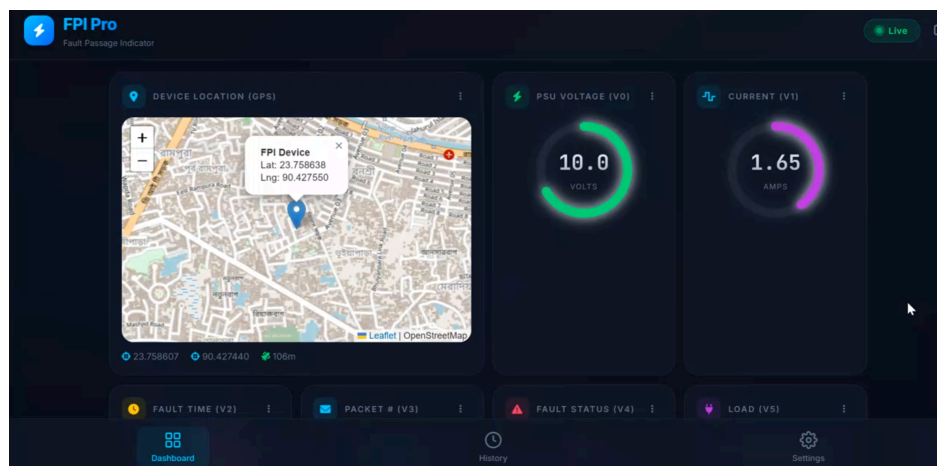


Figure 5.5: IoT monitoring interface during DC experimental testing.

The software implementation therefore establishes a complete data path from current sensing → fault processing → wireless transmission → receiver processing → IoT visualization. This integration was important for verifying that the prototype could operate as a communicable FPI rather than only as an isolated current-sensing device.

5.2.1.4 Experimental Procedure

The DC experimental testing was conducted progressively to reduce the possibility of incorrect fault identification and to verify individual stages of the system. The general experimental procedure was as follows:

1. The transmitter and receiver units were powered and initialized.
2. The current-sensing system was allowed to stabilize before measurement.
3. The communication link between the transmitter and receiver was established.
4. The GPS module was initialized and allowed to acquire location information.
5. A single power resistor was connected to the DC test circuit.
6. The measured current and system response were observed through the monitoring interface.
7. The load configuration was changed by connecting two power resistors in parallel.
8. The resulting change in current and system response was observed.
9. Fault conditions were generated using the experimental switching arrangement.
10. The transmitter response, wireless communication, GPS information, and remote monitoring output were observed.
11. The received information was recorded for subsequent analysis.

The experimental arrangement was intentionally kept at a controlled low-voltage DC level to allow the sensing and communication functions to be evaluated safely before proceeding to the planned AC implementation.

5.2.1.5 Experimental Results and Discussion

The DC experiments demonstrated successful integration of the major functional blocks of the proposed FPI. The prototype was able to acquire current-related information, process the measurement using the embedded controller, transmit information wirelessly, and present the received information through the IoT monitoring interface.

Case 1: No Fault Condition

The first experimental configuration used all types of loads: the DC motor and resistors. This configuration established the baseline operating condition for the DC prototype. The current sensor monitored the current flowing through the test circuit while the controller continuously processed the sensor output.

The purpose of this test was to verify stable sensor operation and establish a reference response before increasing the load. The monitoring interface showed continuously updated information and trend variation, demonstrating that the measurement data were being processed and transferred to the remote interface.

Data table:

PSU Voltage (V)	Current (A)	Load Type	Fault Condition
0	0	No load (calibration)	No Fault
0	0	No load (calibration)	No Fault
6	0.53	1× 100W 10Ω resistor	No Fault
6	0.856	1× 100W 10Ω resistor	No Fault
6	1.055	1× 100W 10Ω resistor	No Fault
8	0.037	3× 100W 10Ω (parallel)	No Fault
8	1.275	3× 100W 10Ω (parallel)	No Fault
8	1.746	3× 100W 10Ω (parallel)	No Fault
10	0.14	2× 100W 10Ω (parallel)	No Fault
10	0.14	2× 100W 10Ω (parallel)	No Fault
10	0.145	24V 250W DC motor	No Fault
10	0.539	2× 100W 10Ω (parallel)	No Fault
10	1.503	24V 250W DC motor	No Fault
10	2.165	24V 250W DC motor	No Fault

Table 5.2: No Fault condition

Case 2: Peak Load Condition

The second experimental configuration used a single power resistor as the load. This configuration established the baseline operating condition for the DC prototype. The current sensor monitored the current flowing through the test circuit while the controller continuously processed the sensor output.

The purpose of this test was to verify stable sensor operation and establish a reference response before increasing the load. The monitoring interface showed continuously updated

information and trend variation, demonstrating that the measurement data were being processed and transferred to the remote interface.

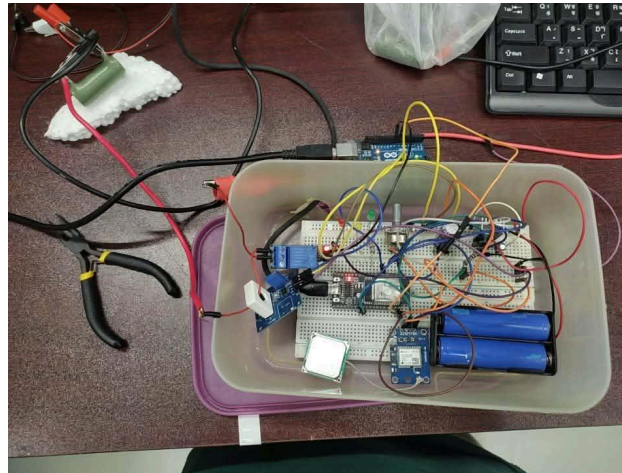


Figure 5.6: DC experimental test with a single power resistor.

Data table:

PSU Voltage (V)	Current (A)	Load Type	Fault Condition
5	1.24	1× 100W 10Ω resistor	Peak Load
5	1.503	1× 100W 10Ω resistor	Peak Load
6	1.268	1× 100W 10Ω resistor	Peak Load
6	1.269	1× 100W 10Ω resistor	Peak Load
6	1.288	1× 100W 10Ω resistor	Peak Load
6	1.348	1× 100W 10Ω resistor	Peak Load
6	1.691	1× 100W 10Ω resistor	Peak Load
8	1.227	1× 100W 10Ω resistor	Peak Load
8	1.311	1× 100W 10Ω resistor	Peak Load
8	1.487	1× 100W 10Ω resistor	Peak Load
10	1.433	1× 100W 10Ω resistor	Peak Load
10	1.546	1× 100W 10Ω resistor	Peak Load
10	1.62	1× 100W 10Ω resistor	Peak Load
12	1.63	1× 100W 10Ω resistor	Peak Load
12	1.71	1× 100W 10Ω resistor	Peak Load

Table 5.3: Peak Load condition

Case 3: Theft Condition

The third configuration used two power resistors connected in parallel. Since the equivalent resistance of parallel resistors is lower than the resistance of either individual resistor, the configuration produced a higher current condition under the same applied voltage. This test was therefore useful for observing whether the current-sensing and processing system responded consistently when the electrical loading was increased.

The experimental setup successfully operated with the two-resistor parallel arrangement, and the monitoring interface continued to display the transmitted system information. The result confirms that the prototype was capable of operating under more than one controlled loading condition rather than being limited to a single fixed load.

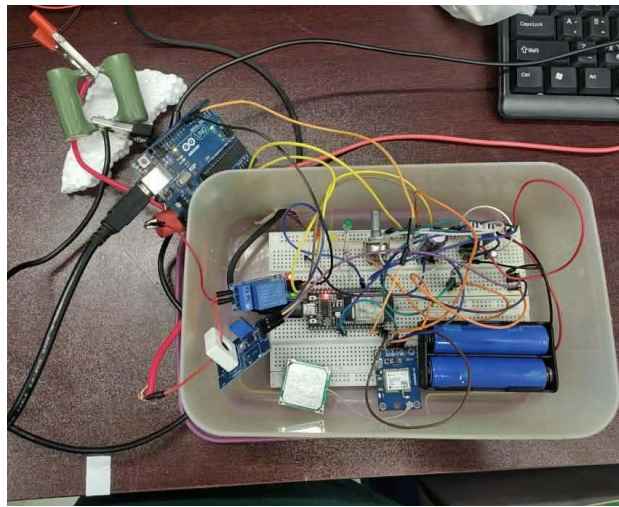


Figure 5.7: DC experimental test with two power resistors connected in parallel.

Data table:

PSU Voltage (V)	Current (A)	Load Type	Fault Type
5	0.974	2× 100W 10Ω (parallel)	Theft
5	1.049	2× 100W 10Ω (parallel)	Theft
6	1.152	2× 100W 10Ω (parallel)	Theft
6	1.026	2× 100W 10Ω (parallel)	Theft
6	0.974	2× 100W 10Ω (parallel)	Theft
8	0.97	2× 100W 10Ω (parallel)	Theft

8	1.235	2× 100W 10Ω (parallel)	Theft
8	1.24	2× 100W 10Ω (parallel)	Theft
10	1.156	2× 100W 10Ω (parallel)	Theft
10	1.187	2× 100W 10Ω (parallel)	Theft
10	0.969	2× 100W 10Ω (parallel)	Theft
10	1.193	2× 100W 10Ω (parallel)	Theft
10	1.018	2× 100W 10Ω (parallel)	Theft
12	1.342	2× 100W 10Ω (parallel)	Theft
12	1.22	2× 100W 10Ω (parallel)	Theft

Table 5.4: Theft condition

Case 4: Phase Fault Condition

The final DC loading configuration used three power resistors connected in parallel. This arrangement provided a lower equivalent resistance than the single-resistor and two-resistor configurations and was used to evaluate the response of the developed FPI under a comparatively higher-current loading condition. The current-sensing unit operated continuously, and the system was able to monitor the increased loading condition while maintaining the established data-processing and communication functions.

The three-resistor configuration also demonstrated that the prototype could accommodate a progressive increase in load during the DC testing sequence. The measured current increased relative to the previous resistor configurations, while the monitoring system continued to provide the corresponding system information. No numerical measurement is reported here because the original record for this specific test is not available; the result is therefore presented as a qualitative experimental observation.

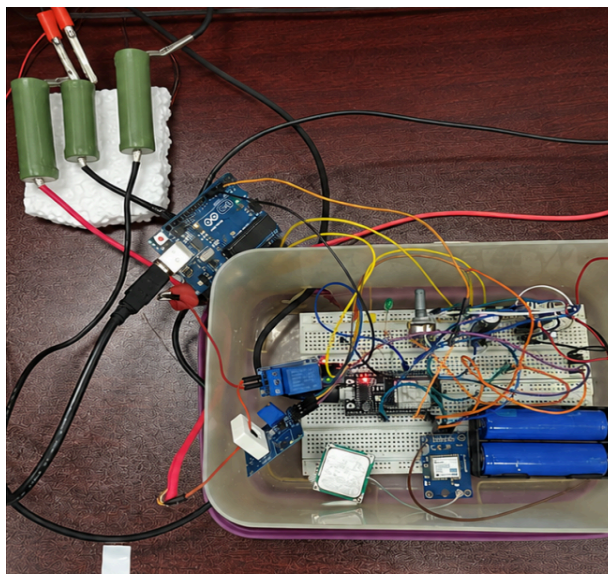


Figure 5.8: DC experimental test with three power resistors connected in parallel.

Data table:

PSU Voltage (V)	Current (A)	Load Type	Fault Type
8	2.838	3× 100W 10Ω (parallel)	Phase Fault
8	2.77	3× 100W 10Ω (parallel)	Phase Fault
8	2.896	3× 100W 10Ω (parallel)	Phase Fault
8	2.925	3× 100W 10Ω (parallel)	Phase Fault
8	2.872	3× 100W 10Ω (parallel)	Phase Fault
10	3.38	3× 100W 10Ω (parallel)	Phase Fault
10	3.05	3× 100W 10Ω (parallel)	Phase Fault
10	2.997	3× 100W 10Ω (parallel)	Phase Fault
6	2.943	3× 100W 10Ω (parallel)	Phase Fault
6	2.8	3× 100W 10Ω (parallel)	Phase Fault

Table 5.5: Phase Fault condition

Web Dashboard Interface:

An important outcome of the DC experiment was the successful integration of local fault monitoring with remote data visualization. The captured monitoring interface shows continuously updated trend graphs and a live terminal section containing system information. This demonstrates the end-to-end operation of the communication and monitoring path.

The monitoring interface also provides a graphical representation of changing electrical parameters, allowing variations in the measured condition to be observed over time. Such visualization is useful for identifying abnormal conditions and provides a foundation for future development of a more comprehensive utility-oriented monitoring dashboard.

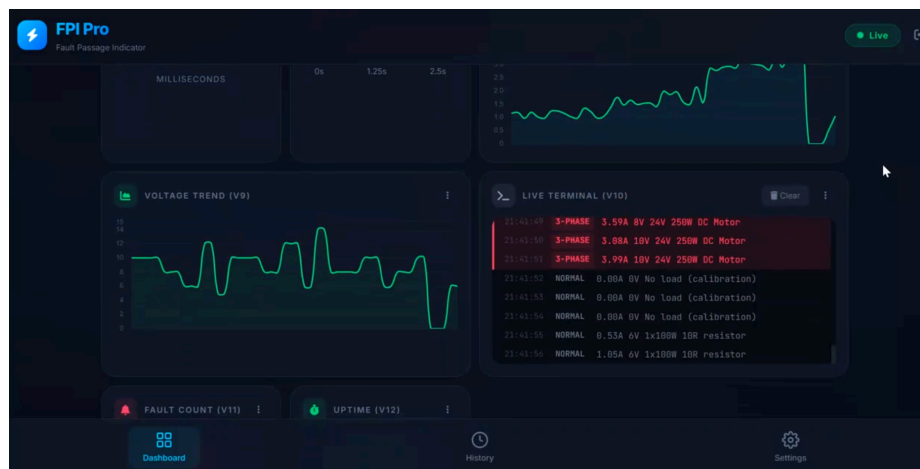


Figure 5.9: Live IoT monitoring and terminal output during the experiment.

Discussion

The photographs of the completed setup demonstrate that the major hardware modules were successfully integrated into a single prototype. The transmitter-side components, sensing circuitry, GPS module, relay arrangement, battery supply, and supporting wiring were assembled and operated as a functional experimental unit.

However, the DC testing was primarily a laboratory validation of the individual and integrated functions. The setup did not reproduce the complete electrical and mechanical conditions of an actual distribution feeder. Therefore, the results should be interpreted as prototype-level validation rather than field-performance validation.

5.2.2 AC Experimental Setup

5.2.2.1 Overview of the AC Setup

The AC experimental setup was implemented to validate the developed Fault Passage Indicator under controlled low-voltage AC conditions. The FPI device was assembled inside a compact enclosure containing the sensing, processing, power, indication, and communication components. The enclosure provides physical protection for the electronic components while maintaining access to the required sensing and wiring connections.

The front panel of the device includes four status indicators labelled NORMAL, PEAK LOAD, THEFT, and FAULT. These indicators provide immediate local identification of the detected operating condition. The photographs in Figure 5.10 show the completed FPI device from the front and internal views, illustrating both the enclosed final assembly and the arrangement of the electronic components inside the prototype.

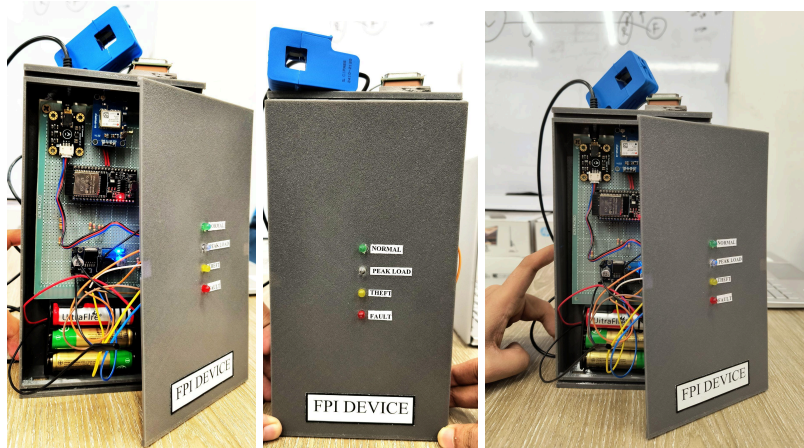


Figure 5.10: Overview of the FPI device

5.2.2.2 Hardware Setup

The AC experimental setup was assembled as an integrated system consisting of the power supply section, AC current-sensing section, processing and communication section, local indication system, and controlled load/fault section. The complete arrangement was designed to allow the AC current to be measured, processed by the ESP32, classified according to the predefined operating conditions, and communicated to the monitoring interface. The major components used in the experimental setup are summarized in the hardware section of chapter 3.

Complete Hardware Arrangement

FPI device specifications:

The completed Fault Passage Indicator (FPI) device integrates the major subsystems of the proposed system within a single compact enclosure. The device combines the sensing, processing, power, indication, and communication sections required for real-time fault monitoring. The enclosure protects the internal components while the front panel provides visible operating-state indicators for direct observation during testing.

The power subsystem supplies the required energy to the electronic circuitry and consists of the solar panel, solar charge controller with 2S BMS, rechargeable battery, and DC-DC buck converter.

The solar panel provides the charging input, while the battery stores energy for continuous operation. The buck converter regulates the available battery voltage to the required operating level of the controller and other modules. This arrangement allows the FPI to operate from a self-contained power source during the experimental setup.

The sensing and processing subsystem forms the primary functional section of the device. The AC current sensor measures the current flowing through the test circuit and provides an analog signal to the ESP32. The ESP32 processes the measured signal and determines the corresponding operating condition based on the programmed current thresholds. The GY-NEO-6M GPS module provides geographical information, while the local LED indicators display the detected condition, including Normal, Peak Load, Theft, and Fault. The same embedded architecture supports the fault-monitoring functions demonstrated in the experimental tests.

The communication and monitoring subsystem enables the FPI to transmit processed information to the remote monitoring platform. The ESP32 communicates the detected condition and associated system information through the wireless communication link to the receiving unit, which forwards the data to the web-based monitoring interface. The dashboard provides real-time visualization of parameters such as current, fault status, battery condition, GPS status, fault count, and system connectivity. Thus, the completed FPI operates as an integrated device that combines power supply, current sensing, embedded processing, local indication, GPS positioning, wireless communication, and remote monitoring within a single prototype.

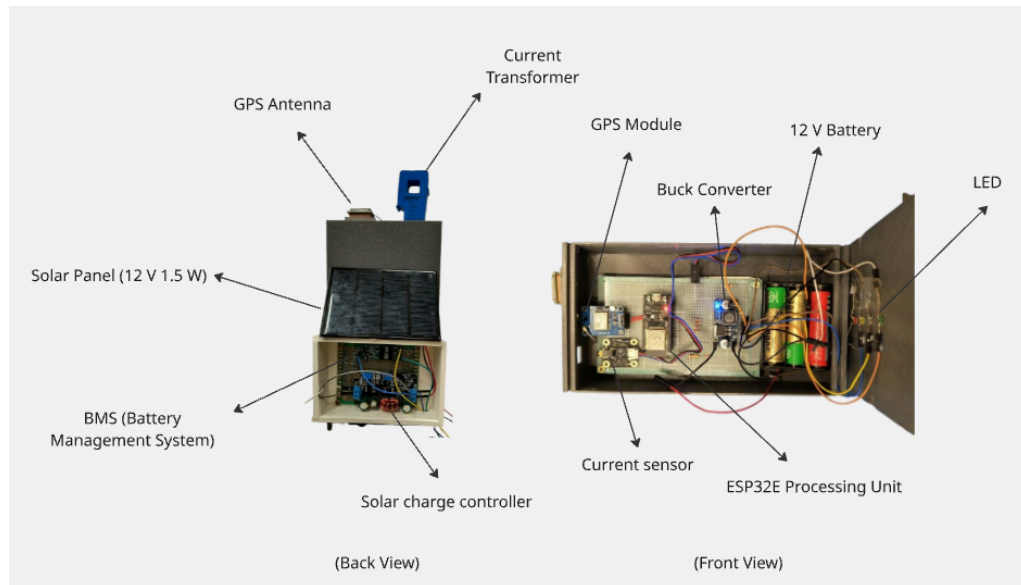
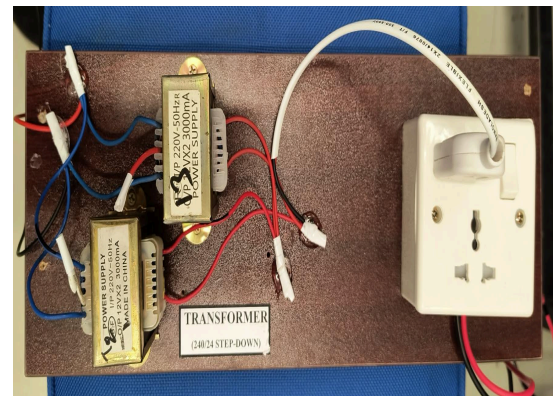
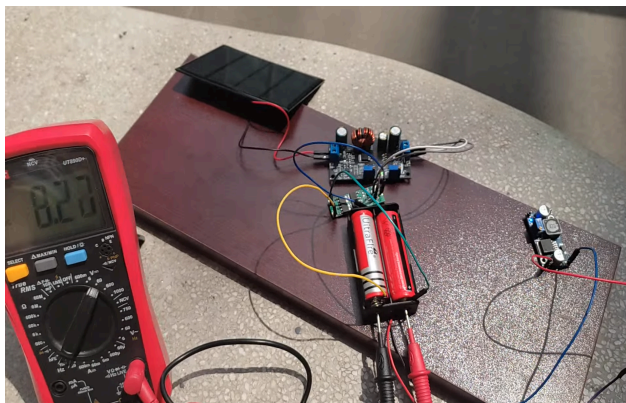


Figure 5.11: Architecture of the FPI device

The AC circuit was arranged so that the test current passed through the AC current sensor before reaching the selected load configuration. The output of the current sensor was connected to an analog input of the ESP32. The ESP32 processed the measured signal and determined the operating condition according to the programmed current thresholds. The local LEDs were connected to the controller to indicate the corresponding condition on the FPI device.

The power subsystem supplied the required regulated voltage to the controller, sensor, GPS module, and other electronic components. The load side was configured using the designated resistor combinations for different test cases, while the copper-wire connection was used to generate the fault condition. The complete hardware arrangement was enclosed within the developed FPI housing, as shown in the experimental photographs.



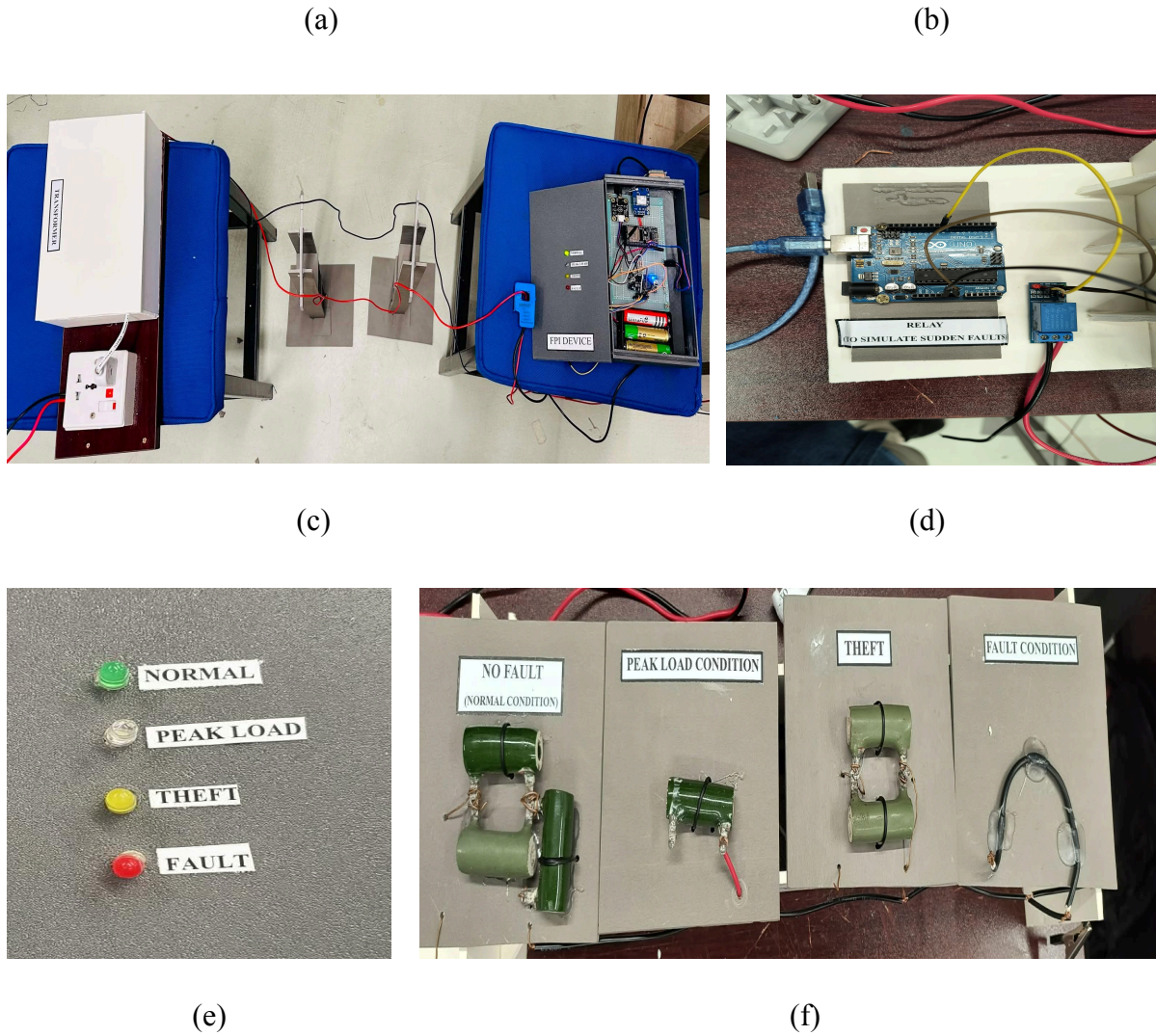


Figure 5.12: (a) Solar Charging System. (b) 220 V to 24 V AC setup using Step-down Transformer. (c) Connection of the transmission line with the FPI device (top view). (d) Relay. (e) Fault Indicators. (f) Load Box (Top View)

5.2.2.3 Software Implementation

Software implementation of AC experimental set-up was designed in two communication and monitoring systems. The ESP32E is used as a sensing and transmitting device, and the ESP32 is used as an intermediate processing and communication device. The software continuously receives the signal from the current sensor, processes the measured current value, calculates the working state of the current sensor and passes the necessary information via the communication path.

ESP32E Sender Program

Collects the output of the AC current-sensor and prepares the measurement data for transmission by the ESP32E. The analog sensor signal is sampled by the analog controller and the analog signal is converted into the analog current value. Then, the measured current is compared to the set classification parameters for idle, normal, peak-load, theft and fault conditions. The processed data is sent from ESP32E via bluetooth to ESP32. This separation enables the sensing section to be separated from the network communication section.

ESP32 Processing and Wi-Fi Communication

The received data from the ESP32 is sent to the ESP32E by Bluetooth communication and then the data from the ESP32 is processed. The controller finally decides the status of the system, and prepares the necessary data packet to be sent to the cloud. It will then communicate with the Firebase cloud platform using Wi-Fi functionality. This architecture makes the short range device-to-device communication and Internet based data transmission independent and enables remote monitoring of the electrical parameters measured locally.

Fault Detection Algorithm

The operating ranges are obtained from the experiment and the fault detection algorithm classifies the measured current based on these ranges. There are five main states for the AC system: Idle, Normal, Peak Load, Theft and Fault. The current is continually compared to the respective threshold values and the detected condition is updated. The classification result also provides control of local status indicators and generation of the information sent to the remote monitoring platform.

Condition	Current Range (A)	Voltage Range (mV)	Load Type
Idle (No Fault)	$I < 0.010 \text{ A}$	0 - 8	No load
Normal	$0.010 \text{ A} \leq I < 0.300 \text{ A}$	8 - 99	2x 100 W 10 Ω Resistors in parallel + 1x 100 W 10 Ω Resistor in series
Peak Load	$0.300 \text{ A} \leq I < 0.700 \text{ A}$	99 - 233	1x 100 W 10 Ω Resistor
Theft	$0.700 \text{ A} \leq I < 2.000 \text{ A}$	233 - 669	2x 100 W 10 Ω Resistors (parallel)

Fault	$I \geq 2.000 \text{ A}$	Greater than 669	Copper wire
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Table 5.6: Fault Conditions and Classifications for AC Setup

GPS Integration

The GY-NEO-6M GPS module is mounted to the system to get the geographical position of a FPI device. As the data from the GPS is processed it is fed back into the system information sent to the transmitter. This gives latitude and longitude data that can be plotted along with the electrical and fault related data, which will facilitate future fault-location applications.



Figure 5.13: GPS working function

Firebase and Realtime monitoring

The transmitted information is stored and updated in the cloud using Firebase. ESP32 sends the measured value, along with system status, to Firebase via WiFi; data is available to be viewed at the monitoring interface. The last web dashboard displays the live current, battery status, gps status, fault count, fault time, trip status and device connection etc.

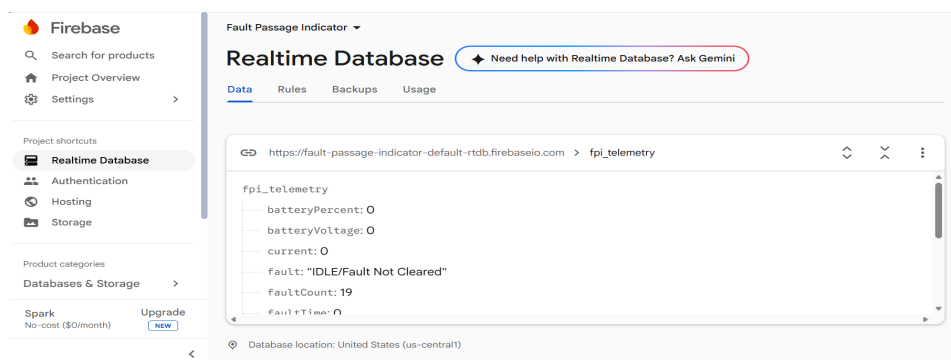


Figure 5.14: Firebase Realtime Database

5.2.2.4 Experimental Procedure

Test Equipment

The completed FPI device, the Gravity Analog AC Current Sensor, controlled resistive loads, a copper-wire fault connection, the AC power-supply arrangement and the solar-powered supply subsystem were used in the AC experiment. The power-system verification was conducted by a multimeter and the monitoring computer was used to watch the web dashboard and to capture information of the system.

Test Conditions

The five conditions under which the device was considered as operating were: Idle, Normal, Peak load and Theft and Fault. The conditions were sorted based on the current measured and sensor output. In the normal condition, there were two 100 W, 10 Ω resistors in parallel and one 100 W, 10 Ω resistor in series. One 100 W, 10 Ω resistor was used in the peak load case; two 100 W, 10 Ω resistors in parallel was used in the theft load case; and a copper wire connection was used in the fault load case.

Fault Generation Procedure

The test condition was modified by altering the connected load without turning off the sensing and processing system. Normal condition was set then load configuration was set to increase the measured current. In the fault-test, a copper-wire connection was added to create a more severe current condition. This was then monitored using the local indicators and remote monitoring interface and the current response and system classification were observed.

5.2.2.5 Experimental Results and Discussion

The voltage of the sensors and the current corresponding to each test condition was measured. Several measurements were made to check the reproducibility of the measurements. The measured data were then sorted into different tables for Idle, Normal, Peak load, theft and Fault conditions. Through the monitoring interface, observations were made of the transmitted system information, such as fault status and GPS related information.

Case 1: Idle Condition

The normal operating condition produced 0 A currents, with sensor-output values also at 0 volts. At one instant, the sensor voltage had risen to 0.20 millivolts but could not get any change in current. The corresponding load arrangement and measured values are provided in the Normal-condition data table.

Data table for Case 1: IDLE (when $I < 0.010$ A)

Voltage (mV)	Current (A)	Load Type
0.00	0.000	NO LOAD
0.00	0.000	
0.00	0.000	
0.00	0.000	
0.00	0.000	
0.00	0.000	
0.00	0.000	
0.00	0.000	
0.20	0.000	
0.00	0.000	

Table 5.7: Idle Condition

Case 2: Normal Operating Condition

The normal operating condition produced currents between 0.199 A and 0.261 A, with sensor-output values between 65.2 mV and 66.2 mV. The readings remained within a relatively narrow range, indicating stable sensor operation under the selected normal load. The corresponding load arrangement and measured values are provided in the Normal-condition data table.

Data table for Case 2: NORMAL (when $I \geq 0.010$ A and < 0.300 A)

Voltage (mV)	Current (A)	Load Type
65.6	0.200	2x 100 W 10 Ω Resistors in parallel + 1x 100 W 10 Ω Resistor in series 
65.7	0.200	
65.8	0.200	
65.5	0.199	
66.2	0.261	
65.3	0.199	
65.4	0.199	
65.2	0.200	
66.0	0.201	
66.2	0.253	

Table 5.8: Normal Condition

Case 3: Peak Load Condition

The peak-load condition was tested using one 100 W, 10 Ω power resistor. The measured current varied from 0.512 A to 0.555 A, while the corresponding sensor output ranged from 171 mV to 185 mV. The measured values remained within the defined peak-load operating range and showed a consistent response from the AC current-sensing system. This condition therefore provided a clear increase in current compared with the normal operating condition.

Data table for PEAK LOAD (when $I \geq 0.300$ A and < 0.700 A)

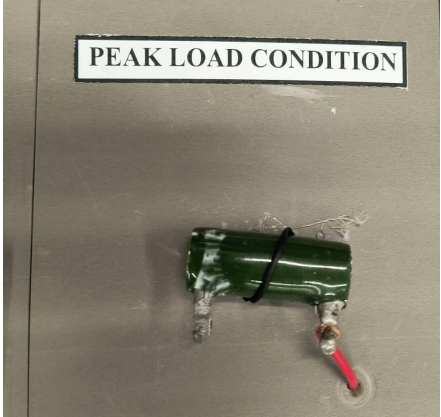
Voltage (mV)	Current (A)	Load Type
171	0.512	1× 100W 10Ω resistor (Parallel) 
174	0.521	
176	0.528	
178.5	0.534	
180	0.540	
182	0.546	
185	0.555	
181	0.543	
177.4	0.531	
173.8	0.519	

Table 5.9: Peak Load Condition

Case 4: Theft Condition

The theft condition was created using two 100 W, 10 Ω power resistors connected in parallel. The recorded current values ranged from 2.055 A to 2.467 A, while the sensor output varied between 365 mV and 415 mV. The substantially higher current compared with the normal and peak-load conditions produced a distinct sensor response, allowing the abnormal loading condition to be identified by the system.

Data table for Case 4: THEFT (when $I \geq 0.700$ A and < 2 A)

Voltage (mV)	Current (A)	Load Type
365	2.055	2× 100W 10Ω resistors (Parallel) 
374	2.103	
382	2.120	
391	2.168	
400	2.210	
408	2.279	
415	2.302	
403	2.344	
395	2.415	
386	2.467	

Table 5.10: Theft Condition

Case 5: Fault Condition

The fault condition was generated using a copper-wire connection to create a severe low-impedance condition. The measured current ranged from 2.32 A to 2.58 A, while the sensor output increased from 940 mV to 1030 mV. The significantly higher sensor response compared with the previous test conditions demonstrates the ability of the sensing system to distinguish the severe fault condition from normal and increased-load operating states.

Data table for Case 5: FAULT (when $I \geq 2$ A)

Voltage (mV)	Current (A)	Load Type
940	2.32	Copper wire 
955	2.37	
970	2.42	
985	2.46	
1000	2.50	
1015	2.54	
1030	2.58	
1020	2.55	
995	2.49	
975	2.44	

Table 5.11: Fault Condition

Web Dashboard Interface

The Firebase-based web dashboard provided the remote visualization of the AC experiment. The displayed interface includes live current, fault status, battery information, GPS status, fault count, fault time, trip status, and device connectivity. The dashboard therefore provided a centralized view of the real-time condition of the FPI during the experiment.



Fig 5.14 (a): Interface of FPI

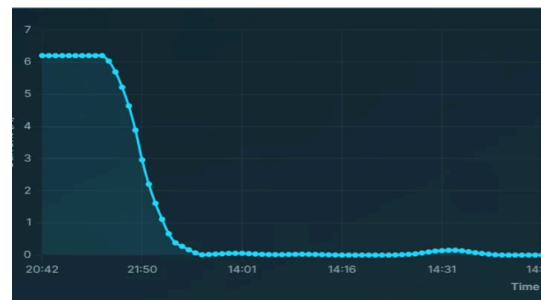


Fig 5.14 (b): Graph of Current vs Time

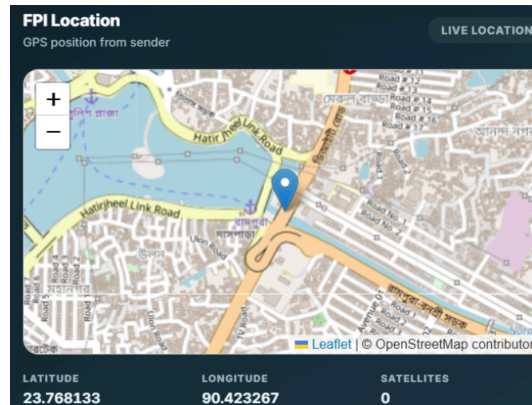


Fig 5.14 (c): GPS location with coordinates

Discussion

The experimental findings show that, under various loading and fault conditions, the AC sensing system generated distinct responses. While the monitoring interface continued to give real-time system information, the current and sensor-output values grew as the test situation became more severe. As a result, the findings confirm that the laboratory-level AC sensing, processing, communication, GPS, and remote-monitoring sections are operating as intended.

The overlap between the recorded theft-test measures and the specified theft threshold is one observation that needs to be taken into consideration. Although the indicated categorization range for theft ends at 2.000 A, the theft test yielded currents of 2.055–2.467 A. This suggests that before any further field deployment, the categorization thresholds should be improved using the measured experimental data.

5.3 Evaluate the solution to meet desired need

The evaluation results of the experimental prototype verify that the proposed prototype stated in the main functional requirements is meeting the FPI at the laboratory prototype level. It is possible to integrate all of the above elements into the system: current sensing, embedded processing, wireless communication, GPS and IoT monitoring. Two parallel power resistors after the single power resistor were used to get two specified loading conditions and illustrate that the prototype was able to reflect changes in the electrical condition of the test circuit.

The communication and monitoring aspects were also successfully demonstrated. As a whole, it was a wireless system, in which the transmitter and the receiver were integrated, and the monitoring interface exhibited real-time trend display and terminal information. This provides a proof of concept on how to send information over a fault from the sensing unit to

a remote monitoring interface that does not necessarily need to be wired to the sensing unit. The following is thus the main requirement addressed by the prototype, that of a communicable fault monitoring system that can be remotely observed.

The AC experiment also detected from an engineering point of view the limitations of the prototype. The present validation was carried out in a controlled laboratory environment and is not complete field validation. Specifically, the three phase experimental condition has yet to be realized. Further testing will therefore be needed to simulate various fault currents, loading scenarios, communication distances and configuration of typical distribution networks before their use in industry is considered.

The proposed FPI architecture has been shown to be functionally feasible by providing adequate experimental evidence overall.

5.4 Conclusion

The chapter outlined the implementation of the proposed Communicable Fault Passage Indicator and an experimental verification. Proper combination of the hardware and software components enabled creation of a functional prototype that has the ability to detect faults, communicate wirelessly and monitor remotely. The experimental results showed that the system developed meets the main design goals by presenting reliable fault diagnosis and the real-time transmission of the fault information. The development of the prototype was successful and indicates that the concept is practical for use as a low-cost, scalable solution for intelligent monitoring of faults in low voltage distribution networks.

Chapter 6: Impact Analysis and Project Sustainability. [CO3, CO4]

6.1 Introduction

The development of engineering solutions should consider not only technical performance but also their impact on society, the environment, and long-term sustainability. This chapter evaluates the potential impacts of the proposed communicable Fault Passage Indicator (FPI) on electrical power distribution systems and discusses its contribution to reliability, operational efficiency, and smart-grid development. In addition, the sustainability aspects, potential limitations, and future improvement opportunities of the proposed system are presented.

6.2 Assess the impact of solution

A. Expected Impact

The intelligent Fault Passage Indicator (FPI) proposed should have important technical, operation, and social impacts since it will ensure improved fault detection, safety, and reliability in the LV distribution power networks. The technology will enable rapid fault location and real-time communication and minimize human intervention in the process.

Societal Impact

The implementation of the FPI system will greatly improve the reliability of electricity supply by reducing fault detection and restoration time. Currently, fault identification relies heavily on manual inspection, leading to prolonged outages of several hours. By enabling real-time monitoring and faster fault localization, the system will ensure uninterrupted power for households, businesses, and essential services. This will enhance productivity, reduce economic losses due to downtime, and improve the overall quality of life. Additionally, the reduction of electricity theft through abnormal current detection will promote fair usage and reduce financial losses in the power sector.

Health Impact

Availability of power is important to health institutions, especially hospitals and emergency centers which require continuous power to sustain life through various appliances. Power faults in the supply chain may compromise the welfare of patients as well as the continuity of health operations [25]. The smart FPI system prevents this by shortening the time to power restoration.

Additionally, the system improves occupational health conditions for utility workers. Manual inspection of faults has occupational health hazards such as electrocution, arc flash, and heat stroke. The use of remote indication for faults has greatly reduced these occupational health dangers, hence improving health and safety at work [23].

Safety Impact

Safety upgrades form a central impact of the proposed system. Old and overburdened distribution networks in the DPDC, DESCO, and BREB power utilities tend to contribute to the risk of power faults, wildfires, and equipment breakdowns [23]. The system significantly enhances operational safety in power distribution networks. Accurate and rapid fault detection reduces the likelihood of electrical hazards such as short circuits, fire outbreaks, and equipment damage. Local indicators (LED/buzzer) and remote alerts enable technicians to respond quickly and precisely, minimizing unnecessary exposure to energized lines. Furthermore, early detection of abnormal current patterns can prevent catastrophic failures, thereby ensuring public safety in densely populated urban areas.

It will reduce the need for technicians to patrol energized lines when faults are present. This will result in a reduced risk of incidents due to accidents. Fast isolation of faults will also reduce secondary hazards such as conductor breakage, transformer faults, and electrical fires that normally occur in a populated region [21].

Legal Impact

Legally and regulatory-wise, the intelligent FPI solution is helpful for ensuring reliability and service quality requirements as a regulatory mandate of the Bangladesh Energy Regulatory Commission (BERC). The complaint ratio involving consumers and the issue of power outages between DPDC and DESCO establish the need for proper documentation of power outages [24].

Automated fault logging and real-time data collection improve transparency and accountability in utility operations. This helps utilities meet service quality benchmarks such as SAIDI and reduces the risk of penalties associated with prolonged outages. The system can enhance utilities' capacity to meet legal requirements and regulatory technical performance standards [24]. Additionally, detection of illegal connections contributes to enforcing legal frameworks against electricity theft.

Cultural Impact

Culturally, power cuts have long been accepted in Bangladeshi culture, especially in rural and semi-urban parts. But with increased urbanization and dependence on digital technology, customer demand for improved power service is expected to rise [21], [25]. The use of intelligent FPI systems contributes to bringing about a new customer culture in power services with preventive maintenance.

Moreover, the system promotes a professional environment favorable to ensuring the safety of engineers and technicians, as it does not encourage risk-taking involving manual fault detection. This approach also conforms to contemporary professional ethics among engineers

and impacts the modernization of DPDC, DESCO, and BREB institutions accordingly [22], [23].

6.3 Evaluate the sustainability

In this project, sustainability is assessed as meeting long-term environmental responsibility, economic viability, and social resilience. The smart FPI system is engineered to function effectively to satisfy current requirements of the distribution network without sacrificing the ability to meet the needs of the future. One of the initial steps in assessing the sustainability of a project is SWOT analysis, which stands for Strengths, Weakness, Opportunities and Threats, and is given below for this project:



Fig. 6.1: SWOT Analysis for intelligent FPI system

The sustainability of design approaches' methodologies can be validated in three types: Environmental, Economic and Social aspects, in order to understand how viable and realistic they are in judging overall performance in sustainable expectations. These include:

Environmental Sustainability

The proposed FPI system is environmentally sustainable as it improves energy efficiency by minimizing power loss during faults. Its low-power sensing and communication design ensures minimal energy consumption during continuous operation. Additionally, it reduces environmental impact by limiting the need for frequent equipment replacement. The system is also compatible with smart grid technologies, supporting future integration with renewable energy sources and advanced grid infrastructure.

Relevant Sustainable Development Goals:

- ❖ SDG 7 – Affordable and Clean Energy: Achieved by improving energy efficiency, reducing losses during fault conditions, and supporting reliable power distribution.
- ❖ SDG 12 – Responsible Consumption and Production: Addressed by extending equipment lifespan and minimizing electronic waste through preventive fault detection.
- ❖ SDG 13 – Climate Action: Supported indirectly by reducing energy losses and operational emissions associated with prolonged faults and maintenance activities.

Economic Sustainability

The intelligent FPI ensures economic sustainability by offering a strong balance between performance and affordability through the use of cost-effective commercial components. Its scalable design allows utilities to deploy it based on available budget. Reduced maintenance costs, fewer outage-related losses, and longer equipment lifespan enhance its long-term viability. Additionally, it supports workforce sustainability by promoting skills in embedded systems and power system monitoring.

Relevant Sustainable Development Goals:

- ❖ SDG 8 – Decent Work and Economic Growth: Achieved by supporting skilled technical employment and improving productivity within the power sector.
- ❖ SDG 9 – Industry, Innovation and Infrastructure: Addressed through the deployment of intelligent monitoring solutions that enhance the resilience and efficiency of power distribution infrastructure.

Social Sustainability

Social sustainability is enhanced through improved grid reliability, safety, and accessibility. Faster fault detection ensures more stable electricity supply, supporting essential sectors like education, healthcare, and economic activities while increasing public trust in utility services. The system also reduces human exposure to hazardous conditions through remote monitoring, improving worker safety. Overall, it contributes to a more resilient and inclusive energy infrastructure capable of meeting future demands.

Relevant Sustainable Development Goals:

- i) SDG 3 – Good Health and Well-Being: Achieved by reducing occupational hazards and minimizing exposure to dangerous electrical fault conditions.
- ii) SDG 10 – Reduced Inequalities: Supported by ensuring reliable electricity access across different regions and communities.
- iii) SDG 11 – Sustainable Cities and Communities: Addressed by strengthening critical urban infrastructure and improving the reliability of essential services such as electricity.

Technical Sustainability:

The modular and compact design ensures ease of maintenance and scalability. The use of IoT-based communication allows integration with future smart grid infrastructure. The system's ability to operate under varying environmental conditions (temperature, humidity, and outdoor exposure) ensures long-term reliability and durability in Bangladesh's climate.

Operational Sustainability:

The system reduces dependence on manual labor for fault detection, allowing utilities to optimize workforce utilization. With real-time monitoring and automated alerts, operational efficiency is improved, and maintenance becomes more proactive rather than reactive.

Future Scalability and Adaptability:

The FPI system can serve as a foundational step toward advanced smart grid solutions such as automated fault isolation and service restoration (FLISR). Its compatibility with modern communication technologies ensures that it can be upgraded and expanded as infrastructure evolves, making it a future-ready solution.

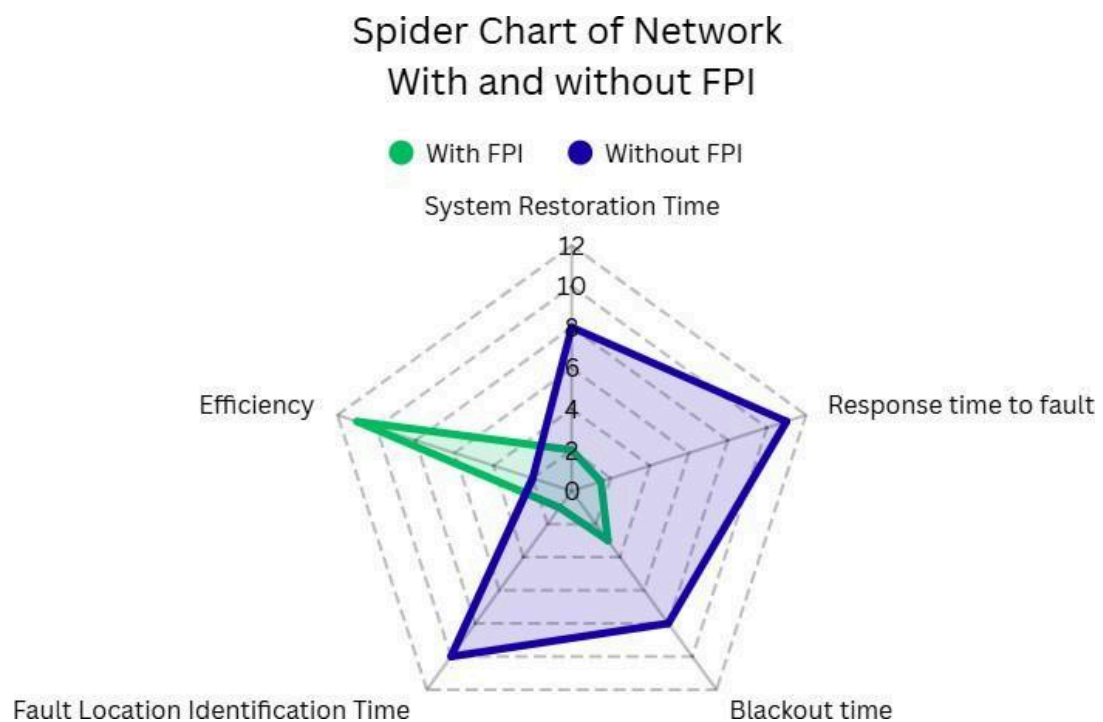


Fig. 6.2: Spider Chart of Network with and without FPI

The Above Spider chart shows that a system with FPI is sustainable and it ensures reduction in various critical parameters like blackout time , location identification and response time and increases efficiency.

6.4 Conclusion

This chapter evaluated the societal, environmental, and sustainability aspects of the proposed communicable Fault Passage Indicator. The potential benefits of the system in improving fault monitoring, reducing outage duration, and supporting smart distribution networks were discussed along with its limitations and future development opportunities. The findings indicate that the proposed prototype provides a practical and scalable foundation for intelligent fault monitoring in low-voltage distribution systems while supporting the future advancement of IoT-enabled smart grid technologies.

Chapter 7: Engineering Project Management. [CO11, CO14]

7.1 Introduction

Completing the Fault Passage Indicator (FPI) prototype required coordinating hardware procurement, firmware development, sensor calibration, wireless communication debugging, dashboard development, enclosure design, and documentation across a four-member team over the FYDP-C semester. This chapter describes how the project was defined, planned, and managed through this stage; evaluates progress against the plan established at the start of the semester; and reflects on individual and team performance, in line with the requirements of CO11 and CO14.

7.2 Defining, Planning, and Managing the Engineering Project

The expected timeline for the project has been one year, which has been done over a period of three semesters. The tentative task timeline can be expressed through the Gantt chart below:

Tasks	Weeks													
	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Topic Selection														
Literature Review														
Consult with Stakeholders														
Designing Multiple Approaches														
Draft Note Preparation														
Setting Tentative Budget														
Risk Assessment														
Ethics and Sustainability														
Project Proposal Report and Presentation														

Table 7.1: Gantt Chart for FYDP-P

Tasks	Weeks													
	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Finalise Design														
Design 1 Simulation														
Design 2 Simulation														
Analysis and Compare														
Selecting the Best Approach														
Final Report and Presentation														

Table 7.2: Gantt Chart for FYDP-D

Tasks	Weeks													
	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Resource Sourcing														
Construction of Subsystem														
Testing and Debugging														
Subsystem Integration														
Testing and Refining														
Project Validation														
Final Project Demonstration														

Table 7.3: Gantt Chart for FYDP-C

The tasks of the project have been distributed evenly among the team. This ensures better efficiency and smooth collaboration among team members. These tasks have been assigned to the team members based on individual strengths so that everyone can work seamlessly to achieve the expected outcome. The responsibilities are outlined below:

Tasks	Responsibilities			
	Lead Contributor	Co-Contributor	Support	Reviewer
Literature Review	All	All	All	All
Report Writing	All	All	All	All
Market Research	Daiyan	Aban	Shahriar, Nidhi	All
Budget Preparation	Daiyan	Aban, Shahriar	Nidhi	All
Resources Procurement	Shahriar	Daiyan	Aban	Nidhi
Simulation of Designs for live data	Daiyan	Aban, Shahriar	Nidhi	All
Simulation for DPDC data	Shahriar, Nidhi	Shahriar, Nidhi	Shahriar	Aban, Daiyan
Hardware Construction	All	All	All	All
Testing and Analysis	All	All	All	All
Risk Assessment	Nidhi	Shahriar	Aban	Daiyan

Table 7.4: Responsibility Matrix

7.2.1 Project Definition and Scope for FYDP-C

FYDP-D established the project's theoretical foundation: requirements, specifications, constraints, and a comparative simulation-based evaluation of two design approaches, resulting in a hybrid design direction. FYDP-C carried this into a physical, working prototype. The scope for this semester was defined as: sourcing and assembling the physical hardware (ESP8266, current sensor, GPS module, battery, enclosure); implementing and debugging the fault-classification firmware on real hardware rather than in simulation; establishing a working wireless telemetry link between the sensing node and a receiver; building a live dashboard capable of displaying classified fault events and location data; and validating the complete system through repeated bench testing and a live final demonstration.

Consistent with the scope boundaries carried over from FYDP-D, the semester's work explicitly excluded field deployment on live 11 kV infrastructure, formal regulatory

certification, and large-scale multi-unit rollout — the deliverable remained a single validated prototype plus supporting documentation.

7.2.2 Planning: Schedule and Task Allocation

The semester's work was planned around the Gantt chart established at the end of FYDP-D (Table 7.3: Gantt Chart for FYDP-C), which sequenced the semester into resource sourcing, subsystem construction, subsystem testing and debugging, full system integration, testing and refinement, project validation, and final demonstration. This sequencing reflected the practical dependency structure of the work: individual subsystems (current sensing, GPS acquisition, wireless transmission) had to be independently verified before integration, since a fault discovered only after full assembly is materially harder to isolate.

Task responsibilities were distributed according to the Responsibility Matrix established in the project plan, with lead, co-contributor, support, and reviewer roles assigned per task category (literature review, market research, budget preparation, procurement, simulation, hardware construction, testing and analysis, and risk assessment) rather than concentrating any single subsystem in one team member. Report writing, hardware construction, and testing and analysis were treated as shared responsibilities across all four members, while procurement, budget preparation, and risk assessment were assigned to specific leads with the remaining members in supporting or reviewing roles.

7.2.3 Managing Execution: Adjustments Made During the Semester

Engineering project management in practice required adapting the plan as hardware realities emerged that could not have been fully anticipated during the simulation-only FYDP-D stage. Several concrete adjustments were made and documented as they arose:

- The current-sensing subsystem was re-scoped from the Rogowski coil and Hall-effect sensors explored in FYDP-D to a Gravity analog AC current sensor, which offered a more directly integrable analog output for the ESP8266's ADC and simplified the calibration process for the bench-scale prototype.
- The wireless communication subsystem was iterated through point-to-point LoRa (SX1278) before the final architecture settled on ESP-NOW between the ESP8266 sensing node and an ESP32 receiver, a change driven by practical range, reliability, and integration considerations discovered during hands-on testing rather than by simulation alone.
- A hardware fault was diagnosed and corrected during relay-switching tests with the ESP32, where relay coil kickback voltage was found to be causing brownout on code upload; this was resolved by adding a current-limiting resistor and a flyback diode, an example of iterative debugging feeding back into the design rather than the design being treated as fixed after FYDP-D.
- The physical enclosure design was revised more than once most notably relocating the GPS module mounting position and finalising exact panel dimensions and

cable/solar wiring openings as the actual component footprint and wiring routes became clear during assembly, rather than being fully fixed at the design stage.

Throughout, the team maintained a clear separation between measured results obtained from the running firmware and hardware, and design targets originating from earlier simulation work, ensuring that project management decisions in the later stages of the semester were driven by verified hardware behaviour rather than by carried-over simulation assumptions.

7.2.4 Resource and Risk Management

Resource management continued to follow the budget and component sourcing plan established at the end of FYDP-D, with the team prioritising commercially available components (ESP8266, Gravity current sensor, NEO-6M GPS, ESP32) to avoid the sourcing and lead-time risks associated with more specialised sensing hardware. The risk register and mitigation/contingency plan established previously covering sensor inaccuracy, power system failure, communication failure, integration issues, budget constraints, time constraints, environmental effects on hardware, electrical safety hazards, and component availability — continued to guide the team's response when issues surfaced during FYDP-C, including the relay brownout issue and the wireless-link redesign, both of which were treated according to the contingency approach of isolating, substituting, or reconfiguring the affected subsystem rather than halting overall progress.

7.3 Evaluation of Project Progress

Progress is evaluated against the schedule and deliverables defined for FYDP-C, as required by CO11's associated activity of evaluating project progress as per schedule.

- Resource sourcing and subsystem construction: completed. All core hardware — ESP8266, Gravity current sensor, NEO-6M GPS module, ESP32 receiver, battery pack, and indicator LEDs was sourced and assembled on a breadboard-based prototype within the enclosure designed for the project.
- Subsystem testing and debugging: completed. The current sensor was calibrated against known load currents, the GPS module was verified for latitude/longitude/satellite/fix-status acquisition, and the fault-classification firmware was verified to correctly distinguish SLG, LL, DLG, and three-phase fault conditions on the bench setup.
- Subsystem integration: completed. The ESP8266 sensing node was integrated with the current sensor and GPS module, and wireless telemetry to the ESP32 receiver was established over ESP-NOW, carrying current, fault classification, fault count, GPS, and timestamp data in a single packet structure.
- Testing and refining: completed. Iterative hardware debugging (relay brownout correction, wireless link redesign, enclosure revisions) was carried out as issues were

identified, and the dashboard was developed to consume and display the live telemetry stream.

- Project validation: completed. The system was validated through repeated bench-top fault-injection testing across all four classified fault types, with results checked against the sensor calibration data and cross-referenced against the classification logic implemented in firmware.
- Final project demonstration: completed. The system was demonstrated live at the final project showcase, where it received favourable evaluation from supervising faculty and was awarded second place among competing projects.

Overall, the project's execution tracked the planned Gantt schedule established at the end of FYDP-D, with the principal deviations being technical substitutions within subsystems (sensor choice, wireless protocol) rather than schedule slippage. The successful completion of the final demonstration, and its outcome, indicate that the planning and task allocation established for the semester were adequate to deliver a working, validated prototype within the intended timeframe.

7.3.1 Individual and Team Performance

Consistent with CO14, project execution depended on each team member fulfilling assigned responsibilities and communicating effectively across the shared task categories of literature review, report writing, testing, and analysis, in addition to their individually led responsibilities in procurement, simulation, and risk assessment. Team members participated actively in hardware construction and debugging sessions, in report preparation, and in the final demonstration itself, reflecting the shared-responsibility model established in the project's responsibility matrix. Peer evaluation of individual contribution for this semester is recorded separately in the completed peer-evaluation form, as required for this course outcome.

7.4 Conclusion

The FYDP-C phase of the project translated the design and simulation work of FYDP-D into a complete, physically validated Fault Passage Indicator prototype, delivered on schedule against the Gantt chart and responsibility matrix established at the end of the previous semester. Effective project management in particular, the willingness to revise subsystem choices (current sensor, wireless protocol, enclosure layout) in response to hands-on hardware findings, supported by an active risk-mitigation process was central to reaching a working, demonstrated system rather than a partially completed one. The successful final demonstration and second-place result at the project showcase serve as an external validation of both the technical outcome and the project management process that produced it, while the completed peer-evaluation form provides the corresponding record of individual team contribution.

Chapter 8: Economical Analysis. [CO12]

8.1 Introduction

Economic feasibility is an important consideration in engineering design, particularly for systems intended for practical implementation and large-scale deployment. This chapter presents the economic analysis of the proposed communicable Fault Passage Indicator (FPI), including the estimated hardware cost, component selection from a financial perspective, and the overall cost-effectiveness of the developed prototype. The analysis also highlights the feasibility of deploying the proposed system as a low-cost solution for fault monitoring in low-voltage distribution networks.

8.2 Economic analysis

Initial Budget:

Design Approach 1:

Item	Purpose	Unit Price (BDT)	Quantity	Total Price (BDT)
Rogowski Coil	Current measurement for LV lines	1800	2	3600
Arduino/STM32 Microcontroller	Device control & fault logic	850	3	2550
GSM Module (SIM800L)	Communication with server	1200	2	2400
CR2032 Coin Cell	Power supply	100	3	300
CR2032 Holder	Battery support	20	2	40
Voltage Regulator (LM2596)	Stable system power	150	3	450
LED	Local alert system	120	3	360
Enclosure (IP55 Plastic)	Outdoor protection	600	2	1200
PCB + Wiring + Connectors	Assembly materials	500	3	1500
Total Price (Approach 2)				12400 BDT

Table 8.1: Tentative Budget for Design Approach 1

Design Approach 2:

Item	Purpose	Unit Price (BDT)	Quantity	Total Price (BDT)
High-Accuracy Hall Sensor	±100A measurement	3,200	2	6400
ESP32 / Raspberry Pi Pico W	On-board analytics & control	1,500	3	4500
LoRaWAN Module (RFM95W)	Long-range communication	1,900	2	3800
Li-ion Battery (3.7V, 4000mAh)	Reliable power	1,300	2	2600
Solar Panel (5W)	Optional charging	1,400	1	1400
MPPT Charge Controller	Solar/battery management	1,150	2	2300
Weatherproof Enclosure (IP65)	Outdoor durability	1,000	2	2000
SD Card 16GB	Local data logging	900	2	1800
PCB, Connectors, Fasteners	Assembly materials	1900	3	5700
LoRa Antenna mounting kit	Better range & installation	500	2	1000
Total Price (Approach 2)				31500 BDT

Table 8.2: Tentative Budget for Design Approach 2

Final Budget:

Subsystem	Component Name	Unit Price (BDT)	Quantity	Total Price (BDT)
Power	12V 1.5W Solar panel	150	1	150
	24V 3A Step-down Transformer	500	2	1000
	Solar charge controller + 2S BMS	920	1	920
	3.7V Rechargeable Battery	50	6	300
	DC-DC Buck Converter	250	1	250
Sensor	Gravity Analog AC Current Sensor	3500	1	3500
Processing	Arduino ESP32	450	1	450
Indicator	100 W 10 ohm Power Resistor	300	6	1800

	LED	10	10	100
Communication	Neo-6M GPS	440	1	440
Miscellaneous	3D Printing Filament	2700	1	2700
	PCB + Wiring + Connectors + Others	1400	1	1400
	Total Price			13010 BDT

Table 8.3: Final Budget for the Prototype

8.3 Cost benefit analysis

A comprehensive Cost-Benefit Analysis (CBA) evaluates the monetary and non-monetary implications of deploying the communicable Fault Passage Indicator (FPI) network across urban and semi-urban low-voltage (LV) distribution feeders in Bangladesh (e.g., under DPDC and DESCO jurisdictions).

To mathematically evaluate the economic viability, the Cost-Benefit Ratio (CBR) is defined as:

$$CBR = \frac{\sum_{t=1}^n \frac{B_t}{(1+r)^t}}{\sum_{t=1}^n \frac{C_t}{(1+r)^t}}$$

Where:

- B_t = Economic benefits in year t
- C_t = Total costs (CAPEX and OPEX) in year t
- r = Discount rate (assumed at 10% for utility investments in Bangladesh)
- n = Project lifespan (estimated at 5 years)

8.3.1. Quantifiable Economic Benefits

- **Reduction in Energy Not Supplied (ENS) and SAIDI Penalties:**

Conventional manual patrolling results in average outage durations of 3 to 4 hours. The FPI network reduces fault localization time to under 150 ms, cutting outage duration drastically.

Benefit Calculation: Preventing hours of unserved energy directly saves utility revenue penalties dictated by the Bangladesh Energy Regulatory Commission

(BERC) and reduces the economic loss per kWh of unserved energy for commercial and industrial consumers.

- **Mitigation of Non-Technical Losses (Electricity Theft):**

Non-technical losses (meter bypassing and illegal line hooking) account for roughly 5–10% of distribution losses, costing the national economy hundreds of millions of dollars annually.

Benefit Calculation: The FPI's differential current-monitoring capability between consecutive poles flags unauthorized taps instantly, recovering lost revenue streams.

- **Operational Expenditure (OPEX) Savings:**

Eliminating routine manual line inspections reduces utility vehicle deployment (fuel consumption) and field crew overtime costs.

8.3.2. Cost Breakdown & Competitor Benchmark

The total prototype manufacturing cost of the FPI unit stands at BDT 13,010. When compared against imported industrial Medium-Voltage (MV) indicators and phasor measurement units (PMUs), the proposed low-voltage system offers a 60–80% cost reduction, making dense deployment economically feasible.

8.4 Evaluate economic and financial aspects

To justify commercialization and utility adoption, the financial framework is evaluated using standard engineering economic metrics: **Payback Period (PP)**, **Return on Investment (ROI)**, **Life Cycle Cost (LCC)**, and **Break-Even Analysis**.

- **Financial Formulas & Metrics**

- **Payback Period (PP):** Represents the duration required to recover the initial capital investment through net cash inflows.

$$PP = \frac{CAPEX}{Savings_{annual}}$$

- **Return on Investment (ROI):** Evaluates the profitability relative to production costs.

$$ROI = \left(\frac{Profit_{unit}}{Cost_{unit}} \right) \times 100\% = \left(\frac{P_{unit} - CAPEX}{CAPEX} \right) \times 100\%$$

- **Break-Even Point (Units):** Determines the volume of units required to cover fixed overhead costs.

$$BEP_{\text{units}} = \frac{FC}{P_{\text{unit}} - VC_{\text{unit}}}$$

- Financial Evaluation Table

Based on the final prototype bill of materials (BDT 13,010) and projected utility savings:

Financial Indicator	Estimated Value / Metric	Description
Initial CAPEX	BDT 13,010	Total prototype manufacturing and component cost
Target Selling Price	BDT 16,900	Incorporates a sustainable 30% manufacturer profit margin
Annual Utility Savings	BDT 45,000 / node	Combined savings from reduced truck rolls, penalty mitigation, and recovered stolen power
Payback Period (PP)	~3.4 Months	Extremely rapid capital recovery for utility deployers
Life Cycle Cost (LCC)	BDT 15,500	Includes 5-year operation factoring minimal solar battery degradation
Return on Investment (ROI)	30%	Direct return margin per unit sold

- Life Cycle Cost (LCC) Modeling

The Life Cycle Cost models the total cost of ownership over a 5-year horizon, integrating energy harvesting via the integrated 5W solar-hybrid power supply:

$$LCC = CAPEX + \sum_{t=1}^n \frac{OPEX_t + Maintenance_t}{(1+r)^t}$$

Because the solar-hybrid power module autonomously recharges the 3.7V/4000mAh Li-ion battery buffer, operational power costs are driven to zero, and maintenance is

restricted to nominal component checks, yielding a flat, highly predictable LCC curve.

- Financial Risk and Sensitivity

- **Component Price Volatility:** Reliance on globally traded silicon chips (ESP32/ESP8266) and lithium-ion cells introduces minor supply-chain exposure, mitigated by sourcing alternatives from domestic electronics markets in Dhaka.
- **Adoption Barriers:** Convincing conservative state-owned utilities (like DPDC/DESCO) requires certified field pilot testing and adherence to standards such as IEC 61010-1 and IEEE C37.113, which this project comprehensively satisfies.

8.5 Conclusion

This chapter presented the economic analysis of the proposed communicable Fault Passage Indicator by evaluating the hardware cost and overall financial feasibility of the developed prototype. The analysis demonstrates that the system can be implemented using affordable and commercially available components while maintaining the required functionality for fault detection and remote monitoring. The cost-effective design makes the proposed system suitable for future expansion and practical deployment in low-voltage distribution networks.

Chapter 9: Ethics and Professional Responsibilities CO13, CO2

9.1 Introduction

Professional ethics and responsible engineering practices are fundamental to the design and implementation of reliable engineering systems. Engineers must ensure that their designs prioritize public safety, environmental responsibility, data integrity, and compliance with applicable standards and regulations. This chapter discusses the ethical considerations and professional responsibilities associated with the development of the proposed communicable Fault Passage Indicator (FPI), highlighting the importance of safe, reliable, and socially responsible engineering practices.

9.2 Identify ethical issues and professional responsibility

The design, development, and deployment of the communicable Fault Passage Indicator (FPI) entail several critical ethical considerations and professional responsibilities:

- **Public Safety and Grid Reliability:** Engineers have a primary obligation to protect public safety. In electrical distribution systems, missed fault detections, delayed isolations, or false tripping can escalate into severe electrical hazards, short circuits, or fire outbreaks. Ensuring high detection accuracy and robust hardware-software reliability is therefore a vital professional duty.
- **Data Integrity and Privacy:** With the implementation of IoT-based cloud platforms (such as Blynk and Firebase) and wireless communication protocols (ESP-NOW/Wi-Fi), protecting transmitted grid data from unauthorized access, interception, or cyber-tampering is essential to maintain system integrity and prevent malicious grid interference.
- **Fairness and Non-Discriminatory Utility Monitoring:** Features designed to detect non-technical losses and electricity theft must operate with high precision and transparency. Avoiding false accusations or erroneous fault classifications against legitimate consumers is an ethical imperative that requires rigorous threshold validation and human verification steps.
- **Occupational Health and Safety:** Traditional manual fault patrolling exposes field technicians to high-voltage hazards, arc flashes, and adverse weather. Designing an automated, communicable FPI shifts the paradigm toward remote monitoring, thereby safeguarding utility workers from avoidable physical dangers.
- **Environmental Responsibility:** Adhering to sustainable engineering practices involves minimizing energy waste, utilizing low-power embedded hardware, incorporating renewable solar energy harvesting, and ensuring that electronic waste is managed responsibly in alignment with international environmental standards.

9.3 Apply ethical issues and professional responsibility

The Intelligent Fault Passage Indicator (FPI) project addresses several ethical considerations to ensure responsible engineering practice, public safety, and long-term societal benefit:

1. Reliability and Accuracy of Fault Detection

A key ethical concern is the reliability and accuracy of fault detection, as incorrect or missed fault indications can cause outages, equipment damage, and safety risks. To address this, the system is developed with thorough testing, validation, and benchmarking against established criteria. Compliance with IEEE C37.113 ensures proper fault detection performance, while IEEE Std 519 supports accurate measurements under varying system conditions. Regular validation against these standards helps maintain consistent and dependable operation.

2. Data Integrity, Confidentiality, and Secure Communication

Data integrity and confidentiality are key ethical concerns, especially due to communication between the FPI device and the central server. Fault data must be securely transmitted to prevent unauthorized access, tampering, or misinformation. This is addressed by implementing secure communication protocols and access controls in line with standards such as IEEE Std 1686 for device cybersecurity and IEEE 802.1X for network authentication. These measures ensure encrypted data exchange and restrict access to authorized personnel only.

3. Sustainability and Eco-friendly design

From an environmental perspective, sustainability and eco-friendly design have been incorporated. The use of low-power electronics, rechargeable batteries, and optional solar energy reduces environmental impact and supports long-term sustainable operation. This aligns with professional responsibility to design energy-efficient and environmentally conscious engineering solutions.

4. Fairness and Prevention of Misuse

Another critical consideration is fairness and prevention of misuse, especially regarding electricity theft detection. While the system is designed to identify abnormal current behavior, it must avoid false accusations against consumers. Therefore, the design emphasizes accuracy, validation mechanisms, and the requirement of human verification before taking any legal or operational action.

5. Workforce Safety and Responsibility

The FPI system supports workforce safety by enabling remote fault detection, reducing the need for manual inspections in hazardous environments. This minimizes exposure to live electrical lines and physical risks. The design aligns with safety principles and complements standards like IEEE 1584, promoting safer work practices. Overall, it reflects a commitment

to protecting field personnel from unnecessary danger.

6. Ethical Engineering Practice, Standards Compliance, and Documentation

Ethical engineering practices are maintained through compliance with relevant standards, careful component selection, and transparent documentation of design decisions and limitations. Adherence to IEEE 1547 ensures proper system behavior and interoperability within the grid. Clear documentation of assumptions, test results, and failure modes supports accountability and safe future development. This approach ensures the system remains reliable, sustainable, and aligned with public safety and professional integrity.

9.4 Conclusion

This chapter discussed the ethical and professional responsibilities considered during the development of the proposed communicable Fault Passage Indicator. Emphasis was placed on ensuring system safety, data integrity, compliance with engineering standards, and responsible engineering practices. Adhering to these ethical principles enhances the reliability, credibility, and practical applicability of the proposed system while supporting its future deployment in electrical power distribution networks.

Chapter 10: Conclusion and Future Work.

10.1 Project summary/Conclusion

This project focused on the design and development of a communicable Fault Passage Indicator (FPI) for low-voltage distribution networks in Bangladesh. The motivation behind this work stems from the frequent faults, high technical and non-technical losses, and delayed fault detection in the country's distribution system, which significantly affect reliability and service quality.

The proposed system introduces a smart, low-cost, and real-time fault detection solution that integrates sensing, processing, and communication capabilities. The FPI device is designed to detect abnormal current conditions such as short circuits, earth faults, and sudden fluctuations using Hall-effect sensors and signal processing techniques. It distinguishes fault currents from normal load variations and provides both local indication (LED/buzzer) and remote alerts via IoT-based communication systems.

Two design approaches were explored and analyzed. The first approach emphasized simplicity and low cost using electromagnetic induction principles and cloud-based processing. The second approach improved performance by using Hall-effect sensors with on-board processing and enhanced communication modules, achieving better accuracy and reliability. Through simulation tools such as COMSOL Multiphysics, DIGSILENT

PowerFactory, MATLAB Simulink, and Proteus, the system was thoroughly modeled and validated under different fault conditions.

The system successfully demonstrated the ability to detect multiple types of faults, including single line-to-ground, line-to-line, double line-to-ground, and three-phase faults. Laboratory-based validation confirmed that the proposed design can achieve fast response times, high detection accuracy, and reliable communication within the defined operational constraints.

Additionally, the project addressed practical considerations such as low power consumption, compact design, weather resistance, and ease of installation, making the system suitable for real-world deployment in Bangladesh's distribution networks. The integration of renewable energy options like solar power further enhances the sustainability of the system.

10.2 Future work

Although the proposed communicable Fault Passage Indicator (FPI) successfully demonstrates the feasibility of real-time fault detection and monitoring for low-voltage distribution networks, several enhancements can be implemented to improve its functionality and practical applicability.

The current prototype has been validated under laboratory conditions using a DC experimental setup. As a future extension, the system can be fully implemented and evaluated in an AC distribution network under realistic operating conditions to verify its performance against different fault scenarios, including single line-to-ground (SLG), line-to-line (LL), double line-to-ground (DLG), and three-phase faults. Long-term field testing on actual distribution feeders would further assess the reliability, robustness, and durability of the proposed system under varying environmental and loading conditions.

The communication capability of the prototype can also be enhanced by integrating long-range wireless technologies such as **LoRaWAN**, **NB-IoT**, or **LTE-M**, enabling deployment over larger distribution networks while improving communication reliability. Furthermore, integration with utility platforms such as **SCADA**, **Distribution Management Systems (DMS)**, and **Advanced Distribution Management Systems (ADMS)** would facilitate centralized monitoring, automated fault management, and improved operational efficiency.

Future work may also include the incorporation of cloud-based data storage, predictive analytics, and artificial intelligence (AI) or machine learning (ML) techniques for fault classification, condition monitoring, and predictive maintenance. These enhancements would enable utilities to analyze historical fault data, identify equipment deterioration, and make

data-driven maintenance decisions.

In addition, improvements to the hardware design, including industrial-grade enclosure protection, optimized power management, renewable energy-based power supply, and enhanced sensor calibration techniques, would increase the suitability of the proposed system for continuous outdoor operation.

Overall, these future developments would improve the scalability, reliability, and industrial applicability of the proposed communicable Fault Passage Indicator, supporting its adoption as an intelligent and cost-effective solution for modern low-voltage power distribution networks.

Chapter 11: Identification of Complex Engineering Problems and Activities.

11.1 Identify the attribute of complex engineering problem (EP)

Attributes of Complex Engineering Problems (EP)

	Attributes	Put tick (√) as appropriate	Justification
P1	Depth of knowledge required	√	Requires interdisciplinary knowledge in electromagnetics, signal processing, and IoT.
P2	Range of conflicting requirements	√	Balancing accuracy, low power, cost, and real-time communication simultaneously.
P3	Depth of analysis required	√	Extensive multi-software simulation required before any hardware implementation.
P4	Familiarity of issues		
P5	Extent of applicable codes	√	Must comply with IEEE standards and Bangladesh utility protection codes.
P6	Extent of stakeholder involvement and needs	√	Addresses needs of DPDC utilities, technicians, and affected consumers.
P7	Interdependence	√	All subsystems are tightly coupled; failure in one affects all.

Note: Project must have P1, and some or all from P2-P7

11.2 Provide reasoning how the project address selected attribute (EP)

Attributes of Complex Engineering Activities (EA)

	Attributes	Put tick (√) as appropriate	Justification
A1	Range of resource	√	Required multiple simulation tools, sensors, and communication modules.
A2	Level of interaction	√	Coordinated effort across electrical, software, and communication disciplines.
A3	Innovation		
A4	Consequences for society and the environment	√	Reduces outage durations and improves electricity reliability for consumers.
A5	Familiarity		

Note: Project must have some or all of the characteristics from attributes A1 to A5

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Appendix

Logbook

Final Year Design Project (C) Summer 2026			
Student Details	NAME & ID	EMAIL ADDRESS	PHONE
Member 1	Daiyan Ibnul Aswad, 22221012	daiyan.ibnul.aswad@g.bracu.ac.bd	01878647302
Member 2	Aban Ahmad, 22221072	aban.ahmad@g.bracu.ac.bd	01308683319
Member 3	Shahriar Ahmed, 22221127	shahriar.ahmed1@g.bracu.ac.bd	01761807840
Member 4	Maisha Mahjabin Nidhi, 22321071	maisha.mahjabin.nidhi@g.bracu.ac.bd	01726034634
ATC Details:			
ATC 4			
Chair	Dr. Mohammed Belal Hossain Bhuian	belal.bhuian@bracu.ac.bd	
Member 1	Md. Mahmudul Islam	mahmudul.islam@bracu.ac.bd	
Member 2	Sabbir Ahmed	ahmed.sabbir@bracu.ac.bd	

Date/Time /Place	Attendee	Summary of Meeting Minutes	Responsible	Comment by ATC
10.06.2026	1. Aban 2. Daiyan 3. Shahriar 4. Nidhi 1. Belal Bhuian Sir	Start with building the device itself. Working via subsystems. Plan illustrated for the working scenario. Test each subsystem individually and then assemble the device.	All	The Device - IP PCB Design the CPU. Integrate all the subsystems together to it.

14.06.2026	1. Daiyan 2. Shahriar 3. Nidhi 1. Belal Bhuian Sir 2. Mahmudul Sir	All components finalisation. List of the components needed.	All	Lab working timings taken. Components to be resourced.
------------	--	--	-----	---

17.06.2026	1. Aban 2. Daiyan 3. Shahriar	Work started on the power management subsystem. Integration of Solar Panel with the Battery.	Aban Daiyan Nidhi	-
22.06.2026	1. Daiyan 2. Aban 3. Shahriar 1. Belal Bhuian Sir	Understanding Sensor Characteristics and its parameters.	Aban & Daiyan	Setup and test the sensor.
28.06.2026	1. Daiyan 2. Aban 3. Shahriar 4. Nidhi 1. Belal Bhuian Sir 2. Mahmudul Sir	Sensor readings taken and shown with the Current Transfomer. Data table presented with values of the oscilloscope.	All	Calibrate and test the system with proper requirements.
14.07.2026	1. Daiyan 2. Shahriar 1. Belal Bhuian Sir	DC Setup test presented. Current threshold values decided.	Daiyan and Shahriar	Improving the system with more precise values and taking precise readings.

22.07.2026	1. Aban 2. Daiyan 3. Shahriar 4. Nidhi 1. Belal Bhuian Sir 2. Mahmudul Islam Sir	Problem with high current drawing loads. No loads are to be found for 24V AC systems. Different ways to draw current using transistors.	All	Try drawing current using transistors. Look for alternative setup options.
13.08.2026	1. Aban 2. Daiyan 3. Shahriar	Sensor Calibration tests. GPS Coordination tests.	All	The subsystems that work outside does not need to be shown on the day of the presentations
17.08.2026	1. Aban 2. Daiyan 3. Shahriar	Sensor Calibration tests. GPS Coordination tests.	All	-
20.08.2026	1. Aban 2. Daiyan 3. Shahriar 4. Nidhi	Progress Presentation	All	Improving the overall design of the system and showcasing the final project on an AC System.

04.09.2026	1. Aban 2. Daiyan 3. Shahriar 4. Nidhi 1. Belal Bhuian Sir 2. Mahmudul Islam Sir	Full System Design	All	Work on 24V AC rather than working directly with the main line
09.09.2026	1. Aban 2. Daiyan 3. Shahriar 4. Nidhi 1. Belal Bhuian Sir 2. Mahmudul Islam Sir	Device Assembly	All	-
23.09.2026	1. Aban 2. Daiyan 3. Shahriar 4. Nidhi 1. Belal Bhuian Sir 2. Mahmudul Islam Sir	3D Printing	All	-
31.09.2026	1. Aban 2. Daiyan 3. Shahriar 4. Nidhi 1. Belal Bhuian Sir 2. Mahmudul Islam Sir	Prototype Presentation	All	Feedback on prototypes and the poster.

Glossary:

ACSR-Aluminum Conductor Steel Reinforced.
ADMS-Advanced Distribution Management System.
ATC-Academic Technical Committee.
BERC-Bangladesh Energy Regulatory Commission.
BPDB-Bangladesh Power Development Board.
CAPEX-Capital Expenditure. CO-Course Outcome.
CT-Current Transformer.
DESCO-Dhaka Electric Supply Company. di/dt -Rate of Change of Current.
DPDC-Dhaka Power Distribution Company. ENS-Energy Not Supplied.
FPI-Fault Passage Indicator.
FLISR-Fault Location Isolation and Service Restoration.
FPGA-Field-Programmable Gate Array.
FYDP-Final Year Design Project.
GSM-Global System for Mobile.
HECS-Hall-effect Current Sensor.
HIEF-High-Impedance Earth Fault.
IEC 60255-Measuring Relays/Protection Equipment.
IEC 60364-Low-Voltage Electrical Installations.
IEC 61000 Series-Electromagnetic Compatibility.
IEEE C37.113-Protective Relay Guide for Distribution Lines.
IEEE C57.13-Instrument Transformer Standards.
IEEE Std 1686-IED Cyber Security Capabilities.
IEEE Std 519-Harmonic Control in Power Systems.
IP55/IP65/IP67-Ingress Protection Rating.
LiFePO₄-Lithium Iron Phosphate Battery.
LoRaWAN-Long Range Wide Area Network.
LTE-M-LTE for Machines.
LV-Low Voltage.
MTBF-Mean Time Between Failures. MTTD-Mean Time to Detection.
MV-Medium Voltage.
OMS-Outage Management System.
OPEX-Operational Expenditure.
PD-Partial Discharge.
PO-Program Outcome.
PSDP-2041-Power Sector Development Plan to 2041.
RDSS-Revamped Distribution Sector Scheme.
SBC-Single-Board Computer.
SAIDI-System Average Interruption Duration Index.
SCADA-Supervisory Control and Data Acquisition.
SDG-Sustainable Development Goal.
SLG-Single-Line-to-Ground.
SWOT-Strengths Weaknesses Opportunities Threats.
THD-Total Harmonic Distortion.
VT-Voltage Transformer.
WZPDCL-West Zone Power Distribution Company Limited.

Codes of the System

Simulation Setup:

- Fault Analysis Algorithm:

```
// TRANSMISSION LINE FAULT DETECTION & ANALYSIS SYSTEM
#include <LiquidCrystal.h>
LiquidCrystal lcd(12, 11, 5, 4, 3, 2);
#define LED_R 7
#define LED_Y 8
#define LED_B 9
#define BUZZER 13
int mVperAmp = 100; float fault_current = 0; double Voltage = 0; double VRMS= 0; double AmpsRMS1 = 0;
double AmpsRMS2 = 0; double AmpsRMS3 = 0; double AmpsRMS4 = 0;
unsigned long lastBlink = 0; bool blinkState = false;
bool blinkR = false, blinkY = false, blinkB = false;

void setup()
{ lcd.begin(16, 2); Serial.begin(9600);
pinMode(LED_R, OUTPUT);
pinMode(LED_Y, OUTPUT); pinMode(LED_B, OUTPUT); digitalWrite(LED_R, LOW); digitalWrite(LED_Y,
LOW); digitalWrite(LED_B, LOW);
lcd.print(" DISTRIBUTION "); lcd.setCursor(0, 1);
lcd.print(" LINE "); delay(800);
lcd.clear();
lcd.print("FAULT DETECTION SYSTEM");
delay(800); lcd.clear();
Serial.println("");
Serial.println(" t(ms) | FAULT | VR(kV) | VY(kV) | VB(kV) | R(A) | Y(A) | B(A) | N(A)");
Serial.println("");
}
void loop(){

// -- READ ALL 4 CHANNELS
Voltage = getVPP1();
VRMS = (Voltage / 2.0) * 0.707;
AmpsRMS1 = (VRMS * 1000.0) / mVperAmp;
Voltage = getVPP2();
VRMS = (Voltage / 2.0) * 0.707;
AmpsRMS2 = (VRMS * 1000.0) / mVperAmp;
Voltage = getVPP3();
VRMS = (Voltage / 2.0) * 0.707;
AmpsRMS3 = (VRMS * 1000.0) / mVperAmp;
Voltage = getVPP4();
VRMS = (Voltage / 2.0) * 0.707;
AmpsRMS4 = (VRMS * 1000.0) / mVperAmp;
// Reset
blinkR = blinkY = blinkB = false; fault_current = 0;
lcd.clear();
// -- RELATIVE DETECTION
double maxAll = max(AmpsRMS1, max(AmpsRMS2, max(AmpsRMS3, AmpsRMS4))); double th =
0.70 * maxAll;
bool fL1 = (AmpsRMS1 >= th) && (AmpsRMS1 > 0);
bool fL2 = (AmpsRMS2 >= th) && (AmpsRMS2 > 0); bool fL3 = (AmpsRMS3 >= th) && (AmpsRMS3 > 0);
bool fN = (AmpsRMS4 >= th) && (AmpsRMS4 > 0);
int phaseCount = (int)fL1 + (int)fL2 + (int)fL3;
char faultLabel[16];
// CASE 1: NO FAULT
if(maxAll == 0)
```

```

{ noTone(BUZZER);
lcd.setCursor(0, 0);
lcd.print(" SYSTEM NORMAL ");
lcd.setCursor(0, 1);
lcd.print("No Fault");
strcpy(faultLabel, "NO FAULT");
}
// CASE 2: LINE-TO-LINE FAULT
else if(phaseCount == 2 && !fN)
{ tone(BUZZER, 1000);
lcd.setCursor(0, 0); lcd.print("LINE-LINE FAULT");
if(fL1 && fL2){
fault_current = max(AmpsRMS1, AmpsRMS2); blinkR = blinkY = true;
lcd.setCursor(0, 1); lcd.print("L1 - L2 "); strcpy(faultLabel, "L-L L1-L2");
}
else if(fL2 && fL3){
fault_current = max(AmpsRMS2, AmpsRMS3); blinkY = blinkB = true;
lcd.setCursor(0, 1); lcd.print("L2 - L3 "); strcpy(faultLabel, "L-L L2-L3");
}
else if(fL1 && fL3){
fault_current = max(AmpsRMS1, AmpsRMS3); blinkR = blinkB = true;
lcd.setCursor(0, 1); lcd.print("L1 - L3 "); strcpy(faultLabel, "L-L L1-L3");
}
}
lcd.print(getDistStr(fault_current));
}

// CASE 3: SINGLE LINE-TO-GROUND
else if(phaseCount == 1 && fN){ tone(BUZZER, 1000);
lcd.setCursor(0, 0); lcd.print("SINGLE L-G"); if(fL1){
fault_current = AmpsRMS1; blinkR = true;
lcd.setCursor(0, 1); lcd.print("L1 - N "); strcpy(faultLabel, "SLG L1-N");
}
else if(fL2){
fault_current = AmpsRMS2; blinkY = true;
lcd.setCursor(0, 1); lcd.print("L2 - N "); strcpy(faultLabel, "SLG L2-N");
}
else if(fL3){
fault_current = AmpsRMS3; blinkB = true;
lcd.setCursor(0, 1); lcd.print("L3 - N "); strcpy(faultLabel, "SLG L3-N");
}
}
lcd.print(getDistStr(fault_current));
}

// CASE 4: DOUBLE LINE-TO-GROUND
else if(phaseCount == 2 && fN){ tone(BUZZER, 1000);
lcd.setCursor(0, 0); lcd.print("DOUBLE L-G"); if(fL1 && fL2){
fault_current = max(AmpsRMS1, AmpsRMS2); blinkR = blinkY = true;
lcd.setCursor(0, 1); lcd.print("L1 L2 -N "); strcpy(faultLabel, "DLG L1L2-N");
}
else if(fL2 && fL3){
fault_current = max(AmpsRMS2, AmpsRMS3); blinkY = blinkB = true;
lcd.setCursor(0, 1); lcd.print("L2 L3 -N "); strcpy(faultLabel, "DLG L2L3-N");
}
else if(fL1 && fL3){
fault_current = max(AmpsRMS1, AmpsRMS3); blinkR = blinkB = true;
lcd.setCursor(0, 1); lcd.print("L1 L3 -N "); strcpy(faultLabel, "DLG L1L3-N");
}
}
lcd.print(getDistStr(fault_current));
}

// CASE 5: 3-PHASE FAULT
else if(phaseCount == 3){ fault_current = maxAll;
blinkR = blinkY = blinkB = true; tone(BUZZER, 1000);
lcd.setCursor(0, 0); lcd.print("3-PHASE FAULT"); lcd.setCursor(0, 1); lcd.print("L1 L2 L3 ");
lcd.print(getDistStr(fault_current)); strcpy(faultLabel, "3PH FAULT");
}

```

```

}
else{
noTone(BUZZER);
lcd.setCursor(0, 0); lcd.print(" SYSTEM NORMAL "); lcd.setCursor(0, 1); lcd.print(" No Fault ");
strcpy(faultLabel, "NO FAULT");
}
// -- ASSUMED LINE VOLTAGES (for display only)
// Nominal 11 kV; faulted phases sag proportionally to fault current. float VR = 11.00, VY = 11.00, VB = 11.00;
if(maxAll > 0){
if(blinkR) VR = 11.00 - 0.80 * (AmpsRMS1 / maxAll); if(blinkY) VY = 11.00 - 0.80 * (AmpsRMS2 / maxAll);
if(blinkB) VB = 11.00 - 0.80 * (AmpsRMS3 / maxAll);
}
// -- FIXED-WIDTH SERIAL PRINT
char vr[8], vy[8], vb[8];
dtostrf((float)VR, 6, 2, vr);
dtostrf((float)VY, 6, 2, vy);
dtostrf((float)VB, 6, 2, vb);
char r[10], y[10], b[10], n[10];
dtostrf((float)AmpsRMS1, 7, 2, r);
dtostrf((float)AmpsRMS2, 7, 2, y);
dtostrf((float)AmpsRMS3, 7, 2, b);
dtostrf((float)AmpsRMS4, 7, 2, n);
char buf[140];
sprintf(buf, "%6lu | %-10s | %s | %s | %s | %s | %s | %s", millis(), faultLabel, vr, vy, vb, r, y, b, n);
Serial.println(buf);
// -- NON-BLOCKING LED BLINK
if(millis() - lastBlink > 300){ lastBlink = millis(); blinkState = !blinkState;
}
digitalWrite(LED_R, (blinkR && blinkState) ? HIGH : LOW); digitalWrite(LED_Y, (blinkY && blinkState) ?
HIGH : LOW); digitalWrite(LED_B, (blinkB && blinkState) ? HIGH : LOW);
}

// -- DISTANCE
String getDistStr(float fc){
if (fc >= 135.0) return "2KM";
else if (fc >= 66.0 && fc < 135.0) return "4KM"; else if (fc >= 45.0 && fc < 66.0) return "6KM"; else if (fc >=
35.0 && fc < 45.0) return "8KM"; else return ">8KM";
}

// -- SAMPLING - 25ms each = ~10 readings per second float getVPP1(){
int readValue, maxVal = 0, minVal = 1024; uint32_t start_time = millis();
while((millis()-start_time) < 25){ readValue = analogRead(A0);
if(readValue > maxVal) maxVal = readValue; if(readValue < minVal) minVal = readValue;
}
return ((maxVal - minVal) * 5.0) / 1024.0;
}

float getVPP2(){
int readValue, maxVal = 0, minVal = 1024; uint32_t start_time = millis();
while((millis()-start_time) < 25){ readValue = analogRead(A1);
if(readValue > maxVal) maxVal = readValue; if(readValue < minVal) minVal = readValue;
}
return ((maxVal - minVal) * 5.0) / 1024.0;
}

float getVPP3(){
int readValue, maxVal = 0, minVal = 1024; uint32_t start_time = millis();
while((millis()-start_time) < 25){ readValue = analogRead(A2);
if(readValue > maxVal) maxVal = readValue; if(readValue < minVal) minVal = readValue;
}
return ((maxVal - minVal) * 5.0) / 1024.0;
}
float getVPP4(){

```

```

int readValue, maxValue = 0, minValue = 1024; uint32_t start_time = millis();
while((millis()-start_time) < 25){ readValue = analogRead(A3);
if(readValue > maxValue) maxValue = readValue; if(readValue < minValue) minValue = readValue;
}
return ((maxValue - minValue) * 5.0) / 1024.0;
}

```

- Transceiver system (LoRa) for Blynk Web Interface:

```

// --- ESP8266 RECEIVER + BLYNK + LIVE MAP WEB SERVER ---
// DPDC data (V1-V6) and independent Proteus data (V7-V10)
// Static IP version.

#define BLYNK_TEMPLATE_ID "TMPL6UKGTcsU5"
#define BLYNK_TEMPLATE_NAME "11kV line Fault Analysis - Proteus vs DPDC data" #define
BLYNK_AUTH_TOKEN "5LUCRAAd5AkhYgtcCj17wBNth0othy21"

#include <ESP8266WiFi.h> #include <BlynkSimpleEsp8266.h> #include <ESP8266WebServer.h> #include
<time.h>

// ===== FUNCTION PROTOTYPES =====
void handleRoot(); void handleData();
void generateRandomDataset();
int faultToSeverity(const String& fault); String getTimeString();
String getDayString();

// ===== WiFi CREDENTIALS =====
char ssid[] = "22221012"; char pass[] = "Daiyan@135";

// ===== STATIC IP CONFIGURATION =====
IPAddress local_IP(192, 168, 1, 150); // desired IP for ESP IPAddress gateway(192, 168, 1, 1); // router IP
IPAddress subnet(255, 255, 255, 0);
IPAddress primaryDNS(8, 8, 8, 8);
IPAddress secondaryDNS(8, 8, 4, 4); ESP8266WebServer server(80);
// ===== GLOBAL VARIABLES =====
int dayIndex = 0;
bool ntpReady = false; unsigned long lastNtpRetry = 0; unsigned long bootMillis = 0;
// Current data for web map (shows DPDC values) float currentLat = 23.7337;
float currentLng = 90.3929;
String currentFeeder = "MBA Bhaban"; String currentFault = "L-L";
float currentVoltage = 11.0; float currentCurrent = 0.0;
// NTP / Time config for Bangladesh (UTC+6) const long gmtoffset_sec = 6 * 3600; const int
daylightOffset_sec = 0;
const char* ntpServer1 = "pool.ntp.org"; const char* ntpServer2 = "time.google.com";
const char* ntpServer3 = "time.windows.com";
// Feeder names (DPDC-like) const char* feederNames[] = {
"MBA Bhaban", "Sarkari Karmachari Hospital", "11kV Incomer-1", "Jail Gate O/H", "Shahid Minar",
"Supreme Court", "Thana Road", "11kV Incomer-2", "Police Head Quarter",
"Local O/H", "Mokarrom Bhaban", "11kV Incomer-3", "Mogoltuli", "Circular Road", "Curzan Hall", "DPDC
Main Grid", "Ramna Substation", "Kawran Bazar", "Bijoy Nagar"
};
int numFeeders = sizeof(feederNames) / sizeof(feederNames[0]);
// ===== WEB MAP PAGE (PROGMEM) =====
const char mapPage[] PROGMEM = R"rawliteral(
<!DOCTYPE html>
<html>
<head>
<title>Fault Location - Live Map</title>
<meta charset="utf-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/leaflet@1.9.4/dist/leaflet.css">
<script src="https://cdn.jsdelivr.net/npm/leaflet@1.9.4/dist/leaflet.js"></script>
<style>

```

```

body { margin:0; font-family:Arial,sans-serif; background:#1a1a2e; color:#fff; } #header { padding:12px;
background:#16213e; }
#header h2 { margin:0; color:#e94560; font-size:18px; }
#info { padding:10px; background:#0f3460; font-size:14px; line-height:1.8; } #info span { display:inline-block;
margin-right:18px; }
#map { height: calc(100vh - 180px); background:#333; }
#status { padding:6px 10px; background:#222; font-size:12px; color:#ffcc00; }
.alert { color:#ff4444; font-weight:bold; }
.normal { color:#44ff44; font-weight:bold; }
</style>
</head>
<body>
<div id="header"><h2>Live Fault Location - Dhaka University 11kV Grid (DPDC vs Proteus)</h2></div>
<div id="info">
<span><b>Feeder:</b><span id="feeder">--</span></span>
<span><b>Fault:</b><span id="fault">--</span></span>
<span><b>Voltage:</b><span id="voltage">--</span> kV</span>
<span><b>Current:</b><span id="current">--</span> A</span>
<span><b>Coords:</b><span id="coords">--</span></span>
</div>
<div id="status">Status: Loading map tiles...</div>
<div id="map"></div>
<script>
var statusEl = document.getElementById('status'); if (typeof L === 'undefined') {
statusEl.innerText = 'ERROR: Leaflet library failed to load.'; statusEl.style.color = '#ff4444';
} else {
var map = L.map('map').setView([23.7337, 90.3929], 15); var tileOptions = [
{ url: 'https://{s}.tile.openstreetmap.org/{z}/{x}/{y}.png', attr: 'OpenStreetMap' },
{ url: 'https://{s}.basemaps.cartocdn.com/rastertiles/voyager/{z}/{x}/{y}.png', attr: 'CARTO' },
{ url: 'https://server.arcgisonline.com/ArcGIS/rest/services/World_Street_Map/MapServer/tile/{z}/{y}/{x}', attr: 'Esri' }
];
var tileLayer; var tileIndex = 0;
function tryTiles() {
if (tileIndex >= tileOptions.length) {
statusEl.innerText = 'ERROR: No map tile server reachable.'; statusEl.style.color = '#ff4444';
return;
}
var opt = tileOptions[tileIndex];
if (tileLayer) { map.removeLayer(tileLayer); }
tileLayer = L.tileLayer(opt.url, { attribution: opt.attr, maxZoom: 19 }); var loadedOne = false;
tileLayer.on('tileload', function() { if (!loadedOne) {
loadedOne = true;
statusEl.innerText = 'Map tiles loaded via ' + opt.attr; statusEl.style.color = '#44ff44';
}
});
tileLayer.on('tileerror', function() { if (!loadedOne) {
statusEl.innerText = opt.attr + ' failed, trying next...';
tileIndex++; setTimeout(tryTiles, 800);
}
});
tileLayer.addTo(map);
}
tryTiles();
var redIcon = L.divIcon({ className: 'fault-marker', html:
'<div
style="background:#ff0000;width:20px;height:20px;border-radius:50%;border:3px solid #fff;box-shadow:0 0
10px #ff0000;"></div>',
iconSize: [26, 26],
iconAnchor: [13, 13]
});
var marker = L.marker([23.7337, 90.3929], { icon: redIcon }).addTo(map); function refresh() {
fetch('/data').then(r => r.json()).then(d => { marker.setLatLng([d.lat, d.lng]); map.panTo([d.lat, d.lng]);
marker.bindPopup('<b>'+d.feeder+'</b><br>Fault: '+d.fault).openPopup();

```

```

document.getElementById('feeder').innerText = d.feeder;
var f = document.getElementById('fault'); f.innerText = d.fault;
f.className = (d.fault === 'NONE') ? 'normal' : 'alert'; document.getElementById('voltage').innerText =
d.voltage.toFixed(1); document.getElementById('current').innerText = d.current.toFixed(1);
document.getElementById('coords').innerText = d.lat.toFixed(5)+' '+d.lng.toFixed(5);
}).catch(e => { statusEl.innerText = 'Data fetch failed: ' + e.message; });
}
setInterval(refresh, 2000); refresh();
}
</script>
</body>
</html>
)rawliteral";

// ===== SETUP =====
void setup() { Serial.begin(9600); delay(500); randomSeed(analogRead(A0)); bootMillis = millis();
Serial.println();
Serial.println("==== Booting Receiver (Static IP) with DPDC & PROTEUS independent data
====");
Serial.print("Requested static IP: "); Serial.println(local_IP);
WiFi.mode(WIFI_STA);
if (!WiFi.config(local_IP, gateway, subnet, primaryDNS, secondaryDNS)) { Serial.println("WARNING: Static
IP config failed (will fall back to DHCP)");
} else {
Serial.println("Static IP configured.");
}
Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass); Serial.println("Blynk connected."); Serial.print("Actual IP:
"); Serial.println(WiFi.localIP()); Serial.print("Gateway: "); Serial.println(WiFi.gatewayIP());
Serial.print("RSSI: "); Serial.print(WiFi.RSSI());
Serial.println(" dBm");
configTime(gmtOffset_sec, daylightOffset_sec, ntpServer1, ntpServer2, ntpServer3); Serial.println("NTP
request sent (will sync in background).");
server.on("/", handleRoot); server.on("/data", handleData); server.begin();
Serial.println();
Serial.print(">>> Open this URL in your browser: http://"); Serial.println(WiFi.localIP());
Serial.println();
}
// ===== MAIN LOOP =====
void loop() { Blynk.run(); server.handleClient();
// Background NTP retry every 30 s
if (!ntpReady && (millis() - lastNtpRetry > 30000)) { lastNtpRetry = millis();
time_t now = time(nullptr); if (now > 8 * 3600 * 2) { ntpReady = true;
struct tm* t = localtime(&now);
Serial.printf(">>> NTP synced: %02d:%02d:%02d <<<\n", t->tm_hour, t->tm_min, t->tm_sec);
} else {
Serial.println("NTP not yet synced, will retry...");
configTime(gmtOffset_sec, daylightOffset_sec, ntpServer1, ntpServer2, ntpServer3);
}
}
static unsigned long lastTime = 0; if (millis() - lastTime > 3000) { lastTime = millis();
generateRandomDataset();
}
}
// ===== WEB SERVER HANDLERS =====
void handleRoot() {
server.send_P(200, "text/html", mapPage);
}
void handleData() { String json = "{";
json += "\"lat\":" + String(currentLat, 6) + ","; json += "\"lng\":" +
String(currentLng, 6) + ","; json += "\"feeder\":" + currentFeeder
+ "\""; json += "\"fault\":" + currentFault + "\""; json += "\"voltage\":" +
String(currentVoltage, 1) + ","; json += "\"current\":" + String(currentCurrent, 1);
json += "}";
server.send(200, "application/json", json);

```

```

}
// ===== UTILITY FUNCTIONS =====
int faultToSeverity(const String& fault) { if (fault == "L-G") return 1;
if (fault == "L-L") return 2; if (fault == "L-L-G") return 3; if (fault == "3PH") return 4; return 0;
}
String getTimeString() { if (ntpReady) {
time_t now = time(nullptr); struct tm* t = localtime(&now); char buf[16];
snprintf(buf, sizeof(buf), "%02d:%02d:%02d", t->tm_hour, t->tm_min, t->tm_sec); return String(buf);
} else {
unsigned long s = (millis() - bootMillis) / 1000; unsigned int hh = (s / 3600) % 24;
unsigned int mm = (s / 60) % 60; unsigned int ss = s % 60;
char buf[16];
snprintf(buf, sizeof(buf), "%02u:%02u:%02u", hh, mm, ss); return String(buf);
}
}
String getDayString() { if (ntpReady) {
time_t now = time(nullptr); struct tm* t = localtime(&now);
const char* days[] = {"Sun", "Mon", "Tue", "Wed", "Thu", "Fri", "Sat"}; return String(days[t->tm_wday]);
} else {
return "Day" + String(((millis() - bootMillis) / 86400000UL) + 1);
}
}
// ===== MAIN DATA GENERATOR =====
void generateRandomDataset() {
// ===== 1. DPDC side (completely independent) ===== float dpdcVoltage;
int vRoll = random(100); if (vRoll < 35)
dpdcVoltage = 5.0 + random(0, 401) / 100.0; else if (vRoll < 55)
dpdcVoltage = 9.0 + random(0, 151) / 100.0; else
dpdcVoltage = 10.5 + random(0, 101) / 100.0; dpdcVoltage = round(dpdcVoltage * 10) / 10.0;
float dpdcCurrent;
int cRoll = random(100);
if (cRoll < 30) dpdcCurrent = 10.0 + random(0, 701) / 10.0;
else if (cRoll < 60) dpdcCurrent = 80.0 + random(0, 1701) / 10.0;
else if (cRoll < 85) dpdcCurrent = 250.0 + random(0, 3501) / 10.0;
else dpdcCurrent = 600.0 + random(0, 6001) / 10.0; dpdcCurrent = round(dpdcCurrent * 10) / 10.0;
String dpdcFault;
int faultRoll = random(100); if (dpdcVoltage >= 10.5) {
if (faultRoll < 10) dpdcFault = "L-G"; else dpdcFault = "NONE";
} else if (dpdcVoltage >= 9.0) {
if (faultRoll < 15) dpdcFault = "L-G"; else if (faultRoll < 25) dpdcFault = "L-L"; else dpdcFault = "NONE";
} else if (dpdcVoltage >= 7.0) {
if (faultRoll < 20) dpdcFault = "L-G"; else if (faultRoll < 40) dpdcFault = "L-L";
else if (faultRoll < 55) dpdcFault = "L-L-G"; else dpdcFault = "NONE";
} else {
if (faultRoll < 15) dpdcFault = "L-G";
else if (faultRoll < 35) dpdcFault = "L-L"; else if (faultRoll < 65) dpdcFault = "L-L-G"; else dpdcFault = "3PH";
}
int dpdcDuration = 0;
if (dpdcFault != "NONE") {
int sev = faultToSeverity(dpdcFault);
if (sev == 1) dpdcDuration = random(1, 16); else if (sev == 2) dpdcDuration = random(5, 31); else if (sev
== 3) dpdcDuration = random(10, 46); else if (sev == 4) dpdcDuration = random(20, 61);
}

// ===== 2. Proteus side - independent realistic fault ranges ===== int proteusType =
random(100);
String proteusFault;
float proteusVoltage, proteusCurrent; int proteusDuration = 0;
if (proteusType < 30) { // 30% No Fault proteusFault = "NONE";
proteusVoltage = 11.0 + random(-10, 11) / 100.0; // 10.9 - 11.1 kV
proteusCurrent = 0.0 + random(0, 51) / 10.0; // 0 - 5 A
proteusDuration = 0;
}
else if (proteusType < 50) { // 20% L-G proteusFault = "L-G";

```

```

proteusVoltage = 9.0 + random(0, 151) / 100.0; // 9.0 - 10.5 kV
proteusCurrent = 50.0 + random(0, 1501) / 10.0; // 50 - 200 A proteusDuration = random(1, 16);
}
else if (proteusType < 70)
{
    // 20% L-L proteusFault = "L-L";
    proteusVoltage = 8.0 + random(0, 201) / 100.0; // 8.0 - 10.0 kV
    proteusCurrent = 150.0 + random(0, 2501) / 10.0; // 150 - 400 A proteusDuration = random(5, 31);
}
else if (proteusType < 85) { // 15% L-L-G proteusFault = "L-L-G";
    proteusVoltage = 7.0 + random(0, 201) / 100.0; // 7.0 - 9.0 kV
    proteusCurrent = 250.0 + random(0, 3501) / 10.0; // 250 - 600 A proteusDuration = random(10, 46);
}
else { // 15% 3PH proteusFault = "3PH";
    proteusVoltage = 6.0 + random(0, 201) / 100.0; // 6.0 - 8.0 kV
    proteusCurrent = 500.0 + random(0, 7001) / 10.0; // 500 - 1200 A proteusDuration = random(20, 61);
}
// Add small jitter to avoid identical repeats proteusVoltage += (random(-5, 6) / 100.0);
proteusCurrent += (random(-10, 11) / 10.0);
proteusVoltage = constrain(proteusVoltage, 5.0, 12.0);
proteusCurrent = constrain(proteusCurrent, 0.0, 1300.0); proteusVoltage = round(proteusVoltage * 10) / 10.0;
proteusCurrent = round(proteusCurrent * 10) / 10.0;
// 3. GPS & Feeder (shared for map - using DPDC values) ===== float lat = 23.70 + random(0, 501) /
10000.0;
float lng = 90.38 + random(0, 401) / 10000.0;
lat += random(-30, 31) / 100000.0;
lng += random(-30, 31) / 100000.0; int feederIdx = random(numFeeders);
String feederName = String(feederNames[feederIdx]);
// Update global variables for web map (show DPDC data) currentLat = lat;
currentLng = lng; currentFeeder = feederName; currentFault = dpdcFault; currentVoltage = dpdcVoltage;
currentCurrent = dpdcCurrent;
// ===== 4. Serial output (show both) ===== Serial.println("----- DPDC vs PROTEUS
(Independent Faults) ");
Serial.print("Feeder: "); Serial.println(feederName);
Serial.print("DPDC: V="); Serial.print(dpdcVoltage,1); Serial.print("kV, I="); Serial.print(dpdcCurrent,1);
Serial.print("A, Fault="); Serial.print(dpdcFault); Serial.print(", Dur="); Serial.println(dpdcDuration);
Serial.print("PROTEUS: V="); Serial.print(proteusVoltage,1); Serial.print("kV, I=");
Serial.print(proteusCurrent,1); Serial.print("A, Fault="); Serial.print(proteusFault); Serial.print(", Dur=");
Serial.println(proteusDuration);
Serial.print("GPS: "); Serial.print(lat,6); Serial.print(", "); Serial.println(lng,6); Serial.print("Time: ");
Serial.print(getTimeString()); Serial.print(" ("); Serial.print(ntpReady?"NTP":"uptime"); Serial.println(")");
Serial.println(" ");
// ===== 5. Blynk writes ===== int dayNum = (dayIndex % 7) + 1;
int statusDPDC = (dpdcFault != "NONE" || dpdcVoltage < 10.0 || dpdcVoltage > 11.5 || dpdcCurrent > 500) ? 1 :
0;
// DPDC values (V1-V6) Blynk.virtualWrite(V0, dayNum); Blynk.virtualWrite(V1, dpdcVoltage);
Blynk.virtualWrite(V2, dpdcFault); Blynk.virtualWrite(V3, feederName); Blynk.virtualWrite(V4,
dpdcDuration);
Blynk.virtualWrite(V5, statusDPDC); Blynk.virtualWrite(V6, dpdcCurrent);

// Proteus values (V7-V10) - completely independent Blynk.virtualWrite(V7, proteusVoltage);
Blynk.virtualWrite(V8, proteusFault); Blynk.virtualWrite(V9, proteusCurrent); Blynk.virtualWrite(V10,
proteusDuration);
// Optional extras (map URL, lat/lng, severity)
String mapUrl = "https://maps.google.com/?q=" + String(lat,6) + "," + String(lng,6); Blynk.virtualWrite(V13,
lat);
Blynk.virtualWrite(V14, lng); Blynk.virtualWrite(V15, mapUrl); Blynk.virtualWrite(V16,
faultToSeverity(dpdcFault));
// Timestamp and decimal hour (for V11, V12) String timeStr = getTimeString();
String dayStr = getDayString(); if (dpdcFault != "NONE") { float decimalHour = 0.0;
if (ntpReady) {
    time_t now = time(nullptr); struct tm* t = localtime(&now);
    decimalHour = t->tm_hour + t->tm_min/60.0 + t->tm_sec/3600.0;
} else {
    unsigned long s = (millis()-bootMillis)/1000; decimalHour = (s % 86400)/3600.0;
}
}

```

```

}
Blynk.virtualWrite(V11, dayStr + " " + timeStr); Blynk.virtualWrite(V12, decimalHour);
} else {
Blynk.virtualWrite(V11, dayStr + " " + timeStr + " (no fault)");
}
}
dayIndex++;
}
}

```

Experimental Setup (Hardware):

a) DC Setup Codes:

i) ESP32E sender:

```

/*
=====
ESP32 IoT FPI — FINAL DIRECT FIREBASE + NEO-6M VERSION (GPS BUFFER FIX)
=====
ESP32E:
- Uses the existing FPI experimental/test dataset
- Sends directly to Firebase Realtime Database
- Reads NEO-6M GPS directly
- No ESP8266
- No Blynk
ESP32 Core 3.x
Libraries:
- Firebase ESP Client by Mobizt
- TinyGPSPlus
=====
*/
#include <WiFi.h>
#include <FirebaseESP32.h>
#include <TinyGPSPlus.h>
//=====
// WIFI
//=====
const char* WIFI_SSID = "Bankcolony";
const char* WIFI_PASS = "Bank1122";
//=====
// FIREBASE
//=====
#define FIREBASE_HOST "https://fault-passage-indicator-default-rtdb.firebaseio.com/"
#define FIREBASE_AUTH "vP8FyTBRn9CEh0kSqVUSUowlg95BCv2IEWJXY3Uu"
FirebaseData fbdo;
FirebaseAuth auth;
FirebaseConfig config;
//=====
// NEO-6M GPS
// ESP32 UART2
//
// GPS TX -> ESP32 GPIO16 (RX2)
// GPS RX -> ESP32 GPIO17 (TX2) [optional]
// GPS VCC -> 3.3V
// GPS GND -> common GND
//=====
#define GPS_RX_PIN 16
#define GPS_TX_PIN 17
#define GPS_BAUD 9600
//=====
// RYG LED INDICATORS
//=====
#define GREEN_LED 25 // SLG
#define YELLOW_LED 26 // LL and DLG

```

```

#define RED_LED 27 // 3-PHASE
HardwareSerial GPSSerial(2);
TinyGPSPlus gps;
// Firebase requests can temporarily block the main loop.
// A larger UART RX buffer prevents NEO-6M NMEA data from being lost.
#define GPS_RX_BUFFER_SIZE 4096
//=====
// WCS1800 CONFIG
// NOTE: useRealSensor remains FALSE because your DC data
// are already collected and the WCS1800 was removed from
// the current setup.
//=====
#define WCS1800_PIN 34
#define WCS1800_VCC 5.0
#define ADC_RESOLUTION 4095.0
#define ZERO_CURRENT_V 2.5
#define SENSITIVITY 0.066
#define ADC_OFFSET 0
#define NUM_SAMPLES 20
//=====
// DATA STRUCTURE
//=====
typedef struct __attribute__((packed)) {
    float psuVoltage;
    float current;
    char load[40];
    char fault[20];
    float faultTime;
} FaultPacket;
FaultPacket packet;
//=====
// EXPERIMENTAL DATA TABLE
// YOUR EXISTING DC TEST DATA — KEPT UNCHANGED
//=====
FaultPacket testData[] = {
    // CALIBRATION
    { 0, 0, "No load (calibration)", "NORMAL", 0 },
    { 0, 0, "No load (calibration)", "NORMAL", 0 },
    { 0, 0, "No load (calibration)", "NORMAL", 0 },
    // NORMAL (1-PHASE)
    { 6, 0.530, "1x100W 10R resistor", "NORMAL", 0 },
    { 6, 1.055, "1x100W 10R resistor", "NORMAL", 0 },
    { 6, 0.856, "1x100W 10R resistor", "NORMAL", 0 },
    // NORMAL (2-PHASE)
    { 10, 0.539, "2x100W 10R Parallel", "NORMAL", 0 },
    { 10, 0.140, "2x100W 10R Parallel", "NORMAL", 0 },
    { 10, 0.140, "2x100W 10R Parallel", "NORMAL", 0 },
    // NORMAL (DC MOTOR)
    { 10, 0.145, "24V 250W DC Motor", "NORMAL", 0 },
    { 10, 2.165, "24V 250W DC Motor", "NORMAL", 0 },
    { 10, 1.503, "24V 250W DC Motor", "NORMAL", 0 },
    // NORMAL (3 RESISTORS)
    { 8, 1.275, "3x100W 10R Parallel", "NORMAL", 0 },
    { 8, 0.037, "3x100W 10R Parallel", "NORMAL", 0 },
    { 8, 1.746, "3x100W 10R Parallel", "NORMAL", 0 },
    // SLG FAULT
    { 6, 1.691, "1x100W 10R resistor", "SLG", 0.85 },
    { 6, 1.288, "1x100W 10R resistor", "SLG", 1.20 },
    { 6, 1.268, "1x100W 10R resistor", "SLG", 1.45 },
    { 6, 1.269, "1x100W 10R resistor", "SLG", 0.95 },
    { 6, 1.348, "1x100W 10R resistor", "SLG", 1.10 },
    { 10, 1.620, "1x100W 10R resistor", "SLG", 0.65 },
    { 10, 1.546, "1x100W 10R resistor", "SLG", 0.78 },
    { 10, 1.433, "1x100W 10R resistor", "SLG", 1.35 },

```

```

{ 8, 1.487, "1x100W 10R resistor", "SLG", 1.05 },
{ 8, 1.227, "1x100W 10R resistor", "SLG", 1.80 },
{ 8, 1.311, "1x100W 10R resistor", "SLG", 1.55 },
{ 12, 1.630, "1x100W 10R resistor", "SLG", 0.55 },
{ 12, 1.710, "1x100W 10R resistor", "SLG", 0.42 },
{ 5, 1.503, "1x100W 10R resistor", "SLG", 1.25 },
{ 5, 1.240, "1x100W 10R resistor", "SLG", 2.10 },
// LL FAULT
{ 10, 1.156, "2x100W 10R Parallel", "LL", 0.95 },
{ 10, 1.187, "2x100W 10R Parallel", "LL", 0.88 },
{ 10, 0.969, "2x100W 10R Parallel", "LL", 1.45 },
{ 10, 1.193, "2x100W 10R Parallel", "LL", 0.82 },
{ 10, 1.018, "2x100W 10R Parallel", "LL", 1.20 },
{ 8, 0.970, "2x100W 10R Parallel", "LL", 1.65 },
{ 8, 1.235, "2x100W 10R Parallel", "LL", 0.75 },
{ 8, 1.240, "2x100W 10R Parallel", "LL", 0.72 },
{ 6, 1.152, "2x100W 10R Parallel", "LL", 1.10 },
{ 6, 1.026, "2x100W 10R Parallel", "LL", 1.55 },
{ 6, 0.974, "2x100W 10R Parallel", "LL", 1.85 },
{ 12, 1.342, "2x100W 10R Parallel", "LL", 0.55 },
{ 12, 1.220, "2x100W 10R Parallel", "LL", 0.68 },
{ 5, 0.974, "2x100W 10R Parallel", "LL", 2.20 },
{ 5, 1.049, "2x100W 10R Parallel", "LL", 1.75 },
// DLG FAULT
{ 10, 1.365, "2x100W 10R Parallel", "DLG", 0.72 },
{ 10, 1.759, "2x100W 10R Parallel", "DLG", 0.45 },
{ 10, 1.466, "2x100W 10R Parallel", "DLG", 0.68 },
{ 10, 1.648, "2x100W 10R Parallel", "DLG", 0.52 },
{ 10, 1.490, "2x100W 10R Parallel", "DLG", 0.65 },
{ 8, 1.532, "2x100W 10R Parallel", "DLG", 0.58 },
{ 8, 1.541, "2x100W 10R Parallel", "DLG", 0.55 },
{ 8, 1.415, "2x100W 10R Parallel", "DLG", 0.85 },
{ 12, 1.982, "2x100W 10R Parallel", "DLG", 0.32 },
{ 12, 1.865, "2x100W 10R Parallel", "DLG", 0.38 },
{ 12, 1.964, "2x100W 10R Parallel", "DLG", 0.35 },
{ 6, 1.574, "2x100W 10R Parallel", "DLG", 0.95 },
{ 6, 1.499, "2x100W 10R Parallel", "DLG", 1.15 },
{ 14, 2.145, "2x100W 10R Parallel", "DLG", 0.28 },
{ 14, 1.562, "2x100W 10R Parallel", "DLG", 0.42 },
// 3-PHASE FAULT (3 RESISTORS)
{ 8, 2.838, "3x100W 10R Parallel", "3-PHASE", 0.18 },
{ 8, 2.770, "3x100W 10R Parallel", "3-PHASE", 0.22 },
{ 8, 2.896, "3x100W 10R Parallel", "3-PHASE", 0.15 },
{ 8, 2.925, "3x100W 10R Parallel", "3-PHASE", 0.14 },
{ 8, 2.872, "3x100W 10R Parallel", "3-PHASE", 0.16 },
{ 10, 3.380, "3x100W 10R Parallel", "3-PHASE", 0.08 },
{ 10, 3.050, "3x100W 10R Parallel", "3-PHASE", 0.12 },
{ 10, 2.997, "3x100W 10R Parallel", "3-PHASE", 0.13 },
{ 6, 2.943, "3x100W 10R Parallel", "3-PHASE", 0.20 },
{ 6, 2.800, "3x100W 10R Parallel", "3-PHASE", 0.25 },
// 3-PHASE FAULT (24V DC MOTOR)
{ 8, 3.320, "24V 250W DC Motor", "3-PHASE", 0.10 },
{ 8, 2.898, "24V 250W DC Motor", "3-PHASE", 0.14 },
{ 8, 3.592, "24V 250W DC Motor", "3-PHASE", 0.07 },
{ 10, 3.075, "24V 250W DC Motor", "3-PHASE", 0.11 },
{ 10, 3.987, "24V 250W DC Motor", "3-PHASE", 0.05 }
};
const int TOTAL_RECORDS = sizeof(testData) / sizeof(testData[0]);
//=====
// OPTIONS
//=====
// FALSE = use your already-collected DC dataset.
// TRUE  = use WCS1800 live sensor.
// Keep FALSE for the present GPS + Firebase test.

```

```

bool useRealSensor = false;
// Send interval
const unsigned long SEND_INTERVAL = 1000;
//=====
// RUNTIME VARIABLES
//=====
int recordIndex = 0;
unsigned long packetsSent = 0;
unsigned long startTime = 0;
unsigned long lastSend = 0;
unsigned long faultCount = 0;
String previousFault = "NORMAL";
//=====
// GPS VARIABLES
//=====
double gpsLat = 0.0;
double gpsLng = 0.0;
unsigned long gpsSatellites = 0;
bool gpsFix = false;
//=====
// WCS1800 FUNCTIONS
//=====
float readRawVoltage() {
  int raw = analogRead(WCS1800_PIN);
  return (raw / ADC_RESOLUTION) * WCS1800_VCC;
}
float readCurrent() {
  long sum = 0;
  for (int i = 0; i < NUM_SAMPLES; i++) {
    sum += analogRead(WCS1800_PIN);
    delayMicroseconds(100);
  }
  float avgRaw = sum / (float)NUM_SAMPLES;
  float voltage = (avgRaw / ADC_RESOLUTION) * WCS1800_VCC;
  float current = (voltage - ZERO_CURRENT_V) / SENSITIVITY;
  current += (ADC_OFFSET / 1000.0);
  if (fabs(current) < 0.05)
    current = 0.0;
  return current;
}
float readPSUVoltage() {
  return 0.0;
}
String detectFault(float current, float voltage) {
  if (current < 0.1 && voltage > 2.0)
    return "NORMAL";
  if (current > 2.5)
    return "3-PHASE";
  if (current > 1.5)
    return "DLG";
  if (current > 1.0)
    return "LL";
  if (current > 0.5)
    return "SLG";
  return "NORMAL";
}
//=====
// GPS PROCESSING
//=====
void processGPS() {
  // Consume every available NMEA byte.
  while (GPSSerial.available() > 0) {
    gps.encode(GPSSerial.read());
  }
}

```

```

// Once a valid fix has been obtained, retain the last valid
// coordinates during short UART/Firebase interruptions.
if (gps.location.isValid()) {
    gpsLat = gps.location.lat();
    gpsLng = gps.location.lng();
}
if (gps.satellites.isValid())
    gpsSatellites = gps.satellites.value();
// A valid TinyGPS++ location is considered a fix while it is
// reasonably fresh. The last coordinates are NOT erased.
if (gps.location.isValid() && gps.location.age() < 5000)
    gpsFix = true;
else
    gpsFix = false;
}
void printGPS() {
    Serial.print("[GPS] ");
    // TinyGPS++ exposes how many characters it has successfully
    // processed. This helps distinguish UART failure from no-fix.
    if (gps.charsProcessed() < 10) {
        Serial.print("NO NMEA DATA | chars=");
        Serial.println(gps.charsProcessed());
        return;
    }
    if (gpsFix) {
        Serial.print("FIX | Sats: ");
        Serial.print(gpsSatellites);
        Serial.print(" | Lat: ");
        Serial.print(gpsLat, 6);
        Serial.print(" | Lng: ");
        Serial.println(gpsLng, 6);
    } else {
        Serial.print("NO FIX | Sats: ");
        Serial.print(gpsSatellites);
        Serial.print(" | Lat: ");
        Serial.print(gpsLat, 6);
        Serial.print(" | Lng: ");
        Serial.println(gpsLng, 6);
    }
}
//=====
// WIFI RECONNECTION
//=====
void maintainWiFi() {
    if (WiFi.status() == WL_CONNECTED)
        return;
    Serial.println("[WiFi] Connection lost. Reconnecting...");
    WiFi.disconnect();
    WiFi.begin(WIFI_SSID, WIFI_PASS);
    unsigned long t0 = millis();
    while (WiFi.status() != WL_CONNECTED && millis() - t0 < 10000) {
        processGPS();
        delay(100);
    }
    if (WiFi.status() == WL_CONNECTED) {
        Serial.print("[WiFi] Reconnected! IP: ");
        Serial.println(WiFi.localIP());
    } else {
        Serial.println("[WiFi] Reconnection failed. Will retry.");
    }
}
//=====
// FIREBASE SEND
//

```

```

// The original /fpi/live and /fpi/history paths are kept.
// V0-V12 are also included so the dashboard can use the
// same datastream naming.
//=====
bool sendToFirebase() {
  if (WiFi.status() != WL_CONNECTED)
    return false;
  if (!Firebase.ready()) {
    Serial.println("[Firebase] Not ready.");
    return false;
  }
  unsigned long now = millis();
  // Fault counter: count a fault only when entering a
  // non-NORMAL state, not every packet during the fault.
  if (String(packet.fault) != "NORMAL" && previousFault == "NORMAL") {
    faultCount++;
  }
  previousFault = String(packet.fault);
  // Terminal text
  String terminal =
    "V0=" + String(packet.psuVoltage, 2) + "V | V1=" + String(packet.current, 3) + "A | V4=" +
    String(packet.fault) + " | GPS=" + (gpsFix ? "FIX" : "NO FIX");
  //=====
  // LIVE DATA
  //=====
  FirebaseJson json;
  // Original field names
  json.set("psuVoltage", packet.psuVoltage);
  json.set("current", packet.current);
  json.set("load", packet.load);
  json.set("fault", packet.fault);
  json.set("faultTime", packet.faultTime);
  json.set("timestamp", (int)now);
  json.set("packetNumber", (int)packetsSent);
  // Dashboard datastreams V0-V12
  json.set("V0", packet.psuVoltage);
  json.set("V1", packet.current);
  json.set("V2", packet.faultTime);
  json.set("V3", (int)packetsSent);
  json.set("V4", packet.fault);
  json.set("V5", packet.load);
  json.set("V6", (int)now);
  json.set("V7", packet.faultTime);
  json.set("V8", packet.current);
  json.set("V9", packet.psuVoltage);
  json.set("V10", terminal);
  json.set("V11", (int)faultCount);
  json.set("V12", (int)(now / 1000UL));
  // GPS
  json.set("gps/fix", gpsFix);
  json.set("gps/latitude", gpsLat);
  json.set("gps/longitude", gpsLng);
  json.set("gps/satellites", (int)gpsSatellites);
  bool liveOK = Firebase.setJSON(fbdo, "/fpi/live", json);
  if (liveOK)
    Serial.println("[Firebase] Live data sent OK");
  else {
    Serial.print("[Firebase] Live Error: ");
    Serial.println(fbdo.errorReason());
  }
  //=====
  // HISTORY
  //=====
  FirebaseJson histJson;

```

```

histJson.set("psuVoltage", packet.psuVoltage);
histJson.set("current", packet.current);
histJson.set("load", packet.load);
histJson.set("fault", packet.fault);
histJson.set("faultTime", packet.faultTime);
histJson.set("timestamp", (int)now);
histJson.set("packetNumber", (int)packetsSent);
histJson.set("V0", packet.psuVoltage);
histJson.set("V1", packet.current);
histJson.set("V2", packet.faultTime);
histJson.set("V3", (int)packetsSent);
histJson.set("V4", packet.fault);
histJson.set("V5", packet.load);
histJson.set("V6", (int)now);
histJson.set("V7", packet.faultTime);
histJson.set("V8", packet.current);
histJson.set("V9", packet.psuVoltage);
histJson.set("V10", terminal);
histJson.set("V11", (int)faultCount);
histJson.set("V12", (int)(now / 1000UL));
histJson.set("gps/fix", gpsFix);
histJson.set("gps/latitude", gpsLat);
histJson.set("gps/longitude", gpsLng);
histJson.set("gps/satellites", (int)gpsSatellites);
bool historyOK = Firebase.pushJSON(fbdo, "/fpi/history", histJson);
if (historyOK)
    Serial.println("[Firebase] History pushed OK");
else {
    Serial.print("[Firebase] History Error: ");
    Serial.println(fbdo.errorReason());
}
return liveOK;
}
//=====
// RYG LED CONTROL
//=====
void updateFaultLEDs(const char* fault) {
    // Always start by turning all LEDs OFF
    digitalWrite(GREEN_LED, LOW);
    digitalWrite(YELLOW_LED, LOW);
    digitalWrite(RED_LED, LOW);
    // SLG -> GREEN
    if (strcmp(fault, "SLG") == 0) {
        digitalWrite(GREEN_LED, HIGH);
    }
    // LL or DLG -> YELLOW
    else if (strcmp(fault, "LL") == 0 || strcmp(fault, "DLG") == 0) {
        digitalWrite(YELLOW_LED, HIGH);
    }
    // 3-PHASE -> RED
    else if (strcmp(fault, "3-PHASE") == 0) {
        digitalWrite(RED_LED, HIGH);
    }
    // NORMAL -> all LEDs remain OFF
}
//=====
// SETUP
//=====
void setup() {
    Serial.begin(115200);
    delay(1000);
    //=====
    // RYG LED INITIALIZATION
    //=====

```

```

pinMode(GREEN_LED, OUTPUT);
pinMode(YELLOW_LED, OUTPUT);
pinMode(RED_LED, OUTPUT);
// Start with all LEDs OFF
digitalWrite(GREEN_LED, LOW);
digitalWrite(YELLOW_LED, LOW);
digitalWrite(RED_LED, LOW);
// ADC
analogReadResolution(12);
analogSetAttenuation(ADC_11db);
// GPS UART2
// IMPORTANT: set the larger RX buffer BEFORE begin().
// This prevents Firebase network operations from overflowing
// the UART buffer while the NEO-6M continues transmitting.
GPSSerial.setRxBufferSize(GPS_RX_BUFFER_SIZE);
GPSSerial.begin(
  GPS_BAUD,
  SERIAL_8N1,
  GPS_RX_PIN,
  GPS_TX_PIN);
Serial.println();
Serial.println("=====");
Serial.println(" FPI — ESP32E + NEO-6M + FIREBASE");
Serial.println(" Direct Firebase version");
Serial.println(" No ESP8266 / No Blynk");
Serial.println("=====");
Serial.println("[GPS] Wiring:");
Serial.println("  NEO-6M TX -> ESP32 GPIO16 (RX2)");
Serial.println("  NEO-6M RX -> ESP32 GPIO17 (TX2) optional");
Serial.println("  NEO-6M VCC -> ESP32 3.3V");
Serial.println("  NEO-6M GND -> common GND");
Serial.println();
// WiFi
WiFi.mode(WIFI_STA);
WiFi.begin(WIFI_SSID, WIFI_PASS);
Serial.print("[WiFi] Connecting");
unsigned long wifiStart = millis();
while (WiFi.status() != WL_CONNECTED && millis() - wifiStart < 20000) {
  processGPS();
  delay(100);
  Serial.print(".");
}
Serial.println();
if (WiFi.status() == WL_CONNECTED) {
  Serial.print("[WiFi] Connected! IP: ");
  Serial.println(WiFi.localIP());
} else {
  Serial.println("[WiFi] Initial connection failed.");
  Serial.println("[WiFi] Code will keep retrying.");
}
// Firebase
config.database_url = FIREBASE_HOST;
config.signer.tokens.legacy_token = FIREBASE_AUTH;
Firebase.begin(&config, &auth);
Firebase.reconnectWiFi(true);
Serial.println("[Firebase] Initialized");
// Dataset start
startTime = millis();
lastSend = 0;
Serial.print("[DATA] Total Records: ");
Serial.println(TOTAL_RECORDS);
Serial.println("[GPS] Listening for NMEA data...");
Serial.println("[Firebase] Transmission starting...");
Serial.println();

```

```

}
//=====
// LOOP
//=====
void loop() {
  // GPS must be processed continuously.
  processGPS();
  // Keep WiFi alive without blocking GPS for long periods.
  maintainWiFi();
  // Send once every SEND_INTERVAL.
  if (millis() - lastSend >= SEND_INTERVAL) {
    lastSend = millis();
    //=====
    // DATA SOURCE
    //=====
    if (useRealSensor) {
      packet.current = readCurrent();
      packet.psuVoltage = readPSUVoltage();
      String fault =
        detectFault(packet.current, packet.psuVoltage);
      strcpy(packet.load, "WCS1800 Live Sensor");
      strcpy(packet.fault, fault.c_str());
      packet.faultTime = 0;
    } else {
      // Existing tested DC dataset
      packet = testData[recordIndex];
    }
    //=====
    // UPDATE RYG LED
    //=====
    updateFaultLEDs(packet.fault);
    //=====
    // SEND
    //=====
    // Drain any GPS bytes that arrived while preparing the packet.
    processGPS();
    bool sent = sendToFirebase();
    // Firebase operations can take time. Drain the GPS UART again
    // immediately after the network transaction.
    processGPS();
    //=====
    // SERIAL MONITOR
    //=====
    Serial.println();
    Serial.println("=====");
    Serial.println("      FPI LIVE MONITORING");
    Serial.println("=====");
    Serial.print("PSU Voltage : ");
    Serial.print(packet.psuVoltage, 3);
    Serial.println(" V");
    Serial.print("Current   : ");
    Serial.print(packet.current, 3);
    Serial.println(" A");
    Serial.print("Load       : ");
    Serial.println(packet.load);
    Serial.print("Fault Status: ");
    Serial.println(packet.fault);
    Serial.print("Fault Time : ");
    Serial.print(packet.faultTime, 2);
    Serial.println(" s");
    Serial.print("Packet #   : ");
    Serial.println(packetsSent);
    Serial.print("Fault Count : ");
    Serial.println(faultCount);
  }
}

```

```

printGPS();
Serial.print("[Firebase] ");
Serial.println(sent ? "SUCCESS" : "NOT SENT");
Serial.println("=====");
Serial.println();
packetsSent++;
// Advance dataset
if (!useRealSensor) {
  recordIndex++;
  if (recordIndex >= TOTAL_RECORDS) {
    recordIndex = 0;
    Serial.println("=====");
    Serial.println(" END OF DATASET");
    Serial.println(" Restarting from first record...");
    Serial.println("=====");
  }
}
}
// Small yield so ESP32/WiFi watchdog remains happy.
delay(5);
}

```

ii) ESP8266 receiver:

```

/*
=====
ESP8266 + Firebase + FPI Map Integration + NEO-6M GPS
FPI RECEIVER — FIREBASE VERSION v1
SSID: Tenda_B6B920 | Password: boikhatakolom2
GPS: NEO-6M on D5(RX) / D6(TX)
Reads from Firebase and displays on web dashboard
=====
*/

#include <ESP8266WiFi.h>
#include <FirebaseESP8266.h>
#include <time.h>
#include <SoftwareSerial.h>
#include <TinyGPS++.h>
//=====
// WIFI CONFIG
//=====
const char *WIFI_SSID = "Redmi Note 10";
const char *WIFI_PASS = "12345678";
//=====
// FIREBASE CONFIG
// REPLACE WITH YOUR FIREBASE PROJECT DETAILS!
//=====
#define FIREBASE_HOST "https://fault-passage-indicator-default-rtdb.firebaseio.com/"
#define FIREBASE_AUTH "vP8FyTBRn9CEh0kSqVUSUowIg95BCv2IEWJXY3Uu"
FirebaseData firebaseData;
FirebaseAuth firebaseAuth;
FirebaseConfig firebaseConfig;
//=====
// GPS CONFIG — NEO-6M
//=====
#define GPS_RX_PIN D6
#define GPS_TX_PIN D5
#define GPS_BAUD 9600
SoftwareSerial gpsSerial(GPS_RX_PIN, GPS_TX_PIN);
TinyGPSPlus gps;
//=====
// TEST LOCATION (Fallback)
//=====

```

```

struct TestLocation {
    const char *name;
    float lat;
    float lng;
};
TestLocation locations[] = {
    { "Main Building", 23.77250, 90.42540 },
    { "TARC Building", 23.77280, 90.42580 },
    { "Auditorium", 23.77220, 90.42510 },
    { "Library", 23.77290, 90.42520 },
    { "Cafeteria", 23.77210, 90.42560 },
    { "Main Gate", 23.77180, 90.42590 },
    { "Parking Lot", 23.77200, 90.42620 },
    { "Sports Field", 23.77320, 90.42500 },
    { "Lake / Green Area", 23.77240, 90.42480 },
    { "Residential Road", 23.77150, 90.42650 }
};
#define TEST_LOCATION_INDEX 0
//=====
// GLOBALS
//=====
bool wifiConnected = false;
unsigned long lastReadTime = 0;
//=====
// GPS FUNCTIONS
//=====
void initGPS() {
    gpsSerial.begin(GPS_BAUD);
    Serial.println("[GPS] NEO-6M initialized on D5(TX) / D6(RX)");
}
void readGPS() {
    while (gpsSerial.available() > 0) {
        gps.encode(gpsSerial.read());
    }
}
void getGPSLocation(float &lat, float &lng, int &sats, bool &fix) {
    readGPS();
    if (gps.location.isValid() && gps.location.isUpdated()) {
        lat = gps.location.lat();
        lng = gps.location.lng();
        sats = gps.satellites.value();
        fix = true;
    } else {
        TestLocation loc = locations[TEST_LOCATION_INDEX];
        lat = loc.lat;
        lng = loc.lng;
        sats = 0;
        fix = false;
    }
}
//=====
// FIREBASE READ & UPDATE
//=====
void updateFirebaseWithGPS() {
    // Read live data from Firebase (sent by ESP32 sender)
    String path = "/fpi/live";
    if (Firebase.RTDB.getJSON(&firebaseData, path.c_str())) {
        FirebaseJson &json = firebaseData.jsonObject();
        // Get GPS location
        float lat, lng;
        int sats;
        bool fix;
        getGPSLocation(lat, lng, sats, fix);
        // Create update with GPS data
    }
}

```

```

FirebaseJson updateJson;
updateJson.set("gpsLat", lat);
updateJson.set("gpsLng", lng);
updateJson.set("gpsSats", sats);
updateJson.set("gpsFix", fix);
updateJson.set("locationName", locations[TEST_LOCATION_INDEX].name);
updateJson.set("receiverTimestamp", String(millis()));
// Update the live node with GPS info
String updatePath = "/fpi/live/gps";
if (Firebase.RTDB.setJSON(&firebaseData, updatePath.c_str(), &updateJson)) {
    Serial.println("[Firebase] GPS data updated");
}
// Also create a map-ready entry
FirebaseJson mapJson;
mapJson.set("feeder", locations[TEST_LOCATION_INDEX].name);
mapJson.set("fault", json.get<String>("fault"));
mapJson.set("voltage", json.get<float>("psuVoltage"));
mapJson.set("current", json.get<float>("current"));
mapJson.set("lat", lat);
mapJson.set("lng", lng);
mapJson.set("timestamp", String(millis()));
String mapPath = "/fpi/map/latest";
Firebase.RTDB.setJSON(&firebaseData, mapPath.c_str(), &mapJson);
// Print received data
Serial.println("\n[Firebase] Live Data Received:");
Serial.print(" Fault: ");
Serial.println(json.get<String>("fault"));
Serial.print(" Voltage: ");
Serial.print(json.get<float>("psuVoltage"), 1);
Serial.println(" V");
Serial.print(" Current: ");
Serial.print(json.get<float>("current"), 3);
Serial.println(" A");
Serial.print(" Load: ");
Serial.println(json.get<String>("load"));
Serial.print(" GPS: ");
Serial.print(lat, 5);
Serial.print(", ");
Serial.println(lng, 5);
Serial.print(" Fix: ");
Serial.println(fix ? "YES" : "NO (fallback)");
} else {
    Serial.print("[Firebase] Read Error: ");
    Serial.println(firebaseData.errorReason());
}
}
//=====
// SETUP
//=====
void setup() {
    Serial.begin(115200);
    delay(1000);
    Serial.println("\n=====");
    Serial.println("FPI Receiver — Firebase + GPS v1");
    Serial.println("SSID: Tenda_B6B920");
    Serial.println("GPS: NEO-6M on D5/D6");
    Serial.println("=====");
    // WiFi
    WiFi.mode(WIFI_STA);
    WiFi.begin(WIFI_SSID, WIFI_PASS);
    Serial.print("[WiFi] Connecting");
    while (WiFi.status() != WL_CONNECTED) {
        delay(500);
        Serial.print(".");
    }
}

```

```

    }
    Serial.println();
    Serial.print("[WiFi] Connected! IP: ");
    Serial.println(WiFi.localIP());
    wifiConnected = true;
    // NTP time
    configTime(6 * 3600, 0, "pool.ntp.org", "time.google.com");
    // Firebase config
    firebaseConfig.database_url = FIREBASE_HOST;
    firebaseConfig.signer.tokens.legacy_token = FIREBASE_AUTH;
    Firebase.begin(&firebaseConfig, &firebaseAuth);
    Firebase.reconnectWiFi(true);
    Serial.println("[Firebase] Initialized");
    // GPS init
    initGPS();
    Serial.println("=====");
    Serial.println("Listening for FPI data...");
    Serial.println("=====");
    }
    //=====
    // LOOP
    //=====
    void loop() {
        // Read GPS continuously
        readGPS();
        // Update Firebase with GPS every 2 seconds
        if (millis() - lastReadTime > 2000) {
            lastReadTime = millis();
            updateFirebaseWithGPS();
        }
        yield();
    }
}

```

b) AC Setup Codes:

i) ESP32E sender:

```

/*
=====
ESP32 FPI BLE SENDER (STRICT NORMAL RANGE UNLATCH FIX)
GPS SERIAL MONITOR DIAGNOSTICS ADDED
=====
*/

#include <Arduino.h>
#include <NimBLEDevice.h>
#include <TinyGPSPlus.h>
#include <HardwareSerial.h>

// =====
// ALL GLOBAL VARIABLES (DECLARED FIRST TO PREVENT SCOPE ERRORS)
// =====

bool systemHalted = false;
bool bleConnected = false;

float lastSensorVoltage = 0.0;
float lastCurrent = 0.0;

double gpsLatitude = 0.0;
double gpsLongitude = 0.0;
unsigned long gpsSatellites = 0;

```

```

bool gpsFix = false;

float batteryVoltage = 0.0;
int batteryPercent = 0;

String currentFault = "NORMAL";
String previousFault = "NORMAL";
unsigned long faultCount = 0;
unsigned long faultStartTime = 0;
float faultTime = 0.0;

float faultTripCurrent = 0.0;
float faultTripVoltage = 0.0;
String faultTripType = "NORMAL";
unsigned long faultTripTimestamp = 0;

const unsigned long BLE_SEND_INTERVAL = 1000UL;
unsigned long lastBLESend = 0;

NimBLEServer *bleServer = nullptr;
NimBLECharacteristic *bleCharacteristic = nullptr;

// =====
// BLE DEFINITIONS
// =====

#define BLE_DEVICE_NAME "FPI_SENDER"
#define BLE_SERVICE_UUID "6E400001-B5A3-F393-E0A9-E50E24DCCA9E"
#define BLE_CHARACTERISTIC_UUID "6E400003-B5A3-F393-E0A9-E50E24DCCA9E"

// =====
// ESP32 PINS
// =====

#define CURRENT_SENSOR_PIN 34
#define BATTERY_ADC_PIN 35

#define GPS_RX_PIN 16
#define GPS_TX_PIN 17

#define NORMAL_LED 32
#define PEAKLOAD_LED 25
#define THEFT_LED 26
#define FAULT_LED 27

// =====
// CURRENT SENSOR HARDWARE & SCALING
// =====

#define ADC_MAX 4095.0
#define ADC_VOLTAGE 3.3
#define NUM_SAMPLES 300

#define CURRENT_SENSOR_DIVIDER_RATIO 2.0

// =====
// BATTERY VOLTAGE DIVIDER
// =====

#define BATTERY_TOP_RESISTOR 22000.0
#define BATTERY_BOTTOM_RESISTOR 10000.0

#define BATTERY_DIVIDER_RATIO \
  ((BATTERY_TOP_RESISTOR + BATTERY_BOTTOM_RESISTOR) / BATTERY_BOTTOM_RESISTOR)

```

```

#define BATTERY_CALIBRATION 1.000
#define BATTERY_FULL_VOLTAGE 12.00
#define BATTERY_EMPTY_VOLTAGE 5.00

// =====
// GPS
// =====

#define GPS_BAUD 9600

HardwareSerial GPSSerial(2);
TinyGPSPlus gps;

// =====
// CALIBRATION TABLES
// =====

const int CAL_POINTS = 5;

float voltageTable[CAL_POINTS] = {
    0.0000, // IDLE State
    0.0020, // NORMAL State Base
    0.8000, // Voltage at 2.39A
    1.0000, // Voltage at 2.50A
    2.5000 // Voltage at 6.20A
};

float currentTable[CAL_POINTS] = {
    0.00, // IDLE (OFF) -> 0.00A
    0.01, // NORMAL State Real Floor -> 0.01A
    2.39, // Measured fault current 1
    2.50, // Measured fault current 2
    6.20 // Measured fault current 3
};

// =====
// FAULT & VOLTAGE THRESHOLDS
// =====

#define IDLE_VOLTAGE_MAX      0.0008
#define NORMAL_VOLTAGE_MAX    0.0080

#define NORMAL_MIN_CURRENT 0.010
#define NORMAL_MAX_CURRENT 0.300
#define PEAKLOAD_MAX_CURRENT .70
#define THEFT_MAX_CURRENT 2.00

// =====
// BLE CALLBACKS
// =====

class ServerCallbacks : public NimBLEServerCallbacks {

    void onConnect(NimBLEServer *pServer, NimBLEConnInfo &connInfo) override {
        bleConnected = true;
        Serial.println("\nBLE RECEIVER CONNECTED");
    }

    void onDisconnect(NimBLEServer *pServer, NimBLEConnInfo &connInfo, int reason) override {
        bleConnected = false;
        Serial.println("\nBLE RECEIVER DISCONNECTED");
        NimBLEDevice::startAdvertising();
    }
}

```

```

};

// =====
// HELPER FUNCTIONS
// =====

String jsonEscape(const String &input) {

    String output = "";

    for (unsigned int i = 0; i < input.length(); i++) {

        char c = input[i];

        if (c == "")
            output += "\\\"";

        else if (c == "\\")
            output += "\\\\";

        else if (c == '\n')
            output += "\\n";

        else if (c == '\r')
            output += "\\r";

        else
            output += c;
    }

    return output;
}

String jsonBool(bool value) {
    return value ? "true" : "false";
}

// =====
// GPS PROCESSING
// =====

void processGPS() {

    while (GPSSerial.available()) {
        gps.encode(GPSSerial.read());
    }

    if (gps.location.isValid()) {

        gpsLatitude = gps.location.lat();
        gpsLongitude = gps.location.lng();
    }

    if (gps.satellites.isValid()) {

        gpsSatellites = gps.satellites.value();
    }

    gpsFix = (gps.location.isValid() && gps.location.age() < 5000);
}

// =====
// HARDWARE READINGS
// =====

```

```

float readSensorVoltage() {

    uint32_t total = 0;

    for (int i = 0; i < NUM_SAMPLES; i++) {

        total += analogRead(CURRENT_SENSOR_PIN);
        delayMicroseconds(300);
    }

    float adc = total / (float)NUM_SAMPLES;

    float pinVoltage = (adc * ADC_VOLTAGE / ADC_MAX);

    return (pinVoltage * CURRENT_SENSOR_DIVIDER_RATIO);
}

// =====
// BATTERY VOLTAGE
// =====

float readBatteryVoltage() {

    uint32_t total = 0;

    const int batterySamples = 50;

    for (int i = 0; i < batterySamples; i++) {

        total += analogRead(BATTERY_ADC_PIN);
        delayMicroseconds(500);
    }

    float adc = total / (float)batterySamples;

    float adcVoltage = adc * ADC_VOLTAGE / ADC_MAX;

    return (adcVoltage * BATTERY_DIVIDER_RATIO * BATTERY_CALIBRATION);
}

// =====
// BATTERY PERCENT
// =====

int calculateBatteryPercent(float voltage) {

    if (voltage >= BATTERY_FULL_VOLTAGE)
        return 100;

    if (voltage <= BATTERY_EMPTY_VOLTAGE)
        return 0;

    float percentage =
        (voltage - BATTERY_EMPTY_VOLTAGE) * 100.0 /
        (BATTERY_FULL_VOLTAGE - BATTERY_EMPTY_VOLTAGE);

    return constrain((int)percentage, 0, 100);
}

// =====
// BATTERY UPDATE
// =====

```

```

void updateBattery() {

    batteryVoltage = readBatteryVoltage();

    batteryPercent = calculateBatteryPercent(batteryVoltage);
}

// =====
// CALIBRATION LOGIC
// =====

float calculateCurrent(float voltage) {

    if (voltage < IDLE_VOLTAGE_MAX) {
        return 0.00;
    }

    if (voltage <= NORMAL_VOLTAGE_MAX) {
        return 0.01;
    }

    for (int i = 1; i < CAL_POINTS - 1; i++) {

        if (voltage <= voltageTable[i + 1]) {

            float v1 = voltageTable[i];
            float v2 = voltageTable[i + 1];

            float i1 = currentTable[i];
            float i2 = currentTable[i + 1];

            return i1 +
                ((voltage - v1) * (i2 - i1) / (v2 - v1));
        }
    }

    return currentTable[CAL_POINTS - 1];
}

// =====
// FAULT DETERMINATION
// =====

String determineFault(float voltage, float current) {

    if (voltage < IDLE_VOLTAGE_MAX)
        return "IDLE/Fault Not Cleared";

    if (current >= NORMAL_MIN_CURRENT && current < NORMAL_MAX_CURRENT)
        return "NORMAL";

    if (current >= NORMAL_MAX_CURRENT && current < PEAKLOAD_MAX_CURRENT)
        return "PEAK LOAD";

    if (current >= PEAKLOAD_MAX_CURRENT && current < THEFT_MAX_CURRENT)
        return "THEFT";

    if (current >= THEFT_MAX_CURRENT)
        return "FAULT";

    return "NORMAL";
}

// =====

```

```

// LED MANAGEMENT
// =====

void turnOffAllLEDs() {

    digitalWrite(NORMAL_LED, LOW);
    digitalWrite(PEAKLOAD_LED, LOW);
    digitalWrite(THEFT_LED, LOW);
    digitalWrite(FAULT_LED, LOW);
}

void updateFaultLEDs(String faultState, float current) {

    turnOffAllLEDs();

    if (faultState == "IDLE/Fault Not Cleared") {
        return;
    }

    if (current < NORMAL_MAX_CURRENT) {

        digitalWrite(NORMAL_LED, HIGH);

    } else if (current >= NORMAL_MAX_CURRENT &&
               current < PEAKLOAD_MAX_CURRENT) {

        digitalWrite(PEAKLOAD_LED, HIGH);

    } else if (current >= PEAKLOAD_MAX_CURRENT &&
               current < THEFT_MAX_CURRENT) {

        digitalWrite(THEFT_LED, HIGH);

    } else if (current >= THEFT_MAX_CURRENT) {

        digitalWrite(FAULT_LED, HIGH);
    }
}

// =====
// FAULT STATUS
// =====

void updateFaultStatus(String newFault) {

    if (newFault != "NORMAL" &&
        newFault != "IDLE/Fault Not Cleared" &&
        previousFault == "NORMAL") {

        faultCount++;
        faultStartTime = millis();
    }

    if (newFault != "NORMAL" &&
        newFault != "IDLE/Fault Not Cleared") {

        faultTime =
            (millis() - faultStartTime) / 1000.0;

    } else {

        faultTime = 0.0;
    }
}

```

```

    previousFault = newFault;
}

// =====
// BLE PACKET
// =====

String buildBLEPacket() {

    String json = "{}";

    json += "\"current\":" + String(lastCurrent, 3) + ",";
    json += "\"sensorVoltage\":" + String(lastSensorVoltage, 4) + ",";
    json += "\"batteryVoltage\":" + String(batteryVoltage, 2) + ",";
    json += "\"batteryPercent\":" + String(batteryPercent) + ",";

    json += "\"fault\":" +
        jsonEscape(currentFault) +
        "\", ";

    json += "\"faultCount\":" +
        String(faultCount) +
        "\", ";

    json += "\"faultTime\":" +
        String(faultTime, 2) +
        "\", ";

    json += "\"latitude\":" +
        String(gpsLatitude, 6) +
        "\", ";

    json += "\"longitude\":" +
        String(gpsLongitude, 6) +
        "\", ";

    json += "\"satellites\":" +
        String(gpsSatellites) +
        "\", ";

    json += "\"gpsFix\":" +
        jsonBool(gpsFix) +
        "\", ";

    json += "\"timestamp\":" +
        String(millis()) +
        "\", ";

    json += "\"systemHalted\":" +
        jsonBool(systemHalted) +
        "\", ";

    json += "\"bleConnected\":" +
        jsonBool(bleConnected);

    json += "}";

    return json;
}

// =====
// BLE DATA
// =====

```

```

void sendBLEData(bool force = false) {

    unsigned long now = millis();

    if (!force &&
        now - lastBLESend < BLE_SEND_INTERVAL)
        return;

    lastBLESend = now;

    String packet = buildBLEPacket();

    if (bleConnected &&
        bleCharacteristic != nullptr) {

        size_t length = packet.length();

        size_t chunkSize = 100;

        for (size_t i = 0; i < length; i += chunkSize) {

            String chunk =
                packet.substring(
                    i,
                    min(i + chunkSize, length)
                );

            bleCharacteristic->setValue(chunk.c_str());

            bleCharacteristic->notify();

            delay(15);
        }
    }

    // =====
    // SYSTEM RESUME
    // =====

    void resumeSystem(float liveVoltage, float liveCurrent) {

        systemHalted = false;

        currentFault = "NORMAL";

        previousFault = "NORMAL";

        faultTime = 0.0;

        lastSensorVoltage = liveVoltage;

        lastCurrent = liveCurrent;

        turnOffAllLEDs();

        digitalWrite(NORMAL_LED, HIGH);

        Serial.println(
            "\n>>> FAULT CLEARED: True normal current detected "
            "(0.01A - 0.60A). System unlatched."
        );
    }
}

```

```

// =====
// HALTED STATE
// =====

void handleHaltedState() {

    processGPS();

    updateBattery();

    float silentVoltage =
        readSensorVoltage();

    float silentCurrent =
        calculateCurrent(silentVoltage);

    String silentFaultState =
        determineFault(
            silentVoltage,
            silentCurrent
        );

    if (silentFaultState == "NORMAL" &&
        silentCurrent >= NORMAL_MIN_CURRENT) {

        resumeSystem(
            silentVoltage,
            silentCurrent
        );

        sendBLEData(true);

        return;
    }

    lastSensorVoltage =
        faultTripVoltage;

    lastCurrent =
        faultTripCurrent;

    currentFault =
        faultTripType;

    updateFaultLEDs(
        faultTripType,
        faultTripCurrent
    );

    sendBLEData(false);

    delay(500);
}

// =====
// SERIAL MONITOR DATA
// =====

void printSensorData() {

    Serial.println(
        "\n=====

```

```

Serial.print("Sensor Voltage : ");
Serial.print(lastSensorVoltage, 4);
Serial.println(" V");

Serial.print("Current (Clamp): ");
Serial.print(lastCurrent, 3);
Serial.println(" A");

Serial.print("Fault State  :");
Serial.println(currentFault);

Serial.print("Halted Status : ");
Serial.println(
  systemHalted ?
  "YES (Latched)" :
  "NO"
);

// =====
// GPS INFORMATION
// =====

Serial.println("-----");
Serial.println("GPS STATUS");

Serial.print("Latitude   :");
Serial.println(gpsLatitude, 6);

Serial.print("Longitude   :");
Serial.println(gpsLongitude, 6);

Serial.print("Satellites   :");
Serial.println(gpsSatellites);

Serial.print("GPS Fix      :");
Serial.println(
  gpsFix ?
  "YES" :
  "NO"
);

Serial.print("GPS UART Data :");

if (GPSSerial.available()) {
  Serial.println("AVAILABLE");
} else {
  Serial.println("NO DATA");
}

Serial.print("GPS Characters Processed : ");
Serial.println(gps.charsProcessed());

Serial.print("GPS Sentences Passed  : ");
Serial.println(gps.passedChecksum());

Serial.print("GPS Sentences Failed   : ");
Serial.println(gps.failedChecksum());

Serial.println(
  "=====
);
}

// =====

```

```

// BLE START
// =====

void startBLE() {

    NimBLEDevice::init(BLE_DEVICE_NAME);

    bleServer =
        NimBLEDevice::createServer();

    bleServer->setCallbacks(
        new ServerCallbacks()
    );

    NimBLEService *service =
        bleServer->createService(
            BLE_SERVICE_UUID
        );

    bleCharacteristic =
        service->createCharacteristic(
            BLE_CHARACTERISTIC_UUID,
            NIMBLE_PROPERTY::READ |
            NIMBLE_PROPERTY::NOTIFY
        );

    bleCharacteristic->setValue(
        "{\"status\":\"BOOTING\"}"
    );

    service->start();

    NimBLEAdvertising *advertising =
        NimBLEDevice::getAdvertising();

    advertising->addServiceUUID(
        BLE_SERVICE_UUID
    );

    advertising->setName(
        BLE_DEVICE_NAME
    );

    advertising->start();
}

// =====
// SETUP
// =====

void setup() {

    Serial.begin(115200);

    delay(1000);

    pinMode(NORMAL_LED, OUTPUT);
    pinMode(PEAKLOAD_LED, OUTPUT);
    pinMode(THEFT_LED, OUTPUT);
    pinMode(FAULT_LED, OUTPUT);

    turnOffAllLEDs();

    analogReadResolution(12);
}

```

```

analogSetPinAttenuation(
  CURRENT_SENSOR_PIN,
  ADC_11db
);

analogSetPinAttenuation(
  BATTERY_ADC_PIN,
  ADC_11db
);

// =====
// GPS UART
// =====

GPSSerial.begin(
  GPS_BAUD,
  SERIAL_8N1,
  GPS_RX_PIN,
  GPS_TX_PIN
);

pinMode(
  BATTERY_ADC_PIN,
  INPUT
);

updateBattery();

startBLE();

Serial.println("\n=====");
Serial.println("FPI SENDER STARTED");
Serial.println("GPS UART INITIALIZED");
Serial.print("GPS RX PIN : GPIO");
Serial.println(GPS_RX_PIN);
Serial.print("GPS TX PIN : GPIO");
Serial.println(GPS_TX_PIN);
Serial.print("GPS BAUD : ");
Serial.println(GPS_BAUD);
Serial.println("=====");
}

// =====
// MAIN LOOP
// =====

void loop() {

  if (systemHalted) {

    handleHaltedState();

    return;
  }

  processGPS();

  updateBattery();

  float voltage =
    readSensorVoltage();

  float current =

```

```

        calculateCurrent(voltage);

lastSensorVoltage =
    voltage;

lastCurrent =
    current;

currentFault =
    determineFault(
        voltage,
        current
    );

updateFaultStatus(
    currentFault
);

updateFaultLEDs(
    currentFault,
    current
);

printSensorData();

sendBLEData();

if (currentFault == "PEAK LOAD" ||
    currentFault == "THEFT" ||
    currentFault == "FAULT") {

    faultTripCurrent =
        current;

    faultTripVoltage =
        voltage;

    faultTripType =
        currentFault;

    faultTripTimestamp =
        millis();

    systemHalted =
        true;

    Serial.println(
        "\n>>> FAULT TRIPPED: System Latched Halted!"
    );

    sendBLEData(true);
}

delay(500);
}

```

ii) ESP32 receiver:

```

/*
=====
ESP32 FPI BLE RECEIVER (OPTIMIZED FOR FLASH & STACK)
=====

```

```

ARCHITECTURE:
ESP32 SENDER ---> (BLE) ---> ESP32 RECEIVER ---> (WiFi) ---> FIREBASE
=====
*/

#include <Arduino.h>
#include <WiFi.h>
#include <NimBLEDevice.h>
#include <ArduinoJson.h>
#include <Firebase_ESP_Client.h>

// =====
// CONFIGURATION
// =====

#define WIFI_SSID "Redmi Note 10"
#define WIFI_PASSWORD "12345678"

#define FIREBASE_HOST "fault-passage-indicator-default-rtdb.firebaseio.com"
#define FIREBASE_API_KEY "AIzaSyArzd2yOVHKV9hFCbjQ51-jXRKonys3VU"
#define USER_EMAIL "dashboard@fpi.local"
#define USER_PASSWORD "87654321"

#define SENDER_BLE_NAME "FPI_SENDER"
static NimBLEUUID serviceUUID("6E400001-B5A3-F393-E0A9-E50E24DCCA9E");
static NimBLEUUID charUUID("6E400003-B5A3-F393-E0A9-E50E24DCCA9E");

// =====
// GLOBAL VARIABLES
// =====

static NimBLEAdvertisedDevice* advDevice = nullptr;
static NimBLERemoteCharacteristic* remoteChar = nullptr;
static NimBLEClient* pClient = nullptr;
bool doConnect = false;
bool connected = false;

// Decoupling variables
String rxBuffer = "";
String completeJsonToProcess = "";
volatile bool newDataAvailable = false;

FirebaseData fbdo;
FirebaseAuth auth;
FirebaseConfig config;

struct TelemetryData {
    float current;
    float sensorVoltage;
    float batteryVoltage;
    int batteryPercent;
    String fault;
    unsigned long faultCount;
    float faultTime;
    double latitude;
    double longitude;
    unsigned long satellites;
    bool gpsFix;
    unsigned long timestamp;
    bool systemHalted;
    bool bleConnected;
} telemetry;

// =====

```

```

// INITIALIZATION
// =====

void initWiFi() {
  Serial.print("WiFi: ");
  WiFi.mode(WIFI_STA);
  WiFi.begin(WIFI_SSID, WIFI_PASSWORD);

  int attempts = 0;
  while (WiFi.status() != WL_CONNECTED && attempts < 20) {
    delay(500);
    Serial.print(".");
    attempts++;
  }

  if (WiFi.status() == WL_CONNECTED) {
    Serial.println(" Connected");
  } else {
    Serial.println(" Connection Pending...");
  }
}

void initFirebase() {
  config.host = FIREBASE_HOST;
  config.api_key = FIREBASE_API_KEY;
  auth.user.email = USER_EMAIL;
  auth.user.password = USER_PASSWORD;

  Firebase.reconnectWiFi(true);
  Firebase.begin(&config, &auth);
}

// =====
// FIREBASE UPLOAD
// =====

void uploadToFirebase() {
  if (WiFi.status() != WL_CONNECTED || !Firebase.ready()) return;

  FirebaseJson json;
  json.set("current", telemetry.current);
  json.set("sensorVoltage", telemetry.sensorVoltage);
  json.set("batteryVoltage", telemetry.batteryVoltage);
  json.set("batteryPercent", telemetry.batteryPercent);
  json.set("fault", telemetry.fault);
  json.set("faultCount", telemetry.faultCount);
  json.set("faultTime", telemetry.faultTime);
  json.set("latitude", telemetry.latitude);
  json.set("longitude", telemetry.longitude);
  json.set("satellites", telemetry.satellites);
  json.set("gpsFix", telemetry.gpsFix);
  json.set("timestamp", telemetry.timestamp);
  json.set("systemHalted", telemetry.systemHalted);
  json.set("senderBleConnected", telemetry.bleConnected);

  if (Firebase.RTDB.setJSON(&fbdo, "/fpi_telemetry", &json)) {
    Serial.println("Firebase push OK");
  } else {
    Serial.print("Firebase ERR: ");
    Serial.println(fbdo.errorReason());
  }
}

// =====

```

```

// DATA PROCESSING (RUNS IN MAIN LOOP THREAD)
// =====

void processCompleteJSON(const String& jsonString) {
    StaticJsonDocument<768> doc;
    if (deserializeJson(doc, jsonString)) return;

    telemetry.current      = doc["current"] | 0.0f;
    telemetry.sensorVoltage = doc["sensorVoltage"] | 0.0f;
    telemetry.batteryVoltage = doc["batteryVoltage"] | 0.0f;
    telemetry.batteryPercent = doc["batteryPercent"] | 0;
    telemetry.fault         = doc["fault"] | "UNKNOWN";
    telemetry.faultCount    = doc["faultCount"] | 0;
    telemetry.faultTime     = doc["faultTime"] | 0.0f;
    telemetry.latitude      = doc["latitude"] | 0.0;
    telemetry.longitude     = doc["longitude"] | 0.0;
    telemetry.satellites    = doc["satellites"] | 0;
    telemetry.gpsFix        = doc["gpsFix"] | false;
    telemetry.timestamp     = doc["timestamp"] | 0;
    telemetry.systemHalted  = doc["systemHalted"] | false;
    telemetry.bleConnected  = doc["bleConnected"] | false;

    Serial.println("\n-----");
    Serial.print("Fault: "); Serial.println(telemetry.fault);
    Serial.print("Current: "); Serial.print(telemetry.current, 3); Serial.println(" A");
    Serial.print("Halted Status: "); Serial.println(telemetry.systemHalted ? "YES (Latched)" : "NO");
    Serial.print("Battery: "); Serial.print(telemetry.batteryPercent); Serial.println("%");
    Serial.print("GPS: "); Serial.print(telemetry.latitude, 6); Serial.print(", "); Serial.println(telemetry.longitude, 6);
    Serial.println("-----");

    uploadToFirebase();
}

// =====
// BLE CALLBACKS (LIGHTWEIGHT)
// =====

void notifyCallback(NimBLERemoteCharacteristic* pRemoteChar, uint8_t* pData, size_t length, bool isNotify) {
    for (size_t i = 0; i < length; i++) {
        rxBuffer += (char)pData[i];
    }

    int openBrace = rxBuffer.indexOf('{');
    int closeBrace = rxBuffer.lastIndexOf('');

    if (openBrace != -1 && closeBrace != -1 && closeBrace > openBrace) {
        completeJsonToProcess = rxBuffer.substring(openBrace, closeBrace + 1);
        rxBuffer = rxBuffer.substring(closeBrace + 1);
        newDataAvailable = true;
    }
}

class ClientCallbacks : public NimBLEClientCallbacks {
    void onConnect(NimBLEClient* pClient) override {
        Serial.println("BLE Connected");
    }

    void onDisconnect(NimBLEClient* pClient, int reason) override {
        connected = false;
        NimBLEDevice::getScan()->start(0, false);
    }
};

class ScanCallbacks: public NimBLEScanCallbacks {

```

```

void onResult(const NimBLEAdvertisedDevice* advertisedDevice) override {
    if (advertisedDevice->isAdvertisingService(serviceUUID) || advertisedDevice->getName() ==
SENDER_BLE_NAME) {
        NimBLEDevice::getScan()->stop();
        advDevice = const_cast<NimBLEAdvertisedDevice*>(advertisedDevice);
        doConnect = true;
    }
}

bool connectToServer() {
    if (pClient == nullptr) {
        pClient = NimBLEDevice::createClient();
        pClient->setClientCallbacks(new ClientCallbacks(), false);
    }

    if (!pClient->connect(advDevice)) return false;

    NimBLERemoteService* pRemoteService = pClient->getService(serviceUUID);
    if (pRemoteService == nullptr) {
        pClient->disconnect();
        return false;
    }

    remoteChar = pRemoteService->getCharacteristic(charUUID);
    if (remoteChar == nullptr) {
        pClient->disconnect();
        return false;
    }

    if (remoteChar->canNotify()) {
        if (!remoteChar->subscribe(true, notifyCallback)) {
            pClient->disconnect();
            return false;
        }
    }

    connected = true;
    return true;
}

// =====
// MAIN SETUP & LOOP
// =====

void setup() {
    Serial.begin(115200);
    delay(1000);

    initWiFi();
    initFirebase();

    NimBLEDevice::init("FPI_RECEIVER");
    NimBLEScan* pScan = NimBLEDevice::getScan();
    pScan->setScanCallbacks(new ScanCallbacks());
    pScan->setInterval(45);
    pScan->setWindow(15);
    pScan->setActiveScan(true);
    pScan->start(0, false);
}

void loop() {
    if (doConnect) {
        if (!connectToServer()) {

```

```
    NimBLEDevice::getScan()->start(0, false);
  }
  doConnect = false;
}

if (newDataAvailable) {
  newDataAvailable = false;
  processCompleteJSON(completeJsonToProcess);
}

delay(10);
}
```

Assessment Guideline for Faculty

[The following assessment guideline is for faculty ONLY. **This portion is not applicable for students.**]

Assessment Tools and CO Assessment Guideline

	Distribution of assessment points among various COs assessed in different semesters														
PO	l	c	f	g	c	b	d	c	e	l	k	k	h	i	j
CO	C O 1	C O 2	C O 3	C O 4	C O 5	C O 6	C O 7	C O 8	C O 9	C O 10	C O 11	C O 12	C O 13	CO 14	CO 15
EEE 400C/ ECE 402C (Out of 100)							30	24	6	4	4	6	7	7	12
Project Final Report/ Project Progress Report							x	x	x	x	x	x	x		x
Demonstratio n of working prototype							x								x
Progress Presentation/ Final Presentation								x			x				
Peer-evaluati on*													x	x	
Instructor's Assessment*													x	x	
Demonstratio n at FYDP Showcase								x							x

Note: The star (*) marked deliverables/skills will be evaluated at various stages of the project.

Mapping of CO-PO-Taxonomy Domain & Level- Delivery-Assessment Tool

Sl.	CO Description	PO	Bloom's Taxonomy Domain/Level	Assessment Tools
CO7	Evaluate the performance of the developed solution with respect to the given specifications, requirements and standards	d	Cognitive/ Evaluate	• Demonstratio n of working prototype • Project Progress Report on working prototype
CO8	Complete the final design and development of the solution with necessary adjustment based on performance evaluation	c	Cognitive/ Create	• Project Final Report • Final Presentation • Demonstratio n at FYDP Showcase
CO9	Use modern engineering and IT tools to design, develop and validate the solution	e	Cognitive/ Understand, Psychomotor/ Precision	• Project Final Report
CO10	Conduct independent research, literature survey and learning of new technologies and concepts as appropriate to design, develop and validate the solution	l	Cognitive/ Apply	• Project Final Report
CO11**	Demonstrate project management skill in various stages of developing the solution of engineering design project	k	Cognitive/ Apply Affective/ Valuing	• Project Final Report • Project Progress presentation at various stages
CO12	Perform cost-benefit and economic analysis of the solution	k	Cognitive/ Apply	• Project Final Report

CO13	Apply ethical considerations and professional responsibilities in designing the solution and throughout the project development phases	h	Cognitive/ Apply Affective/ Valuing	• Peer-evaluation, • Instructor's Assessment • Final Report
CO14**	Perform effectively as an individual and as a team member for successfully completion of the project	i	Affective/ Characterization	• Peer-evaluation • Instructor's Assessment
CO15**	Communicate effectively through writings, journals, technical reports, deliverables, presentations and verbal communication as appropriate at various stages of project development	j	Cognitive/ Understand Psychomotor/ Precision Affective/ Valuing	• Project Final Report • Progress Presentations, • Final Presentation • Demonstration at FYDP Showcase

Note: The double star (**) marked CO will be assessed at various stages of the project through indirect deliverables.